

ON COMPLEXITY, COMPUTATION, AND GRAPH HOMOMORPHISMS

by

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Abstract

A wide range of problems in computer science, including constraint satisfaction, can be framed in the language of graph homomorphisms. There are problems for which the existence of a solution may be proved by means of the Axiom of Choice or a non-principal ultrafilter. Although such solutions exist abstractly, in practice we want solutions that are definable, and so it is natural to impose restrictions. Interestingly, this often makes the problems harder to solve, but the solutions have a concrete description. This thesis studies the cost of insisting on definable solutions. The parallel between computer-science and set-theoretic complexity is a recurring theme in the thesis.

Chapter 2 presents new proofs and results about the complexity of colouring problems for analytic graphs, with the main highlight being a resolution of the descriptive complexity of the 2-Borel-dicolourability problem, where we show that it is Σ_2^1 -complete. This rules out a countable basis for Borel digraphs of dichromatic number at least three. We also present a streamlined proof of the Carroy–Miller–Schrittesser–Vidnyánszky $\mathbb{L}_0$ dichotomy, and the solution to an open problem about infinite games played on a linear order.

In Chapter 3, we study the asymptotic behaviour of computations, motivated by neural networks of increasing depth, using tools from the topology of function spaces and, more specifically, the classification of Rosenthal compacta. We give explicit descriptions of new computational definability classes for the Alva–Dueñez–Iovino–Walton floating-point arithmetic framework, and establish connections to statistical learning theory.

Finally, Chapter 4 studies a group action on graphs introduced by Caro, Hansberg, and Montejano to study balanceability and interpolation properties in extremal combinatorics, and develops a descriptive-set-theoretic theory of these actions on infinite graphs.

Descriptive set theory provides the language for all three substantive chapters: Borel definability for Chapter 2, the Baire-class hierarchy for Chapter 3, and Polish group actions for Chapter 4.

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List of Publications

The contents of this thesis have been published in the following manner:

- **Section 2.2: A concise proof of the $\mathbb{L}_0$ dichotomy**
Contains original proofs first disseminated at the Fields Institute Set Theory Seminar, Toronto (March 28, 2025).
- **Section 2.3: No countable basis for Borel directed graphs of dichromatic number at least three**
This new result completes the picture for all finite Borel chromatic and dichromatic thresholds. Presented at the Caltech Logic Seminar (April 29, 2026).
- **Section 2.4: Uncountable sets and an infinite linear order game**
Adapted from [MWS25], where we provide a solution to a problem originally posed by Brian and Clontz. Presented at the 2023 Summer Meeting of the Canadian Mathematical Society in Ottawa.
- **Chapter 3: Deep Computations**
Primarily based on [DIMW⁺26] and a multi-part series at the Fields Institute Set Theory Seminar (2024–2026).
- **Section 4.2: Border tracing in oriented adjacency graphs of polygonal tilings**
Based on my contributions to [WMW26a].
- **Section 4.3: Recursive constructions**
Extends joint work found in [EHMWV25] and [CGH⁺23]. Presented at the 2022 Summer Meeting of the Canadian Mathematical Society in St John’s.
- **Section 4.4: Amoeba trees**
Based on [GGHMW].

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Chapter 1

Introduction

Consider the following version of the perfect set theorem.

Theorem 1.1 (Cantor). *Let C be a closed subset of the Cantor space 2^ω . Then C is uncountable if and only if there is a continuous injection from 2^ω into C .*

The statement says that the question of whether a closed subset of 2^ω is uncountable has a structurally simple answer: one either embeds the Cantor space itself, or the set must be countable. The Cantor space is the only obstruction to countability.

In the study of function spaces, a phenomenon of similar spirit manifests when comparing the sequential and topological closures of a set of functions.

On the Cantor space $X = 2^\omega$, the n -th coordinate map $p_n: 2^\omega \rightarrow 2$ is continuous for each $n < \omega$; however, $(p_n)_{n \in \mathbb{N}}$ has *no* convergent subsequences in $\mathbb{R}^X$ ([Tod97, Chapter 1.1]). In a sense, this example exhibits the worst failure of sequential convergence possible: the closure of $\{p_n\}$ in $\{0, 1\}^X$ (or in $\mathbb{R}^X$ for that matter) is homeomorphic to the Stone-Čech compactification $\beta\mathbb{N}$ of the (discrete) space of natural numbers, which has cardinality 2^{2^ω} .

The following theorem, proved by Haskell Rosenthal in 1974, is fundamental in functional analysis and captures a sharp division in the behavior of sequences in a Banach space. We denote by $C_p(X)$ the space of continuous real-valued functions on X with the topology of pointwise convergence. Recall also that $\beta\mathbb{N}$ is the Stone-Čech compactification of the discrete space of natural numbers.

Theorem 1.2 (Rosenthal’s Dichotomy, [Ros74]). *If X is Polish and $(f_n)_{n \in \mathbb{N}} \subseteq C_p(X)$ is pointwise bounded, then (f_n) has a convergent subsequence, or a subsequence whose closure in $\mathbb{R}^X$ is homeomorphic to $\beta\mathbb{N}$.*

This result is a precursor to the classification theorems about Rosenthal compacta that we use in Chapter 3 to establish new definability classes for various spaces of computations.

Let G be a graph with vertex set $V(G)$ and edge set $E(G)$. A k -colouring is a mapping from $V(G)$ to a set of k colours such that adjacent vertices are given different colours. We say G is k -colourable if it admits a k -colouring. The least number k of colours for which a k -colouring of G exists is denoted by $\chi(G)$. Throughout this thesis, we will often endow the sets of vertices and edges with a topology and impose additional restrictions on the colouring, such as it being a Borel mapping. In that case, the least number of colours k for which there is a Borel k -colouring is denoted by $\chi_B(G)$.

Graph colourings are special instances of *graph homomorphisms* between two graphs $f: G \rightarrow H$, that is, a mapping between the vertex sets that preserves the adjacencies: whenever x, y form an edge in G , $f(x), f(y)$ form an edge in H . Indeed, a graph G is k -colourable if and only if there exists a homomorphism from G into the complete graph on k vertices, K_k .

Given a finite graph, there is an algorithm that decides whether it is 2-colourable or not in polynomial time in the size of the graph. Informally, traversing the graph through its adjacencies while alternating colours produces either a valid 2-colouring, or a contradiction (the method reaches a vertex adjacent to an already-coloured vertex and is forced to use the same colour as its neighbour); by the limited number of colours, this contradiction implies that no 2-colouring exists. For finite graphs, deciding 3-colourability is NP-complete.

An elementary exercise for finite graphs proves that a graph is 2-colourable if and only if it has no odd cycles. There is a compactness argument that extends this result to infinite graphs, which is essentially the De Bruijn–Erdős theorem [BM08, 14.1.27]. This can be used to show that the *infinite Hamming graph* $\mathcal{H}$, defined in the subsequent paragraph, is 2-colourable.

Consider the Cantor space 2^ω and impose a graph structure on it by declaring two infinite binary sequences x and y adjacent if they differ at exactly one coordinate. The resulting graph, which we denote by $\mathcal{H}$, is 2-colourable as it lacks any odd cycles. Suppose that $A \subseteq 2^\omega$ is an independent set¹. Suppose also that A has the Baire property (for example, every Borel set has the Baire property). If A were not meagre, then it would be comeagre in some basic clopen set $[t] := \{x \in 2^\omega : t \subseteq x\}$. Let $f: 2^\omega \rightarrow 2^\omega$ be the homeomorphism that flips the $(|t| + 1)$ -st² coordinate, so that the image of $[t]$ under f is $[t]$ itself, and every x is adjacent to $f(x)$ in $\mathcal{H}$. Then $f[A]$ is also comeagre in $[t]$, but $A \cap f[A] = \emptyset$ by independence, which contradicts the Baire category theorem applied to the Polish subspace $[t]$. Therefore, every independent set with the Baire property is meagre, and thus 2^ω cannot be covered by countably many such classes.

The preceding argument illustrates the cost of imposing descriptive (computable or measurable) requirements on the solution, in this case of the colouring problem: namely, that the number of colours must be increased. These dichotomies share a common shape, formalized in descriptive set theory as the basis paradigm.

Kechris, Solecki, and Todorcević [KST99] introduced this notion of the *Borel chromatic number* precisely to study this phenomenon. Remarkably, they proved the following result, known as the $\mathbb{G}_0$ dichotomy: there exists a single Borel graph $\mathbb{G}_0$ such that

$$\chi_B(G) > \aleph_0 \iff \mathbb{G}_0 \rightarrow_B G.$$

The symbol $\mathbb{G}_0 \rightarrow_B G$ abbreviates the phrase “there is a Borel homomorphism from $\mathbb{G}_0$ to G .” The graph $\mathbb{G}_0$ forms a one-element *basis* for the class of Borel graphs with uncountable Borel chromatic number. One can interpret the existence of a small (i.e. singleton, or finite, or countable) basis as a sort of measure of complexity of the associated decision problem. For instance, the existence of $\mathbb{G}_0$ says that deciding whether a Borel graph has uncountable Borel chromatic number is simple: one only has to verify one possible ill-behaved property, namely containing a homomorphic copy of $\mathbb{G}_0$, to give a positive answer; and in some sense, finding this homomorphic copy is the *only* way of proving that a Borel graph has uncountable Borel chromatic number. In contrast, deciding

¹An *independent* set of vertices is a set disjoint from the edge relation, i.e. a set of pairwise non-adjacent vertices.

² $|t|$ denotes the length of the finite sequence t , i.e. its domain.

whether a Borel graph has infinite Borel chromatic number is a much more complex task, as no easily describable method exists for deciding this in terms of graph homomorphisms, and so different graphs probably require wildly different proof techniques, if any.

Similarly to the case for finite graphs in the opening paragraphs of the chapter, Carroy, Miller, Schrittemser, and Vidnyánszky [CMSV21] proved the existence of a one-element basis for analytic graphs of Borel chromatic number at least three, that is, there exists a Borel graph $\mathbb{L}_0$ with the following property. The result is stated as a corollary in Section 2.2.

Corollary 1.3 ($\mathbb{L}_0$ dichotomy). *For any analytic graph G on a Polish space, $\chi_B(G) \geq 3$ if and only if there is a continuous mapping $\mathbb{L}_0 \rightarrow_c G$.*

Remark 1.4. Finite graphs are considered trivial examples of analytic graphs. Concretely, for a finite space X and a graph G defined on it, $\chi_B(G) = \chi(G)$ and every set of vertices and every set of edges is both Borel and analytic. Note that if G is finite, then $\chi(G) \geq 3$ is equivalent to G having an odd cycle; and it is not too difficult to show that this condition is equivalent to there being a continuous mapping $\mathbb{L}_0 \rightarrow_c G$. In summary, the preceding result holds for finite graphs as well.

A new proof of Corollary 1.3 is given in Section 2.2, and a new antibasis result for *dicolourings*, a directed analogue of colourings, is established in Section 2.3.

It is worth seeing explicitly that the $\mathbb{G}_0$ dichotomy refines Theorem 1.1. Let $C \subseteq 2^\omega$ be closed and consider the complete graph $K_C = (C \times C) \setminus \Delta_C$, which is closed and in particular analytic. Apply the $\mathbb{G}_0$ dichotomy:

- If $\chi_B(K_C) \leq \aleph_0$, then C partitions into countably many K_C -independent Borel sets. But independent sets in a complete graph contain at most one point, so C is countable.
- Otherwise, there is a continuous homomorphism $\phi: (2^\omega, \mathbb{G}_0) \rightarrow (C, K_C)$, that is, a continuous map $\phi: 2^\omega \rightarrow C$ sending $\mathbb{G}_0$ -edges to pairs of distinct points. The closed equivalence relation $E_\phi = \{(x, y) : \phi(x) = \phi(y)\}$ is then disjoint from $\mathbb{G}_0$ off the diagonal, so each E_ϕ -class is $\mathbb{G}_0$ -independent, hence meagre; a standard Mycielski-style argument therefore yields a continuous injection $\iota: 2^\omega \rightarrow 2^\omega$ on which E_ϕ restricts to equality, and $\phi \circ \iota: 2^\omega \rightarrow C$ is the desired continuous injection.

Cantor's theorem thus appears as the instance of the $\mathbb{G}_0$ dichotomy for complete graphs on closed sets.

A recent paper by Todorćević and Vidnyánszky [TV21] rules out the existence of countable bases for any other finite colour threshold. In Section 2.3, a new result completes the picture for the *dichromatic* number, an analogue of the chromatic number for directed graphs.

This is an illustrative introduction to the dichotomies that occupy the present thesis: the perfect set theorem which foreshadows the $\mathbb{G}_0$ dichotomy, and the Rosenthal-style dichotomies underlying the Deep Computations exposition of Chapter 3, as well as other complexity-theoretic themes that recur throughout this work.

The same dichotomy phenomenon, sequential vs. topological closure, Baire-1 vs. wild, drives Chapter 3, where we identify new definability classes for spaces of computations.

It is worth pausing to note that the basis paradigm where small obstructions characterize a structural property predates descriptive combinatorics by decades. Two of the most celebrated results in graph theory take this form. The following two theorems (see [BM08, Section 10.5]) are

well-known characterizations of *planarity* in graphs in these terms. While the two bases consist of the same graphs, notice that the underlying ordering on graphs differs.

Theorem 1.5 (Wagner, 1937). *The set $\{K_5, K_{3,3}\}$ is a basis for non-planarity under the minor ordering: a finite graph G is non-planar if and only if there exists $B \in \{K_5, K_{3,3}\}$ such that $B \leq_{\text{minor}} G$.*

Theorem 1.6 (Kuratowski, 1930). *The set $\{K_5, K_{3,3}\}$ is a basis for non-planarity under the topological minor ordering: a finite graph G is non-planar if and only if there exists $B \in \{K_5, K_{3,3}\}$ such that G contains a subdivision of B .*

Planar graphs are a particular example of a broader class of graphs for which a much more general finite basis result exists.

Theorem 1.7 (Robertson–Seymour). *Finite graphs are well-quasi-ordered under the minor relation. In particular, for every minor-closed class $\mathcal{C}$ of finite graphs, there exists a finite set $\mathcal{B}$ such that $G \notin \mathcal{C}$ if and only if $B \leq_{\text{minor}} G$ for some $B \in \mathcal{B}$.*

Nevertheless, there are classes of graphs with interesting results that follow a very similar theme for which the preceding theorem does not apply. One such instance, discussed in Section 5.3, concerns asymmetric graphs, which are not subject to the Robertson–Seymour theorem by virtue of asymmetry failing to be minor-closed.

Theorem 1.8 (Schweitzer–Schweitzer [SS17]). *There exists a set $\mathcal{B}$ of exactly 18 finite graphs, the minimal asymmetric graphs, such that every finite asymmetric graph G contains some $B \in \mathcal{B}$ as an induced subgraph; that is, there is an injective homomorphism $\varphi: B \hookrightarrow G$ that preserves non-adjacency. Moreover, the elements of $\mathcal{B}$ are pairwise incomparable under the induced subgraph ordering.*

A precursor to the study of the group action presented in Chapter 4 involves the following dichotomy, which is the main result of our article [CGH⁺23].

Theorem 1.9 (Part of Theorem 4.19). *A graph G with an even number of edges $k > 0$ has constant balancing number³ if and only if there are injective homomorphisms $\frac{k}{2}K_2 \rightarrow G$ and $K_{1,k/2} \rightarrow G$.*

³Constant balancing number is defined in Subsection 4.3.1.

Chapter 2

Descriptive Combinatorics

2.1 Background

Throughout this entire chapter, G denotes an analytic graph on a Polish space $V(G) = X$. That is, X is a separable completely metrizable topological space, and G is a symmetric and irreflexive relation on X which, as a subspace of $X \times X$, is the continuous image of a Polish space.¹

We employ the symbol $H \rightarrow_c G$ (respectively, $H \rightarrow_B G$) to abbreviate the fact that there is a continuous (respectively Borel) homomorphism from the analytic graph H to the analytic graph G .

2.1.1 The Borel chromatic number

Recall the $\mathbb{G}_0$ *dichotomy* from Chapter 1: there exists a Borel graph $\mathbb{G}_0$ such that

$$\chi_B(G) > \aleph_0 \iff \mathbb{G}_0 \rightarrow_B G. \quad (2.1)$$

This dichotomy breaks down when one moves from uncountable to merely infinite Borel chromatic number. In [TV21], Todorćević and Vidnyánszky proved that the set of Borel graphs with infinite Borel chromatic number is Σ_2^1 -complete; their reduction in fact gives the same conclusion for closed subgraphs of the shift graph at the threshold $\chi_B \geq 4$. In particular, no countable family of Borel graphs can serve as a basis for this class. Moreover, their argument rules out the existence of a simple $\mathbb{G}_0$ -type dichotomy substituting $\aleph_0$ in (2.1) for any $k \geq 3$. Interestingly, when substituting the value 2, there again is a single Borel graph that makes a dichotomy like (2.1) hold. This is known as the $\mathbb{L}_0$ dichotomy, a result first established in [CMSV21] and proved concisely in Section 2.2.

A common situation is having an analytic set and the desire to apply to it a statement that holds for Borel sets instead. One way to circumvent this problem is by using Borel separation to “inflate” the analytic set into a Borel set with the same property. This technique is crucial, for instance, for the proof of the $\mathbb{G}_0$ dichotomy found in [Ber24]. Concretely, since the dichotomy is a statement about independent sets of vertices, the proof makes use of the following fact.

¹In contrast with graph-theoretical literature, in this context it is customary to use the variable G as the set of edges. The difference is inessential.

Proposition 2.1. *In the graph G , any analytic independent set of vertices is contained in a Borel independent set.²*

We present two proofs of this fact, original to this exposition, for several reasons. First, because [Ber24] omits it;³ it is an illustrative example of how different notions of separation get used in an argument; and it motivates the introduction of a more powerful technique required in Section 2.2. Specifically, Claim 2.10.1 states an analogous result for odd-walks, instead of merely independent sets. The added complexity is that odd-walks can have different lengths, and so proving the aforementioned claim is not as straightforward without invoking Lemma 2.3.

Proof. Suppose that $I \subseteq V(G)$ is an analytic independent set. We view the edge relation of G as a subset of $V(G) \times V(G)$, which is analytic by assumption.

First, consider the set

$$A_0 := \{x_0 \in V(G) : \exists x_1 \in I (x_0 x_1 \in G)\},$$

which consists of all vertices with at least one neighbour in I . To see that A_0 is analytic, write it as the projection onto the first coordinate of the set

$$(V(G) \times I) \cap G = \{(x_0, x_1) : x_1 \in I \text{ and } x_0 x_1 \in G\}.$$

This is the intersection of the analytic set $V(G) \times I$ with the analytic set G , hence analytic. Since I is independent, $I \cap A_0 = \emptyset$. As I and A_0 are disjoint analytic sets, the Lusin Separation Theorem [Kec95, Theorem 14.7] yields a Borel set $B_0 \subseteq V(G)$ with

$$I \subseteq B_0 \quad \text{and} \quad B_0 \cap A_0 = \emptyset.$$

The second condition means that no vertex of B_0 has any neighbour in I .

Next, let

$$A_1 := \{x_1 \in V(G) : \exists x_0 \in B_0 (x_0 x_1 \in G)\}$$

be the set of all vertices with at least one neighbour in B_0 . Analogously to before, A_1 is the projection onto the second coordinate of

$$(B_0 \times V(G)) \cap G = \{(x_0, x_1) : x_0 \in B_0 \text{ and } x_0 x_1 \in G\},$$

which is the intersection of the Borel set $B_0 \times V(G)$ with the analytic set G , and is therefore analytic. We claim that $I \cap A_1 = \emptyset$. Indeed, if some $x_1 \in I$ had a neighbour $x_0 \in B_0$, then x_0 would be a vertex of B_0 with a neighbour $x_1 \in I$, that is $x_0 \in A_0$, contradicting $B_0 \cap A_0 = \emptyset$. So again we have two disjoint analytic sets, and separation yields a Borel set $B_1 \subseteq V(G)$ with

$$I \subseteq B_1 \quad \text{and} \quad B_1 \cap A_1 = \emptyset.$$

The second condition means that no vertex of B_1 has any neighbour in B_0 .

²In the case that X is finite, this Proposition becomes trivial; see Remark 1.4.

³The fact is indeed left as an exercise to the reader in [Ber24].

Set $B := B_0 \cap B_1$, which is Borel and contains I . It remains to check that B is independent. Suppose for contradiction that $x_0, x_1 \in B$ with $x_0 x_1 \in G$. Then $x_0 \in B_0$, so x_1 is a vertex with a neighbour in B_0 , giving $x_1 \in A_1$. But $x_1 \in B_1$ and $B_1 \cap A_1 = \emptyset$, a contradiction. $\square$

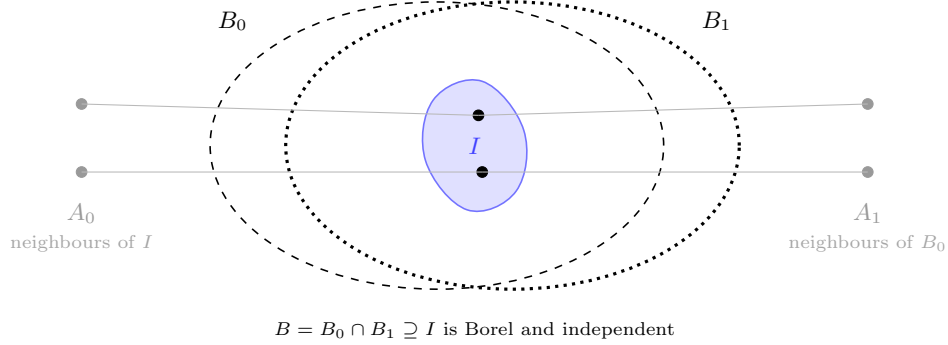


Figure 2.1: An illustration of Borel Separation in vertex sets.

The following notion, taken from [Kec95, Theorem 35.10, p. 285], is a well-known fact about analytic sets.

Definition 2.2. Let X be a Polish space and let $\Phi \subseteq \mathcal{P}(X)$. We say Φ is Π_1^1 on Σ_1^1 if for any Polish Y and any Σ_1^1 set $A \subseteq Y \times X$, the set $A_\Phi := \{y \in Y : A_y \in \Phi\}$, where $A_y := \{x \in X : (y, x) \in A\}$, is Π_1^1 .

Lemma 2.3 (First Reflection Theorem). *If Φ is as in the preceding definition, then for any Σ_1^1 set $A \in \Phi$, there is a Borel set $B \subseteq X$ such that $A \subseteq B$ and $B \in \Phi$.*

For example, the property of being an independent set of vertices is, as expected, Π_1^1 on Σ_1^1 . Indeed, let Φ denote the collection of all independent subsets of $V(G)$, let Y be a Polish space, and let $A \subseteq Y \times V(G)$ be Σ_1^1 . We must show that $A_\Phi = \{y \in Y : A_y \in \Phi\}$ is Π_1^1 . Observe that its complement consists precisely of those $y \in Y$ for which A_y contains an edge:

$$Y \setminus A_\Phi = \{y \in Y : \exists x_0, x_1 \in V(G) (x_0 x_1 \in G \wedge (y, x_0) \in A \wedge (y, x_1) \in A)\}.$$

The set $\{(y, x_0, x_1) : (y, x_0) \in A \wedge (y, x_1) \in A\}$ is Σ_1^1 , being the preimage of $A \times A$ under the continuous map $(y, x_0, x_1) \mapsto ((y, x_0), (y, x_1))$; intersecting with the Σ_1^1 set $\{(y, x_0, x_1) : x_0 x_1 \in G\}$ keeps it Σ_1^1 . Projecting onto Y therefore yields a Σ_1^1 set, so A_Φ is Π_1^1 , as required.

Consequently, Proposition 2.1 is an immediate corollary of Lemma 2.3: given an analytic independent set $I \in \Phi$, the First Reflection Theorem directly produces a Borel set $B \supseteq I$ with $B \in \Phi$. Note that this second proof, unlike the elementary one given above, generalises readily: one only needs to verify the Π_1^1 on Σ_1^1 property for the relevant family Φ , which, for more complex combinatorial properties, is typically a non-trivial step.

2.1.2 The Borel dichromatic number

All of the foregoing has been about undirected graphs; we now turn to the directed setting, where the basis landscape is richer and only partially understood. The *dichromatic number* of a directed graph was introduced by Neumann-Lara [NL82] as a natural directed analogue of the chromatic number.

In a *directed graph*, instead of having a set of edges as unordered pairs of vertices, we consider ordered pairs of vertices, called *arcs*. A *dicolouring* of a directed graph D is a map $c: V(D) \rightarrow k$ such that every colour class induces a subgraph without any directed cycles; the *dichromatic number* $\vec{\chi}(D)$ is the least such k . Equivalently, $\vec{\chi}(D)$ is the minimum number of vertex sets inducing acyclic subgraphs.

The following table summarizes the basis landscape, contrasting the undirected and directed cases at different thresholds.

Class	Existence of a countable basis	Reference
$\chi_B(G) > \aleph_0$	Yes (size 1: $\mathbb{G}_0$)	[KST99]
$\chi_B(G) > 2$	Yes (size 1: $\mathbb{L}_0$)	[CMSV21], Section 2.2
$\chi_B(G) > k, k \geq 3$	No ($\mathbf{\Pi}_2^1$ -complete)	[TV21]
$\vec{\chi}_B(D) > \aleph_0$	No (size $\mathfrak{c}$ basis exists)	[RX24]
$\vec{\chi}_B(D) > k, k \geq 2$	No ($\mathbf{\Pi}_2^1$ -complete)	Prop. 2.14, Rmk. 2.4

The notion of dichromatic number has attracted considerable attention in finite combinatorics. Neumann-Lara [NL82] established several foundational results, including analogues of classical lower bounds for the chromatic number. Bokal, Fijavž, Juvan, Kayll, and Mohar [BFJ⁺04] later introduced the circular dichromatic number of a directed graph and proved that the decision problem of whether a directed graph D satisfies $\vec{\chi}(D) \leq 2$ or not is NP-complete; this is in contrast with the usual chromatic number for (undirected) graphs, where one gets NP-completeness only after posing the question for three colours or more.

Very recently, Raghavan and Xiao [RX24] established a dichotomy for the Borel dichromatic number that generalizes the $\mathbb{G}_0$ dichotomy to directed graphs: they showed that a continuum-size family of directed graphs characterizes when a Borel directed graph has uncountable Borel dichromatic number. This continuum-size basis consists of pairwise Borel-incomparable directed graphs, but it is unknown whether any smaller basis exists; in particular, the question of whether a *countable* basis suffices is open.

The Borel dichromatic number can recover the Borel chromatic number in the following sense.

Remark 2.4. Given an undirected graph G , let $\vec{G}$ denote the directed graph obtained by replacing each edge $\{u, v\}$ with two opposing arcs (u, v) and (v, u) . Then $\vec{\chi}_B(\vec{G}) = \chi_B(G)$. Indeed, a subset of vertices is acyclic in $\vec{G}$ if and only if it is independent in G : any directed cycle in $\vec{G}$ would in particular contain an arc (u, v) with $uv \in G$, so a set containing both u and v is neither acyclic in $\vec{G}$ nor independent in G ; conversely, an independent set in G induces a directed graph in $\vec{G}$ with no arcs at all, hence trivially acyclic. Moreover, the map $\text{code}(G) \mapsto \text{code}(\vec{G})$ is Δ_1^1 , since it is built from the edge set using products and unions (Lemma 2.17(1),(3)). It follows that the Σ_2^1 -completeness of Borel k -colouring of undirected graphs [TV21] implies the Σ_2^1 -completeness of Borel k -dicolouring of directed graphs, for every $k \geq 3$. The case $k = 2$ is not covered by this reduction (since deciding 2-colourability of undirected graphs is trivial algorithmically, hence not Σ_2^1 -complete in the relevant sense), and is precisely the content of Proposition 2.14.

It is natural to ask whether the Todorćević–Vidnyánszky phenomenon also occurs in the directed setting at the remaining finite threshold, or if there is a directed analogue for the $\mathbb{L}_0$ dichotomy:

Is there a countable basis for the class of Borel directed graphs with Borel dichromatic number at least 3?

Section 2.3 gives a negative answer, thereby completing the picture for all finite thresholds of both the Borel chromatic and dichromatic numbers.

2.2 A concise proof of the $\mathbb{L}_0$ dichotomy

There are two natural strategies for proving Cantor’s perfect set theorem (Theorem 1.1); both proceed by analyzing the pruned tree $T \subseteq 2^{<\omega}$ with $C = [T]$.

Iterative derivation. Set $T^0 = T$. Given a countable ordinal α and a subtree T^α , call a node $t \in T^\alpha$ *terminal* if at most one branch of $C^\alpha := [T^\alpha]$ extends t , and let

$$T^{\alpha+1} = \{t \in T^\alpha : t \text{ is not terminal}\}, \quad T^\lambda = \bigcap_{\alpha < \lambda} T^\alpha \text{ for } \lambda \text{ a limit ordinal.}$$

The sequence stabilizes at some countable ordinal α_0 . If $T^{\alpha_0} = \emptyset$, then each $x \in C$ is the unique branch through some node that became terminal at some stage $\alpha < \alpha_0$, and the map $x \mapsto t_x$ (where t_x is the first such node along x) injects C into the countable set $\bigcup_\alpha (T^\alpha \setminus T^{\alpha+1})$; since the set of such nodes is a subset of the countable set $2^{<\omega}$, we conclude that C is countable. Otherwise T^{α_0} is a perfect tree, and $[T^{\alpha_0}] \subseteq C$ is the desired continuous copy of 2^ω .

Condensation. Let

$$T' = \{t \in T : \{x \in C : t \subseteq x\} \text{ is uncountable}\}.$$

If $T' = \emptyset$, then C is countable. Otherwise, every $t \in T'$ admits two incomparable extensions in T' , since the countably many extensions outside T' cannot exhaust the uncountably many $x \in C$ extending t . Thus T' is perfect and $[T']$ provides the desired continuous copy of 2^ω .

At first glance the condensation argument is more economical; one defines T' in a single stroke rather than iterating an operation through the countable ordinals. For trees in $2^{<\omega}$ the two strategies are equivalent in strength: when $T' = \emptyset$, the condition $\emptyset \notin T'$ already forces C to be countable directly, and the iterative-derivation argument injects the deleted nodes into the countable set $2^{<\omega}$, also without invoking choice. The asymmetry only becomes visible when one passes to the analogous perfect set theorem for analytic subsets of an arbitrary Polish space: there, the condensation argument concludes “no condensation points $\Rightarrow$ countable” by means of a Lindelöf-style cover of A by neighbourhoods with countable trace, and collapses the resulting countable union of countable sets, while the iterative-derivation argument generalizes more directly via a canonical injection into a countable index set. This distinction is innocuous over ZFC, but it becomes relevant when generalizing to the projective hierarchy under PD (projective determinacy): the iterative-derivation argument adapts directly there, via the same canonical injection into a countable index set, whereas the condensation argument’s Lindelöf-style cover would need to be revisited in that setting.

The original proof of the $\mathbb{G}_0$ dichotomy by Kechris–Solecki–Todorćević [KST99] follows the iterative-derivation pattern; a condensation argument was already sketched by Miller [Mil12], and later, independently, given in full by Bernshteyn [Ber24]. The natural generalization of the condensation argument to graphs is the homomorphism formulation, in which one looks for a continuous homomorphism from a fixed graph $\mathbb{H}$ into the graph under study. In this section, we give a new

argument for the $\mathbb{L}_0$ dichotomy using condensation; the original proof of [CMSV21] proceeds by iterative derivation. The graph $\mathbb{L}_0$ itself is constructed in Subsection 2.2.3 below, as the special case $c = c_0$ of a family $\{\mathbb{L}_c\}_{c \in \omega^\omega}$ of graphs built from the finite paths L_n^c introduced in Subsection 2.2.1.

Fact 2.5. *If $H \rightarrow_c G$ or, more generally if $H \rightarrow_B G$, then $\chi_B(H) \leq \chi_B(G)$.*

Using a standard greedy algorithm argument, it is clear that any finite graph G of maximum degree Δ satisfies $\chi(G) \leq \Delta + 1$. Interestingly, a similar fact, which we require for the main result of the section, holds for Borel graphs of uniformly bounded degree.

Proposition 2.6 ([KST99, Proposition 4.6]). *If G is a Borel graph on a Polish space all of whose degrees are bounded by the natural number k , then $\chi_B(G) \leq k + 1$.*

The central device of the proof is a σ -ideal of *small* sets of homomorphisms from finite path approximations into the target graph, where smallness is witnessed by a bounded odd-walk condition on vertex projections. The key lemma that largeness is preserved under the doubling operation is established via the First Reflection Theorem, replacing the original transfinite construction with a single Borel reflection argument. The continuous homomorphism from the canonical graph $\mathbb{L}_c$ into the target is then obtained as a limit of shrinking families of copies, in direct analogy with Bernshteyn's proof.

2.2.1 Finite path approximations

Fix the parameter $c \in \omega^\omega$ and a path of length ω on the vertex set $\{p_n\}_{n < \omega}$. That is, the edge set $\{(p_n, p_{n+1}) : n < \omega\}$. Naturally, one could assume that $p_n = n$, but the labelling helps distinguish these vertices from other indices.

Recall that if s and t are finite strings (of, say, natural numbers), then $s \frown t$ is the *concatenation* of s and t . Similarly, t^n is the repeated concatenation of t with itself, and (i) denotes the string of single value i . Next, recursively construct a family of graphs $\{L_n^c : n < \omega\}$ such that for all $n < \omega$: L_n^c is a finite path; if $n > 0$, the endpoints of L_n^c are $e_i^n := (p_0) \frown (0)^{n-1} \frown (i)$, for $i < 2$; and

$$\text{every vertex of } L_n^c \text{ is of the form } (p_k) \frown t \text{ for some } k \text{ and } t \in 2^{\leq n}. \quad (2.2)$$

Unless $t = \emptyset$, call such vertex a *non-path vertex*, and call the vertices of the form (p_k) the *path vertices* of L_n^c .

Start with L_0^c as the graph consisting of a single vertex p_0 and no edges, and set $e_0^0 = e_1^0 = p_0$. Now suppose that L_n^c has been constructed. L_{n+1}^c will be a path obtained from joining two copies of L_n^c by a path of $c(n) + 2$ edges (with $c(n) + 1$ *new* path vertices), but one must update the vertex labels in order to distinguish the two copies. One accomplishes this by simply appending 0 to each vertex of the first copy, and 1 to the other. Formally, start by adding the vertex $v \frown (i)$ for every $i < 2$ and $v \in L_n^c$. Adding the path

$$((e_1^n) \frown (0), p_0, p_1, \dots, p_{c(n)}, (e_1^n) \frown (1))$$

completes the construction of L_{n+1}^c , and its endpoints are

$$e_i^{n+1} = (e_0^n) \frown (i) = (p_0) \frown (0)^n \frown (i),$$

for $i < 2$. Explicitly, with $E(L_0^c) = \emptyset$, the edge set of L_{n+1}^c is

$$\begin{aligned} E(L_{n+1}^c) = & \{(u^\wedge(i), v^\wedge(i)) : (u, v) \in E(L_n^c), i < 2\} \\ & \cup \{((e_1^n)^\wedge(0), p_0), (p_{c(n)}, (e_1^n)^\wedge(1))\} \\ & \cup \{(p_j, p_{j+1}) : 0 \leq j < c(n)\}. \end{aligned}$$

This finishes the recursion. Note that to construct L_{n+1}^c from L_n^c , one only requires knowledge of the value $c(n)$.

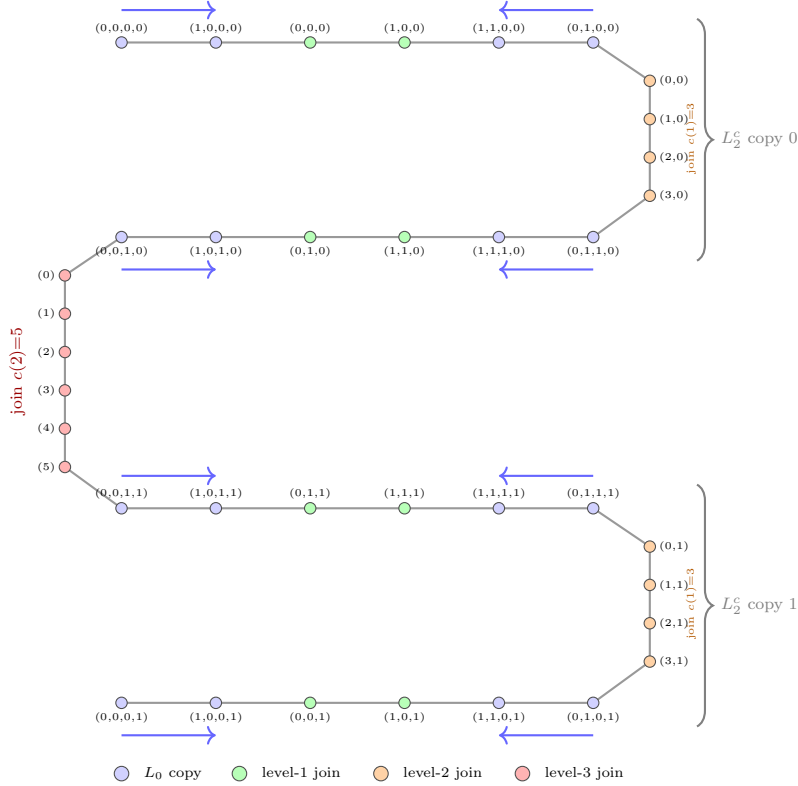


Figure 2.2: L_3^c for $c(0) = 1, c(1) = 3, c(2) = 5$.

To develop some intuition on what (2.2) says about a vertex v in L_n^c : the number $|v| - 1$ indicates that v was added that many stages ago in the recursion, at the position of p_k in the path joining the two copies of the previous paths; the sequence t tells the story, in order, of which of the two copies of the previous path the vertex lies in. In other words, vertices of the form $(p_k)^\wedge t$ for $t \in 2^n$ are precisely those that were copied from the original L_0^c in all possible positions along L_n^c , doubling the number of copies at each stage.

Based on this discussion, a vertex in some L_n^c is completely determined by the stage $n - m$ at which it was added in the position of p_k and then copied according to the binary sequence t .

2.2.2 A notion of smallness for copies

Now, define a notion of smallness analogous to the one presented in [Ber24]. By definition, the set of edges of G is the continuous image of some Polish space E , let us say under π . For a finite graph

H , consider $\text{Hom}(H, G)$ as the set of all maps

$$\begin{aligned}\varphi &: V(H) \rightarrow X \\ \varphi &: E(H) \rightarrow E\end{aligned}$$

that satisfy that for all $uv \in E(H)$, $\pi(\varphi(uv)) = \varphi(u)\varphi(v)$.⁴ Recall that $X = V(G)$ is the Polish space of vertices on which the analytic graph G is defined. Notice that $\text{Hom}(H, G)$ is also Polish space, being a closed subset of the product space $X^{V(H)} \times E^{E(H)}$.

Given $\mathcal{H} \subseteq \text{Hom}(H, G)$, define $\mathcal{H}(u) := \{\varphi(u) : \varphi \in \mathcal{H}\}$, which is analytic whenever $\mathcal{H}$ is Borel. Also, let $\mathcal{H}(uv) := \{\varphi(uv) : \varphi \in \mathcal{H}\}$.

Definition 2.7.

1. Let $k < \omega$. Say that the set A has property $\Phi(A, k)$ if $A \subseteq X$ is an analytic set of vertices of G such that all odd walks of G with endpoints in A have length at most $2k - 1$.
2. A Borel set $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$ is called *tiny* if there is a vertex $u \in V(L_n^c)$ and a natural k such that $\Phi(\mathcal{L}(u), k)$.
3. A set $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$ is *small* if it is the countable union of tiny sets.
4. A set is *large* if it is not small.

Hereditariness of Φ together with item (3) implies that the small sets form a σ -ideal in the powerset of $\text{Hom}(L_n^c, G)$.

When H is the graph with a single vertex $H = \bullet (= L_0^c)$, $\text{Hom}(H, G)$ can be identified with X , and for any $\mathcal{H} \subseteq \text{Hom}(\bullet, G)$, $\mathcal{H}(\bullet)$ is identified with $\mathcal{H}$ under this identification.

Now invoke the following auxiliary result that links the Φ property to Borel 2-colourability. The proof proceeds by induction on n using analytic separation; see [CMSV21, Claims 3.3–3.5] for details. Recall that a set of vertices of G is *invariant* if it is the union of connected components of G . In addition, if $A \subseteq X$, then $[A]_{E_G}$ is the union of all connected components of G that intersect A , that is, the set of vertices that can be reached from some element of A by a finite path. Therefore, a set A is invariant if and only if $A = [A]_{E_G}$.

Lemma 2.8 (Claim 3.5 of [CMSV21]). *If $\Phi(A, n)$, then there is an invariant Borel set $B \supseteq [A]_{E_G}$ that induces a 2-Borel-colourable subgraph of G , that is, there is a Borel proper colouring $c_B: G \upharpoonright_B \rightarrow 2$.*

Proposition 2.9. *If $\chi_B(G) > 2$ then X (via its identification with $\text{Hom}(\bullet, G)$) is large.*

Proof. Suppose that $\text{Hom}(\bullet, G) = \bigcup_{m < \omega} \mathcal{H}_m$ where each $\mathcal{H}_m$ is a tiny Borel set. For every $m < \omega$, Lemma 2.8 produces an invariant Borel set $B_m \supseteq [\mathcal{H}_m]_{E_G}$ with a Borel 2-colouring c_m . Define $c: X \rightarrow 2$ by letting $m(x) := \min\{m < \omega : x \in B_m\}$ and $c(x) = c_{m(x)}(x)$. Then c is a Borel 2-colouring because each B_m is invariant. $\square$

⁴Formally, φ is a pair of two maps, however one can assume that $V(H) \cap E(H) = \emptyset$ and then there is no risk of confusion.

Given that L_{n+1}^c is obtained from two copies of L_n^c joined by a path, it is convenient to have a means to referring to these distinct copies. Keeping property (2.2) in mind, for every $\varphi \in \text{Hom}(L_{n+1}^c, G)$ and $i < 2$, define $\varphi^i \in \text{Hom}(L_n^c, G)$ by $\varphi^i(u) = \varphi(u^\frown(i))$ for all $u \in V(L_n^c)$ and $\varphi^i(uv) = \varphi(u^\frown(i), v^\frown(i))$ for all $(u, v) \in E(L_n^c)$. Thus, given $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$, define $\mathcal{L}^{+c}$ as the set of all $\varphi \in \text{Hom}(L_{n+1}^c, G)$ for which $\varphi^i \in \mathcal{L}$ for both $i < 2$. Notice that if $\mathcal{L}$ is Borel, then so is $\mathcal{L}^{+c}$.

The length of the path added needs to be updated dynamically throughout the proof. Formally, one adjusts the parameter c by moving into a different branch of the Baire space when constructing the graphs L_n^c . Notice that, for two $c, d \in \omega^\omega$ and for every $n < \omega$, the condition $d \upharpoonright_n = c \upharpoonright_n$ implies that $L_k^c = L_k^d$ for all $k \leq n$.

Lemma 2.10.

- (a) Suppose that $n > 0$, that c only takes odd values, that k is a natural and $t \in 2^{<n}$ is such that $(p_k)^\frown t^\frown(i)$, for $i < 2$, are vertices of L_n^c (see (2.2)). Then, they are an odd distance apart in L_n^c .
- (b) For any $n < \omega$, if $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$ is a large Borel set, then for any $N < \omega$ there exists $d \in \omega^\omega$ such that $d \upharpoonright_n = c \upharpoonright_n$, $d(n) \geq N$ is odd, and $\mathcal{L}^{+d}$ is a nonempty subset of $\text{Hom}(L_{n+1}^d, G)$.
- (c) For any $n, N < \omega$, if $\mathcal{L} \subseteq \text{Hom}(L_n^c, G)$ is a large Borel set, then there exists $d \in \omega^\omega$ such that $d \upharpoonright_n = c \upharpoonright_n$, $d(n) \geq N$ is odd, and $\mathcal{L}^{+d}$ is a large subset of $\text{Hom}(L_{n+1}^d, G)$.

Proof. For (a), since the suffix bits of $(p_k)^\frown t^\frown(0)$ and $(p_k)^\frown t^\frown(1)$ differ only in the last (most recently appended) bit, both vertices come from the common vertex $(p_k)^\frown t$ in L_{n-1}^c , and they sit in opposite top-level copies of L_{n-1}^c inside L_n^c . Their distance in L_n^c is therefore twice the distance from $(p_k)^\frown t$ to e_1^{n-1} in L_{n-1}^c , plus the length of the stage- n joining path, which is $c(n-1) + 2$. Since $c(n-1)$ is odd, this distance is odd.

For (b), since $\mathcal{L}$ is large, it is not tiny. Hence, for every vertex $u \in V(L_n^c)$ and every natural k , $\Phi(\mathcal{L}(u), k)$ fails. In particular, fix the gluing vertex $u_0 := e_1^n \in V(L_n^c)$ (recall that $e_1^0 = p_0$). Then for every k there exist $\varphi_0, \varphi_1 \in \mathcal{L}$ and an odd walk $(v_0, v_1, \dots, v_m)$ in G with $v_0 = \varphi_0(u_0)$, $v_m = \varphi_1(u_0)$, and $m > 2k - 1$. Choosing k large enough, it is ensured that $m \geq N + 2$ and m is odd. Since m is odd, $d(n) := m - 2$ is also odd and satisfies $d(n) \geq N$. For each $1 \leq j \leq m$, since $(v_{j-1}, v_j) \in E(G) = \pi[E]$, pick $e_j \in E$ with $\pi(e_j) = (v_{j-1}, v_j)$. Setting $d(x) = c(x)$ for all $x \neq n$ and noting $L_n^c = L_n^d$ (since $d \upharpoonright_n = c \upharpoonright_n$), define $\varphi \in \text{Hom}(L_{n+1}^d, G)$ by $\varphi^0 = \varphi_0$, $\varphi^1 = \varphi_1$, $\varphi(p_i) = v_{i+1}$ for $0 \leq i \leq m - 2 = d(n)$, and the edge assignments $\varphi(u_0^\frown(0), p_0) = e_1$, $\varphi(p_j, p_{j+1}) = e_{j+1}$ for $0 \leq j < m - 2$, and $\varphi(p_{m-2}, u_0^\frown(1)) = e_m$. Then $\varphi \in \mathcal{L}^{+d}$.

Next, prove (c) by contradiction. Recall that constructing L_{n+1}^c from L_n^c only requires knowing the value of $c(n)$. Suppose that for all d that agree with c except possibly in the n -th value, with $d(n) \geq N$ odd, $\mathcal{L}^{+d} = \bigcup_m \mathcal{F}_m^d$ where each $\mathcal{F}_m^d$ is a tiny Borel subset of $\text{Hom}(L_{n+1}^d, G)$. For each m and d , let $w_m^d \in V(L_{n+1}^d)$ and $k_m^d < \omega$ be such that $\Phi(\mathcal{F}_m^d(w_m^d), k_m^d)$.

Since $w_m^d \in V(L_{n+1}^d)$, it is either a *non-path vertex* of the form $u^\frown(i)$ for some $u \in V(L_n^c)$ and $i < 2$, or a *path vertex* p_j for some $0 \leq j \leq d(n)$ in the joining path. We claim that in either case, there exists a vertex $u_m^d \in V(L_n^c)$ and $\ell_m^d < \omega$ such that $\Phi(\mathcal{F}_m^d(u_m^d \frown(i_m^d)), \ell_m^d)$ for some $i_m^d < 2$ where $u_m^d \frown(i_m^d)$ is a non-path vertex of L_{n+1}^d .

Indeed, if $w_m^d = u^\wedge(i)$ is already non-path, take $u_m^d = u$, $i_m^d = i$, $\ell_m^d = k_m^d$. If $w_m^d = p_j$ is a path vertex of L_{n+1}^d , then the joining path connects $e_1^{n\wedge}(0)$ to $e_1^{n\wedge}(1)$ via $p_0, \dots, p_{d(n)}$. Since p_j and $e_1^{n\wedge}(0)$ are at distance $j+1$ in L_{n+1}^d , every homomorphism $\varphi \in \mathcal{F}_m^d$ maps them to vertices connected by a walk of length $j+1$ in G . Given any odd walk $(a_0, \dots, a_r)$ in G with $a_0, a_r \in \mathcal{F}_m^d(e_1^{n\wedge}(0))$, witnessed by $\varphi, \varphi' \in \mathcal{F}_m^d$, the walk $(\varphi(p_j), \dots, \varphi(e_1^{n\wedge}(0))) = a_0, \dots, a_r = \varphi'(e_1^{n\wedge}(0)), \dots, \varphi'(p_j))$ has length $r + 2(j+1)$, which has the same parity as r (hence is also odd), and has endpoints in $\mathcal{F}_m^d(p_j)$. By $\Phi(\mathcal{F}_m^d(p_j), k_m^d)$, $r + 2(j+1) \leq 2k_m^d - 1$, so $r \leq 2(k_m^d - j - 1) - 1$. It follows that $\Phi(\mathcal{F}_m^d(e_1^{n\wedge}(0)), \max\{0, k_m^d - j - 1\})$. Thus, set $u_m^d = e_1^n$, $i_m^d = 0$, $\ell_m^d = \max\{0, k_m^d - j - 1\}$.

Claim 2.10.1. *For each m and d , there is a Borel set $B_m^d \supseteq \mathcal{F}_m^d(u_m^d \wedge (i_m^d))$ such that $\Phi(B_m^d, \ell_m^d)$.*

Proof of Claim. Use the First Reflection Theorem (Lemma 2.3). Fix $\ell = \ell_m^d$. It suffices to verify that $\Phi_\ell := \{A \subseteq X : \Phi(A, \ell)\}$ is Π_1^1 on Σ_1^1 . Let Y be Polish and $A \subseteq Y \times X$ be Σ_1^1 . Fix a Polish space N and a continuous surjection $g : N \rightarrow A$. For each odd $r \geq 2\ell + 1$, define

$$C_r := \{(y, x_0, \dots, x_r, n_0, n_1, \hat{e}_1, \dots, \hat{e}_r) \in Y \times X^{r+1} \times N^2 \times E^r : \\ g(n_0) = (y, x_0) \wedge g(n_1) = (y, x_r) \wedge \bigwedge_{j < r} \pi(\hat{e}_{j+1}) = (x_j, x_{j+1})\}.$$

Then C_r is closed (by the continuity of g and π), and $D_r := \text{proj}_Y(C_r)$ is Σ_1^1 . Hence $D := \bigcup \{D_r : r \text{ odd}, r \geq 2\ell + 1\}$ is Σ_1^1 , and $A_{\Phi_\ell} = \{y \in Y : A_y \in \Phi_\ell\} = Y \setminus D$ is Π_1^1 . Since $\mathcal{F}_m^d(u_m^d \wedge (i_m^d))$ is analytic and satisfies $\Phi(\cdot, \ell)$, the First Reflection Theorem yields the desired Borel. $\dashv$

Take

$$\mathcal{H}_m^d = \{\varphi \in \mathcal{L} : \varphi(u_m^d) \in B_m^d\},$$

which is also a Borel subset of $\text{Hom}(L_n^c, G)$. We claim that $\mathcal{H}_m^d$ is tiny. Indeed, $\mathcal{H}_m^d(u_m^d) = \{\varphi(u_m^d) : \varphi \in \mathcal{L} \wedge \varphi(u_m^d) \in B_m^d\} \subseteq B_m^d$, so $\Phi(\mathcal{H}_m^d(u_m^d), \ell_m^d)$.

Now, the set of pairs (u_m^d, i_m^d) ranges over the finite set $V(L_n^c) \times 2$ (which does not depend on d), and (m, d) range over countably many values. Thus $\bigcup_{m,d} \mathcal{H}_m^d$ is a countable union of tiny sets, hence small. The set $\mathcal{L}_- := \mathcal{L} \setminus \bigcup_{m,d} \mathcal{H}_m^d$ is therefore large (and Borel). By part (b) applied to $\mathcal{L}_-$ with parameter N , there exist φ and d with $d(n) \geq N$ odd such that $\varphi \in \mathcal{L}_-^{+d} \subseteq \mathcal{L}^{+d}$. Then $\varphi \in \mathcal{F}_m^d$ for some m . It follows that $\varphi^{i_m^d}(u_m^d) = \varphi(u_m^d \wedge (i_m^d)) \in \mathcal{F}_m^d(u_m^d \wedge (i_m^d)) \subseteq B_m^d$, hence $\varphi^{i_m^d} \in \mathcal{H}_m^d$. But $\varphi \in \mathcal{L}_-^{+d}$ implies $\varphi^{i_m^d} \in \mathcal{L}_-$, contradicting $\mathcal{L}_- \cap \mathcal{H}_m^d = \emptyset$. $\square$

2.2.3 Construction of a minimal graph of Borel chromatic number at least 3

The goal of this subsection is to construct a family of graphs $\mathbb{L}_c$ indexed by $c \in \omega^\omega$.

Fix c and consider X_c as the set of all tuples $(m, k, x) \in \mathbb{N} \times \mathbb{N} \times 2^\mathbb{N}$ such that either $m = 0$ and $k = 0$, or $m \geq 1$ and $k \leq c(m-1)$; this is a closed subspace of the product space, and hence Polish. Define, for each (m, k, x) with $m \leq n$, $\pi_n(m, k, x)$ as the vertex of L_n^c determined by these parameters. Formally, recalling the concatenation notation of Subsection 2.2.1 and (2.2), $\pi_n(m, k, x) := (p_k)^\wedge x \upharpoonright_{n-m}$, i.e. the vertex of L_n^c obtained by extending p_k with the first $n-m$ bits of x . For example, $\pi_3(1, 2, 0110 \dots) = (p_2, 0)$ can be seen in Figure 2.2 labelled as $(2, 0)$.

Finally, $\mathbb{L}_c$ is the graph on X_c where two (n_i, k_i, x_i) , for $i < 2$, are adjacent if and only if the $\pi_n(n_i, k_i, x_i)$ are adjacent in all L_n^c with $n \geq \max\{n_0, n_1\}$. Some basic properties of $\mathbb{L}_c$ follow.

Lemma 2.11. (a) Suppose that $c \in \omega^\omega$ is unbounded. Two vertices $(n_i, k_i, x_i) \in X_c$, for $i < 2$, are in the same connected component of $\mathbb{L}_c$ if and only if there are $t_i \in 2^{<\omega}$ and $x \in 2^\omega$ such that $|t_0| - |t_1| = n_1 - n_0$ and $x_i = t_i \hat{\ } x$.

(b) $\mathbb{L}_c$ has no cycles and all vertices of $\mathbb{L}_c$ have degree 2 except for the vertex $(0, 0, 0^\omega)$, which has degree 1.

(c) If $c \in \omega^\omega$ is unbounded and takes only odd values, then $\chi_B(\mathbb{L}_c) = 3$.

Proof. For (a), let $(n_i, k_i, x_i) \in X_c$ for $i < 2$. Suppose first that they lie in the same connected component, so there is a finite $\mathbb{L}_c$ -path between them, of some length D . Then for every n for which both projections are defined, the corresponding L_n^c -distance is at most D . Since c is unbounded, the joining path at stage n has length $c(n-1) + 2$, which exceeds D for all sufficiently large n , so neither the $\mathbb{L}_c$ -path nor the realizing L_n^c -paths cross the stage- n joining path; consequently, $\pi_n(n_0, k_0, x_0)$ and $\pi_n(n_1, k_1, x_1)$ lie in the same top-level copy of L_{n-1}^c inside L_n^c . By the construction of L_n^c , two vertices $(p_{k_0})^\wedge x_0 \upharpoonright_{n-n_0}$ and $(p_{k_1})^\wedge x_1 \upharpoonright_{n-n_1}$ are in the same top-level copy of L_{n-1}^c in L_n^c if and only if their last binary digits agree, that is, $x_0(n-n_0-1) = x_1(n-n_1-1)$ for all large n . Writing $x_0 = t_0 \hat{\ } x$ and $x_1 = t_1 \hat{\ } x$ with $|t_0| = n_1 - n_0 + |t_1|$ (assuming $n_0 \leq n_1$), one obtains the stated condition.

Conversely, suppose $x_i = t_i \hat{\ } x$ with $|t_0| - |t_1| = n_1 - n_0$, and assume without loss of generality that $n_0 \leq n_1$. Set $N := \max\{n_0 + |t_0|, n_1 + |t_1|\} = n_0 + |t_0|$, and observe that for every $n \geq N$ the suffixes $x_i \upharpoonright_{n-n_i}$ both end with the same tail $x \upharpoonright_{n-N}$. By the recursive description of L_n^c , this common tail places both projections $\pi_n(n_0, k_0, x_0)$ and $\pi_n(n_1, k_1, x_1)$ in a single copy of L_N^c inside L_n^c (the copy specified by the bits of $x \upharpoonright_{n-N}$). Consequently, the L_n^c -distance between them equals their L_N^c -distance, a fixed value D independent of $n \geq N$. Let $v_0, v_1, \dots, v_D$ be the unique path of length D in L_N^c between $\pi_N(n_0, k_0, x_0)$ and $\pi_N(n_1, k_1, x_1)$. Each v_j pulls back uniquely to an element $u_j \in X_c$ (with $u_0 = (n_0, k_0, x_0)$ and $u_D = (n_1, k_1, x_1)$); since the v_j, v_{j+1} remain adjacent in every L_n^c with $n \geq N$ (their relative position is preserved by the doubling construction once they sit within a common copy of L_N^c), the u_j form a $\mathbb{L}_c$ -path connecting (n_0, k_0, x_0) to (n_1, k_1, x_1) .

For (b), every vertex of L_n^c has degree at most 2, so the same holds for $\mathbb{L}_c$. It follows from acyclicity of the finite paths L_n^c that $\mathbb{L}_c$ is acyclic. The vertex $(0, 0, 0^\omega)$ maps under π_n to $e_0^n = (p_0)^\wedge 0^n$, which is an endpoint of L_n^c , hence has degree 1 in L_n^c for all $n \geq 1$. Thus $(0, 0, 0^\omega)$ has degree 1 in $\mathbb{L}_c$. For any other vertex $(m, k, x) \in X_c$, the image $\pi_n(m, k, x)$ is an interior vertex of L_n^c for all sufficiently large n , so it has degree exactly 2.

(c) That $\chi_B(\mathbb{L}_c) \leq 3$ follows from Proposition 2.6 and part (b). Argue the other inequality by contradiction. Suppose that there is a Borel partition of X_c into two independent sets $X_c = I_0 \cup I_1$. Since X_c is Polish, the Baire category theorem guarantees that some I_ε is non-meagre, hence comeagre in a basic open set $U = \{(m, k, x) \in X_c : x \upharpoonright_r = t\}$ for some m, k, r , and $t \in 2^r$. In particular, for a comeagre set of $x \in 2^\mathbb{N}$ extending t , $(m, k, x) \in I_\varepsilon$. Since $(m, k, t^\wedge(0)^\wedge y)$ and $(m, k, t^\wedge(1)^\wedge y)$ are in the same connected component (by part (a), which applies because c is unbounded) for any $y \in 2^\mathbb{N}$, and I_ε is comeagre in U , one can find y such that both $(m, k, t^\wedge(0)^\wedge y)$ and $(m, k, t^\wedge(1)^\wedge y)$ lie in I_ε . But by Lemma 2.10(a), these two vertices are an odd distance apart in $\mathbb{L}_c$ (as $(p_k)^\wedge t^\wedge(0)$ and $(p_k)^\wedge t^\wedge(1)$ are an odd distance apart in every L_n^c for n large enough), contradicting I_ε being independent. $\square$

The main result is then split into two parts.

Theorem 2.12. *For any analytic graph G , either $\chi_B(G) \leq 2$ or there is a c such that $\mathbb{L}_c \rightarrow_c G$.*

Moreover, c can be chosen to be unbounded and take only odd values.

Theorem 2.13. *Let $c, d \in \omega^\omega$ be unbounded and only taking odd values, then $\mathbb{L}_c$ and $\mathbb{L}_d$ are continuously homomorphically equivalent.*

The proof of Theorem 2.13 is purely combinatorial: one constructs homomorphisms between $\mathbb{L}_c$ and $\mathbb{L}_d$ by iteratively extending partial maps along finite path components, using the “large gap property” guaranteed by unboundedness. The argument does not depend on the method used to establish Theorem 2.12; see [CMSV21, Proposition 4.2 and Claim 4.1] for details.

Combining Theorems 2.12 and 2.13, fixing any unbounded $c_0 \in \omega^\omega$ taking only odd values (e.g., $c_0(0) = 1$ and $c_0(n) = 2n - 1$ for $n > 0$ ⁵; the original $\mathbb{L}_0$ of [CMSV21] corresponds to one such admissible choice), and setting $\mathbb{L}_0 := \mathbb{L}_{c_0}$, the $\mathbb{L}_0$ dichotomy of [CMSV21] is recovered:

Corollary 1.3 ($\mathbb{L}_0$ dichotomy). *For any analytic graph G on a Polish space, $\chi_B(G) \geq 3$ if and only if there is a continuous mapping $\mathbb{L}_0 \rightarrow_c G$.*

2.2.4 Proof of Theorem 2.12

By Lemma 2.11(c) and Fact 2.5, it is clear that both conditions cannot hold simultaneously. Now suppose that G is an analytic graph of Borel chromatic number at least three. The theorem is proved by constructing a valid c and a continuous homomorphism witnessing $\mathbb{L}_c \rightarrow_c G$.

By Proposition 2.9, X is large. Fix compatible complete metrics ρ on X and δ on E . Since $L_0^c = \bullet$, $\text{Hom}(L_0^c, G)$ may be identified with X and set $\mathbb{L}_0 := X$, which is large. Recursively construct a function $c \in \omega^\omega$ (taking only odd values) and a sequence of large Borel sets $\mathcal{L}_n \subseteq \text{Hom}(L_n^c, G)$ such that for all n ,

- (1) $\mathcal{L}_{n+1} \subseteq \mathcal{L}_n^{+c}$;
- (2) for all $u \in V(L_n^c)$, $\text{diam}_\rho(\overline{\mathcal{L}_n(u)}) < 2^{-n}$; and
- (3) for all $(u, v) \in E(L_n^c)$, $\text{diam}_\delta(\overline{\mathcal{L}_n(u, v)}) < 2^{-n}$.

Given $\mathcal{L}_n$ large, apply Lemma 2.10(c) with parameter $N := n+1$ to obtain $d \in \omega^\omega$ with $d|_n = c|_n$, $d(n) \geq n+1$ odd, and $\mathcal{L}_n^{+d}$ large; set $c(n) := d(n)$. This automatically forces c to take only odd values and to be unbounded. To arrange conditions (2) and (3) for $n+1$: since X and E can each be covered by countably many closed sets of ρ -diameter (respectively δ -diameter) less than $2^{-(n+1)}$, and small sets form a σ -ideal, one may intersect with the preimages of these covers and discard the resulting small pieces, passing to a large Borel subset $\mathcal{L}_{n+1} \subseteq \mathcal{L}_n^{+c}$ satisfying (2) and (3) at stage $n+1$.

By (1), for all $n \geq m$,

$$\mathcal{L}_{n+1}(\pi_{n+1}(m, k, x)) \subseteq \mathcal{L}_n(\pi_n(m, k, x));$$

indeed, given that $\pi_{n+1}(m, k, x) = \pi_n(m, k, x) \smallfrown (x(n-m))$, each $\varphi \in \mathcal{L}_{n+1}$ satisfies $\varphi^{x(n-m)} \in \mathcal{L}_n$ and

$$\varphi^{x(n-m)}(\pi_n(m, k, x)) = \varphi(\pi_{n+1}(m, k, x)).$$

⁵It is perhaps worth noting that we picked a *different* parameter for Figure 2.2 purely for expository purposes.

By the same reasoning, if $\pi_n(m_0, k_0, x_0)$ and $\pi_n(m_1, k_1, x_1)$ are adjacent in L_n^c for all large n , then

$$\begin{aligned} & \mathcal{L}_{n+1}(\pi_{n+1}(m_0, k_0, x_0), \pi_{n+1}(m_1, k_1, x_1)) \\ & \subseteq \mathcal{L}_n(\pi_n(m_0, k_0, x_0), \pi_n(m_1, k_1, x_1)). \end{aligned}$$

Thus, the sets $\overline{\mathcal{L}_n(\pi_n(m, k, x))}$ are closed and nested, with diameters tending to zero (see (2) from before). Since ρ is a complete metric, Cantor's intersection theorem applies. For $(m, k, x) \in X_c$, let $f(m, k, x)$ be the unique point in

$$\bigcap_{n=m}^{\infty} \overline{\mathcal{L}_n(\pi_n(m, k, x))} = \bigcap_{n=0}^{\infty} \overline{\mathcal{L}_{n+m}((p_k)^\frown x \upharpoonright_n)}.$$

The map $f: X_c \rightarrow X$ is continuous.

It remains to check that it is a homomorphism from $\mathbb{L}_c$ to G . Suppose that $(m_i, k_i, x_i)_{i < 2}$ are adjacent in $\mathbb{L}_c$, that is, for all $n \geq n_0 := \max\{m_0, m_1\}$, $\pi_n(m_i, k_i, x_i)$ are adjacent in L_n^c . Similar to the preceding paragraph, the fact that δ is a complete metric implies that there is a unique edge $e \in E$ inside

$$\bigcap_{n \geq n_0} \overline{\mathcal{L}_n(\pi_n(m_0, k_0, x_0), \pi_n(m_1, k_1, x_1))}.$$

By the continuity of π , $(f(m_0, k_0, x_0), f(m_1, k_1, x_1)) = \pi(e) \in E(G)$, in other words, the images form an edge of G , and this concludes the proof.

The hereby presented argument shows that the core of the $\mathbb{L}_0$ dichotomy, namely the construction of a continuous homomorphism from an $\mathbb{L}_c$ graph into any analytic graph of Borel chromatic number at least three, admits a proof that closely parallels Bernshteyn's argument for the $\mathbb{G}_0$ dichotomy, with the challenge of adding a dynamically-changing parameter to the construction.

The main new ingredient is the replacement of the independence-based σ (sets whose vertex projections are independent) with the odd-walk-bounded σ -ideal defined by the Φ property. It remains an interesting question whether analogous adaptations can yield simplified proofs of other dichotomy results in Borel combinatorics, or even entirely new ones.

2.3 No countable basis for Borel directed graphs of dichromatic number at least three

This section proves that the Borel directed graphs whose vertex set admits a partition into two Borel acyclic sets form a Σ_2^1 -complete set; equivalently, that deciding whether a Borel directed graph has Borel dichromatic number at least 3 is a Π_2^1 -complete problem. It follows that no countable family of Borel directed graphs can serve as a basis for this class under Borel homomorphism and, more generally, that any basis must be at least as complex as Π_2^1 .

The proof lifts the classical NP-completeness reduction of Bokal, Fijavž, Juvan, Kayll, and Mohar to the Borel setting, using the coding framework of Thornton. Combined with a straightforward reduction from undirected to directed colouring problems, this completes the picture for finite Borel chromatic and dichromatic thresholds: for every finite k , the set of Borel (directed) graphs admitting a Borel k -(di)colouring is Σ_2^1 -complete, and the complement therefore admits no countable basis.

This contrasts with the uncountable threshold, where a single-element basis exists for Borel chromatic number (Kechris–Solecki–Todorćević) and a continuum-size basis exists for Borel dichromatic number (Raghavan–Xiao).

We work throughout with *nice codes* in the sense of Thornton [Tho22a, Tho22b]; see Subsection 2.3.1 for a brief review.

Proposition 2.14. *The set of nice codes for Borel directed graphs admitting a Borel 2-dicolouring is Σ_2^1 -complete. Equivalently, the set of nice codes for Borel directed graphs with Borel dichromatic number at least 3 is Π_2^1 -complete.*

Corollary 2.15. *There is no countable family $\mathbf{B}$ of Borel directed graphs that forms a basis for Borel directed graphs of Borel dichromatic number at least 3 under Borel homomorphism.*

The proof of Proposition 2.14 proceeds by adapting the classical NP-completeness reduction from 2-colouring of 3-uniform hypergraphs to 2-dicolouring of directed graphs [BFJ⁺04] to the Borel setting. The key step is verifying that this reduction is Δ_1^1 in the codes, following the template established by Thornton in [Tho22b, Appendix A] for a similar result about edge 3-colouring. I remark that 2-dicolouring of directed graphs is *not* a CSP in the sense of Thornton’s algebraic framework [Tho22a], since acyclicity is not a constraint expressible as a homomorphism to a fixed finite template; thus Proposition 2.14 does not follow directly from Thornton’s general results.

The picture reveals a notable difference between the undirected and directed settings at the uncountable threshold: a single graph suffices as a basis in the undirected case, while a continuum-size family is needed for directed graphs. With finitely many colours, both settings exhibit the same completeness obstruction to a countable basis, though at different thresholds: the phenomenon begins at $k = 3$ for undirected graphs and already at $k = 2$ for directed graphs, mirroring the classical NP-completeness landscape. The case $k = 2$ for directed graphs, established in this paper, is the last remaining piece of this picture (see Remark 2.4).

Several questions are posed in Section 5.1.

2.3.1 Coding framework

Briefly recall the coding apparatus from [Tho22b, Appendix A]; see also [Tho22a, Section 4]. The reader already familiar with this framework may skip to Subsection 2.3.2.

Fix a good ω -parametrization $U \subseteq \omega \times \mathcal{N}$ of Π_1^1 [Tho22a, Theorem A.5].

Definition 2.16 ([Tho22a, Definition A.6]). A *simple code* for a Δ_1^1 set $B \subseteq \mathcal{N}$ is a pair $(e, i) \in \omega^2$ with $U_e = \mathcal{N} \setminus U_i = B$.

A *nice coding* is a triple $(\mathcal{C}, \mathcal{D}^\Pi, \mathcal{D}^\Sigma)$ where:

- (i) $\mathcal{C} \subseteq \omega$ is Π_1^1 (the set of *codes*);
- (ii) $\mathcal{D}^\Pi \subseteq \omega \times \mathcal{N}$ is Π_1^1 and $\mathcal{D}^\Sigma \subseteq \omega \times \mathcal{N}$ is Σ_1^1 ;
- (iii) for every $e \in \mathcal{C}$, $\mathcal{D}_e^\Pi = \mathcal{D}_e^\Sigma$;
- (iv) every Δ_1^1 set $B \subseteq \mathcal{N}$ has a code $e \in \mathcal{C}$ with $B = \mathcal{D}_e^\Pi$;
- (v) there are recursive functions translating between simple and nice codes in both directions.

Fix nice codings for $\Delta_1^1(\mathcal{N}^k)$ for all k , writing $\mathcal{D}_e$ for $\mathcal{D}_e^\Pi$.

Lemma 2.17 ([Tho22a, Lemma A.7]). *Fix a Δ_1^1 linear order $\preceq$ on $\mathcal{N}$. The following operations are Δ_1^1 in the codes:*

- (1) $(A, B) \mapsto A \cup B$;
- (2) $(A, B) \mapsto A \cap B$;
- (3) $(A, B) \mapsto A \times B$;
- (4) $A \mapsto \text{dom}(A)$, for relations with countable sections;
- (5) $(A, f) \mapsto f(A)$, for countable-to-one f ;
- (6) $A \mapsto$ the enumeration function of the finite sections of A along $\preceq$.

This lemma can typically be used as a black box: to verify that a construction is Δ_1^1 in the codes, it suffices to express it as a composition of the preceding operations.

2.3.2 The classical reduction

Now, I recall the reduction from 2-colouring of 3-uniform hypergraphs to 2-dicolouring of directed graphs. The NP-completeness of 2-dicolouring follows by combining this reduction with the well-known NP-completeness of 2-colouring 3-uniform hypergraphs. The reduction is due to Bokal et al. [BFJ⁺04]; I present it in a form convenient for the Borel lifting.

Fact 2.18. *The problem of deciding whether a 3-uniform hypergraph admits a 2-colouring is NP-complete.*

Definition 2.19 (Template gadget). Let

$$T_{\text{shared}} := \{a, b, c\} \quad T_{\text{aux}} := \{a', b', c'\} \quad T := T_{\text{shared}} \cup T_{\text{aux}}.$$

The *template gadget* F is the directed graph on vertex set T with arc set E_T as depicted in Figure 2.3.

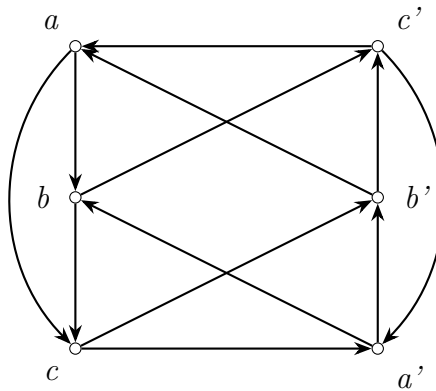


Figure 2.3: The template gadget F . Shared vertices $\{a, b, c\}$ correspond to hypergraph vertices; auxiliary vertices $\{a', b', c'\}$ are local to each gadget copy.

Lemma 2.20 (Gadget property). *A map $\sigma: \{a, b, c\} \rightarrow 2$ extends to a 2-dicolouring of F if and only if σ is not constant (i.e., not all three shared vertices receive the same colour).*

Proof. This is a finite verification on the structure of F ; see [BFJ⁺04, Theorem 3.1]. $\square$

Lemma 2.21 (Classical reduction). *For every 3-uniform hypergraph H , there is a directed graph $D(H)$ such that H is 2-colourable if and only if $D(H)$ is 2-dicolourable.*

Proof. Let $V = V(H)$ and fix a linear order on V . Define

$$E_{\text{sort}} = \{(v_1, v_2, v_3) \in E(H) : v_1 < v_2 < v_3\},$$

a canonical listing of hyperedges as ordered triples. For each

$$e = (v_1, v_2, v_3) \in E_{\text{sort}},$$

place a copy F_e of the template gadget, identifying the shared labels a, b, c with the hypergraph vertices v_1, v_2, v_3 and introducing fresh auxiliary vertices $(e, a'), (e, b'), (e, c')$. The directed graph $D(H)$ has vertex set

$$V_D = V \cup \{(e, t) : e \in E_{\text{sort}}, t \in T_{\text{aux}}\}$$

and arcs given by the union of the arcs of all copies F_e , with each shared label mapped to the corresponding vertex of V .

First, suppose that $c: V \rightarrow 2$ is a proper 2-colouring of the hypergraph H . For each $e \in E_{\text{sort}}$, the induced colouring on the shared vertices of F_e is non-constant, so by Lemma 2.20 it extends to a 2-dicolouring σ_e of F_e . Combining these extensions with c on shared vertices yields a globally well-defined map $\psi: V_D \rightarrow 2$. We now claim that ψ is a 2-dicolouring of $D(H)$. Suppose not, and let C be a monochromatic directed cycle of minimal length. Since every arc between vertices of V is directed according to the linear order $<$ on V , the cycle C cannot be contained entirely in V ; hence C contains an auxiliary vertex. Let F_e be a gadget containing an auxiliary vertex of C . Since σ_e is a 2-dicolouring of F_e , the cycle C cannot be contained entirely in F_e , so C contains a directed segment

$$x \rightarrow \cdots \rightarrow y \rightarrow \cdots \rightarrow z$$

lying inside F_e , with y an auxiliary vertex of F_e , $x, z \in \{a, b, c\}$ shared, and all interior vertices auxiliary in F_e . By minimality of C , this segment cannot be shortcut by an arc $x \rightarrow z$, so $(x, z) \notin E(F_e)$, leaving $(x, z) \in \{(b, a), (c, a), (c, b)\}$. Since the only arcs of F between shared vertices are $a \rightarrow b$, $a \rightarrow c$, $b \rightarrow c$, this forces $(z, x) \in E(F_e)$. The arc $z \rightarrow x$ together with the segment $x \rightarrow \cdots \rightarrow z$ forms a directed cycle in F_e , monochromatic in the colour class of C , contradicting that σ_e is a 2-dicolouring of F_e .

Conversely, let $\psi: V_D \rightarrow 2$ be a 2-dicolouring of $D(H)$ and set $\varphi = \psi|_V$. For any $e = (v_1, v_2, v_3) \in E_{\text{sort}}$, the restriction $\psi|_{V(F_e)}$ is a 2-dicolouring of F_e , so by the contrapositive of Lemma 2.20, $\varphi(v_1), \varphi(v_2), \varphi(v_3)$ are not all equal. Hence φ is a 2-colouring of H . $\square$

Remark 2.22. The converse direction requires no selection or choice: φ is simply the restriction of ψ to the original vertex set V , which is a subset of V_D by construction. This will be important in

Subsection 2.3.3, where it means that no uniformization argument is needed for the reverse direction of the Borel reduction.

2.3.3 The Borel reduction

I now verify that the construction of Subsection 2.3.2 lifts to the Borel setting. The argument follows the template of [Tho22a, Theorem A.9], where a similar verification for edge 3-colouring is carried out.

Fact 2.23 ([Tho22a, Corollary 4.6]). *The set of nice codes for locally finite⁶ Borel 3-uniform hypergraphs admitting a Borel 2-colouring is Σ_2^1 -complete.*

It therefore suffices to produce a Δ_1^1 reduction from the set of codes for Borel 2-colourable 3-uniform hypergraphs to the set of codes for Borel 2-dicolourable directed graphs.

Now I verify that the construction is Δ_1^1 in the codes. Let H be a locally finite Borel 3-uniform hypergraph with vertex set $V \subseteq \mathcal{N}$ and hyperedge set $E \subseteq V^3$. We treat E as a symmetric ternary relation, i.e. closed under permutations of its three coordinates, so that the sorted set E_{sort} defined below extracts a unique sorted triple per hyperedge. We fix a Δ_1^1 linear order $\preceq$ on $\mathcal{N}$ and encode the template gadget F , its label set T , and its arc set E_T by canonical finite sequences in $\mathcal{N}$.

Begin by sorting the hyperedges.

$$E_{\text{sort}} = \{(v_1, v_2, v_3) \in E : v_1 \preceq v_2 \preceq v_3\}.$$

This is the intersection of E with the ternary relation $\{(v_1, v_2, v_3) \in \mathcal{N}^3 : v_1 \preceq v_2 \preceq v_3\}$, which is itself Δ_1^1 as the conjunction of two pullbacks of the graph of $\preceq$. Hence E_{sort} is Δ_1^1 on codes by Lemma 2.17(2).

Then, build the vertex set. Define $V_D = V \cup (E_{\text{sort}} \times T_{\text{aux}})$, which is a union of V with a product, hence Δ_1^1 on codes by Lemma 2.17(1),(3).

Finally, build the arc set. For each $e = (v_1, v_2, v_3) \in E_{\text{sort}}$ and each template arc $(u, v) \in E_T$, define the *substitution map* $\varphi_H : E_{\text{sort}} \times E_T \rightarrow V_D^2$ that sends a pair $(e, (u, v))$ to the corresponding arc of $D(H)$, by replacing each label $\ell \in T$ with its realization in $D(H)$:

- if $\ell \in T_{\text{shared}}$ is the shared label in position i (i.e., $\ell \in \{a, b, c\}$ in position i), it maps to the vertex $v_i \in V$ occurring in $e = (v_1, v_2, v_3)$;
- if $\ell \in T_{\text{aux}}$, it maps to $(e, \ell) \in E_{\text{sort}} \times T_{\text{aux}}$.

Since the gadget is finite and fixed, the substitution map is recursive relative to the code for E_{sort} , obtained by finitely many coordinate projections and pairings. Adding these constructions over the finitely many template arcs $(u, v) \in E_T$ via Lemma 2.17(1) shows that the substitution map φ_H is Δ_1^1 in the code of H (with E_T treated as fixed throughout).

The arc set is $A_D = \varphi_H(E_{\text{sort}} \times E_T)$. To apply Lemma 2.17(5), we need φ_H to be countable-to-one. This holds: for any arc $\alpha \in A_D$, the preimage $\varphi_H^{-1}(\alpha)$ is finite, since E_T is finite and, by local finiteness of H , each vertex of V appears in only finitely many hyperedges in E_{sort} . By Lemma 2.17(1),(3),(5), the arc set A_D is Δ_1^1 on codes.

Applying the simple-to-nice code translation from the coding framework (Definition 2.16(v)), we obtain:

⁶A hypergraph is *locally finite* if every vertex has finitely many incident hyperedges.

Lemma 2.24. *The function that maps the code of H to the code of $D(H)$ is Δ_1^1 .*

Now I show that Borel colourability is preserved.

Claim 2.24.1. *If H is Borel 2-colourable, then $D(H)$ is Borel 2-dicolourable.*

Proof. Let $c: V \rightarrow 2$ be a Borel 2-colouring of H . For each $e \in E_{\text{sort}}$, the restriction of c to the shared vertices of F_e is non-constant, so by the gadget property (Lemma 2.20) there exists at least one extension to a 2-dicolouring of F_e . Denote by $\mathcal{C}_F$ the finite set of maps $T \rightarrow 2$ and define S as the set of all $(e, \sigma) \in E_{\text{sort}} \times \mathcal{C}_F$ such that σ is a 2-dicolouring of F_e extending the shared-vertex colours. The set S is Borel since the defining condition involves only finitely many acyclicity checks and colour-matching conditions, all determined by the Borel colouring c and the fixed finite template. Furthermore, S has finite nonempty sections S_e . By the Luzin–Novikov uniformization theorem, there is a Borel selector $e \mapsto \sigma(e) \in S_e$.

The map $\psi: V_D \rightarrow 2$ defined by

$$\psi(v) = \begin{cases} c(v), & v \in V; \\ \sigma(e)(t), & (e, t) \in E_{\text{sort}} \times T_{\text{aux}} \end{cases}$$

is well-defined (auxiliary vertices of distinct gadgets are disjoint). Since c and $e \mapsto \sigma(e)$ are Borel, and the decomposition of V_D into V and $E_{\text{sort}} \times T_{\text{aux}}$ is Borel, the map ψ is Borel.

We claim each colour class of ψ is acyclic. Suppose not, and let C be a monochromatic directed cycle of minimal length. Since every arc between vertices of V is directed according to the linear order $\prec$ on V , the cycle C cannot be contained entirely in V ; hence C contains an auxiliary vertex. Let F_e be a gadget containing an auxiliary vertex of C . Since $\sigma(e)$ is a 2-dicolouring of F_e , the cycle C cannot be contained entirely in F_e , so C contains a directed segment

$$x \rightarrow \cdots \rightarrow y \rightarrow \cdots \rightarrow z$$

lying inside F_e , with y an auxiliary vertex of F_e , $x, z \in \{a, b, c\}$ shared, and all interior vertices auxiliary in F_e . Once again, the minimality of C implies that this segment cannot be shortcut by an arc $x \rightarrow z$, so $(x, z) \notin E(F_e)$, leaving $(x, z) \in \{(b, a), (c, a), (c, b)\}$. Since the only arcs of F between shared vertices are $a \rightarrow b$, $a \rightarrow c$, $b \rightarrow c$, we have that $(z, x) \in E(F_e)$. The arc $z \rightarrow x$ together with the segment $x \rightarrow \cdots \rightarrow z$ forms a directed cycle in F_e , monochromatic in the colour class of C , contradicting that $\sigma(e)$ is a 2-dicolouring of F_e . $\square$

Claim 2.24.2. *If $D(H)$ is Borel 2-dicolourable, then H is Borel 2-colourable.*

Proof. Let $\psi: V_D \rightarrow 2$ be a Borel 2-dicolouring of $D(H)$ and define $\varphi = \psi|_V$. Since $V \subseteq V_D$ is Borel, φ is Borel. For any hyperedge $e = (v_1, v_2, v_3) \in E_{\text{sort}}$, the gadget F_e is a directed subgraph of $D(H)$, so $\psi|_{V(F_e)}$ is a 2-dicolouring of F_e , and the gadget property implies that $\varphi(v_1), \varphi(v_2), \varphi(v_3)$ are not all equal. Hence φ is a Borel 2-colouring of H . $\square$

Remark 2.25. Note that the reverse direction (Claim 2.24.2) uses neither the local finiteness of H nor any uniformization. Local finiteness of H is invoked at three points in the forward direction: it is a hypothesis of Fact 2.23; it makes the substitution map φ_H countable-to-one (in fact finite-to-one), enabling the application of Lemma 2.17(5) in the Δ_1^1 reduction; the sections S_e are finite (indeed uniformly bounded independently of H), enabling the application of Luzin–Novikov.

I can finally prove the main result of the section.

Proof of Proposition 2.14. By Fact 2.23, the set of nice codes for locally finite Borel 3-uniform hypergraphs with Borel 2-colourings is Σ_2^1 -complete. By Lemma 2.24, the map $\text{code}(H) \mapsto \text{code}(D(H))$ is Δ_1^1 . By Claims 2.24.1 and 2.24.2, H is Borel 2-colourable if and only if $D(H)$ is Borel 2-dicolourable. Thus the set of nice codes for Borel 2-dicolourable directed graphs is Σ_2^1 -hard.

For the upper bound, note that a directed graph D (given by a nice code) admits a Borel 2-dicolouring if and only if there exists a nice code e_c such that $\mathcal{D}_{e_c}$ defines a valid 2-dicolouring of D . The condition “ $\mathcal{D}_{e_c}$ is a 2-dicolouring of D ” is Π_1^1 in the codes e_c and e_D . Totality, that $\mathcal{D}_{e_c}$ defines a function $V(D) \rightarrow 2$, is a universal condition over $V(D)$. Acyclicity of each colour class is Borel (indeed, arithmetic), since a directed graph is acyclic if and only if it contains no finite directed cycle. This is a universal quantification (over finite sequences, coded in $\mathcal{N}$) of a Δ_1^1 matrix. The existential quantification over e_c gives a Σ_2^1 condition. Hence the set of codes for Borel 2-dicolourable directed graphs is Σ_2^1 -complete; its complement, the set of codes for directed graphs with $\bar{\chi}_B(D) \geq 3$, is Π_2^1 -complete. $\square$

Remark 2.26. The convention used in [TV21] and [Tho22a] is to state the Σ_2^1 -completeness of the class of graphs *admitting* a colouring. We follow this convention.

Proof of Corollary 2.15. Suppose a countable basis $\mathbf{B} = \{H_1, H_2, \dots\}$ exists, so that

$$\bar{\chi}_B(D) \geq 3 \iff \exists H_n \in \mathbf{B}, \exists \text{ Borel homomorphism } H_n \rightarrow D.$$

For each fixed H_n , the condition “there exists a Borel homomorphism $H_n \rightarrow D$ ” is Σ_2^1 in the code for D : one existential quantifier over a nice code e_f for a Borel function $f: V(H_n) \rightarrow V(D)$, and the requirement that f preserves arcs is Π_1^1 in the codes.

The countable union $\bigcup_n \{D : H_n \rightarrow_B D\}$ is therefore Σ_2^1 . And Proposition 2.14 asserts that the $\{D : \bar{\chi}_B(D) \geq 3\}$ is Π_2^1 -complete. A set that is simultaneously Σ_2^1 and Π_2^1 -hard would give $\Pi_2^1 \subseteq \Sigma_2^1$, contradicting the fact that the projective hierarchy is strictly ordered. $\square$

Remark 2.27. The argument proves more than the non-existence of a countable basis. If $\mathbf{B}$ is any basis for the class of Borel directed graphs with $\bar{\chi}_B(D) \geq 3$, then the right-hand side of

$$\bar{\chi}_B(D) \geq 3 \iff \exists H \in \mathbf{B}, \exists \text{ Borel homomorphism } H \rightarrow D$$

must define a Π_2^1 -complete set, since the left-hand side is Π_2^1 -complete by Proposition 2.14. If $\mathbf{B} \in \Sigma_2^1$ (viewed as a subset of the code space), then “ $H \in \mathbf{B}$ ” is a Σ_2^1 condition on the code of H , and “there exists a Borel homomorphism $H \rightarrow D$ ” is Σ_2^1 in the codes of H and D (as in the proof of Corollary 2.15). The conjunction of two Σ_2^1 conditions is Σ_2^1 , and existential quantification preserves Σ_2^1 , so the right-hand side would be Σ_2^1 in the code of D . But this contradicts Π_2^1 -completeness of the left-hand side, since $\Sigma_2^1 \neq \Pi_2^1$. Hence $\mathbf{B} \notin \Sigma_2^1$: any basis must be at least Π_2^1 -hard as a subset of the code space, and is therefore at least as complex to describe as the set of all directed graphs without a Borel acyclic 2-colouring. The separation $\Sigma_2^1 \neq \Pi_2^1$ used here is a theorem of ZFC (the classical Hierarchy Theorem for the projective pointclasses, via a universal set and a diagonal argument; see e.g. [Kec95]); no determinacy hypothesis is needed anywhere in this thesis.

Acknowledgments I thank Dilip Raghavan for suggesting that the NP-completeness proof for dichromatic number could be relevant to the Borel setting, my doctoral advisor Spencer Unger for pointing me towards Thornton’s coding framework, and Riley Thornton for confirming that 2-dicolouring is not a CSP⁷ in the sense of [Tho22b], advising that the approach of [Tho22a, Appendix A] should be followed instead, and suggesting several of the questions in Section 5.1. I also thank Alekos Kechris for extending an invitation to present this result at the Caltech Logic Seminar, and pointing out a minor mistake during the presentation.

2.4 Uncountable sets and an infinite linear order game

This section addresses a question by Will Brian and Steven Clontz regarding an infinite game of perfect information with two players, called *Baker’s game*.

Games have many useful applications in fields like topology and set theory. Leaving many details out of this famous example, the Banach–Mazur game (see [Oxt57]) is played by two players who alternate turns picking subsets of ω^ω , forming a descending chain $W_0 \supseteq W_1 \supseteq \dots$. A previously agreed upon set $S \subseteq \omega^\omega$ constitutes the winning condition: Player I wins if there is an element $x \in S$ such that $x \in W_n$ for all n ; otherwise, Player II wins. A well-known result is that Player II has a winning strategy if and only if S is of the first category. In particular, Player II wins whenever S is countable.

The game studied in this section, Baker’s game, is similar: two players take turns selecting real numbers; Player I tries to ensure that some element in the fixed set S remains in between all of these real numbers. In [BC23], the authors show that Player II has a winning strategy if and only if the set S is countable. It makes sense to play this game in more general linear orders, and so they ask if it is possible for Player II to have a winning strategy on an uncountable set S instead, or if this property somehow only holds in the reals, thus providing a characterization of the reals in these terms.

In collaboration with the other author of [MWS25], we showed the existence of a dense linear order of arbitrary infinite cardinality on which Player II has a winning strategy on **every** set S , including uncountable ones. This construction demonstrates that the outcome of Baker’s game is highly sensitive to the order type of the underlying space and not merely a matter of cardinality. Concretely, the Brian–Clontz characterization does not extend to arbitrary dense linear orders.

2.4.1 The Cantor Game

In [Bak07], Matthew Baker introduced the following game on the real numbers. Fix a subset $S \subseteq \mathbb{R}$. Player I starts by picking a real number a_0 . Then, Player II picks a real number b_0 such that $a_0 < b_0$. In the n -th turn, Player I picks a real number a_n such that $a_{n-1} < a_n < b_{n-1}$ and then Player II picks a real number b_n such that $a_n < b_n < b_{n-1}$. After ω -many turns they form two sequences of real numbers $\{a_n\}$ and $\{b_n\}$ such that $a_0 < a_1 < \dots < a_n < \dots < b_n < \dots < b_1 < b_0$. Player I wins if and only if there exists $x \in S$ such that $a_n < x < b_n$ for all $n < \omega$. Otherwise, Player II wins.

It should be noted that the $[a_n, b_n]$ form a nested set of intervals, and thus there always exists real number x with the property that for all $n < \omega$, $a_n < x < b_n$. Proposition 2.31 generalizes the following observation when playing the game in an arbitrary dense linear order.

⁷Constraint satisfaction problem.

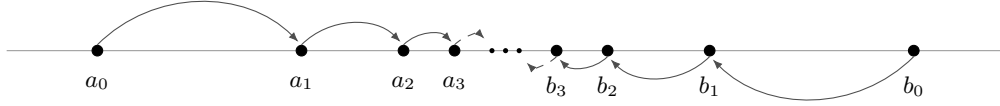


Figure 2.4: The nested construction of intervals in the Cantor Game.

Proposition 2.28 ([Bak07]). *If S is countable, then Player II has a winning strategy in the Cantor Game.*

The so-called *Cantor Game* was introduced as a game-theoretic way of proving that $\mathbb{R}$ is uncountable, which follows by the previous observation. Baker also proved that if S contains a perfect set, then Player I has a winning strategy. Later in [LaD17], M.D. Ladue (a student of Baker) proved that Player I has a winning strategy if and only if the set S contains a perfect set. The question of whether the converse of Proposition 2.28 is true remained open until 2022. In [BC23], Brian and Clontz used elementary submodels to prove the converse statement of proposition 2.28 above.

Question 2.1 ([BC23]). Is there a linear order $(X, <)$ and an uncountable $S \subseteq X$ on which Player II has a winning strategy?

2.4.2 Scattered and well-ordered sets

Given an ordinal α , denote by α^* the converse of α , i.e., the conversely well-ordered set $(\alpha, >)$. Recall that a linearly ordered set is *scattered* if it does not contain a copy of the rationals. A linearly ordered set is said to be σ -*scattered* if it is a countable union of scattered sets. Given an order type γ , say that a linearly ordered set is γ -*free* if it does not contain a copy of γ . The following result can be found in [Tod85]. We include the proof for the reader's convenience.

Theorem 2.29 ([Tod84]). *Let X be a linearly ordered set. The following are equivalent.*

- (i) X is a countable union of well-ordered sets.
- (ii) X is σ -scattered and ω_1^* -free.

Proof. (i) $\Rightarrow$ (ii) is clear, so we focus on (ii) $\Rightarrow$ (i). It suffices to assume that X is scattered (and ω_1^* -free). Define the following relation on X : $x \sim y$ if and only if the closed interval $[x, y]$ is a countable union of well-ordered sets. Routine arguments show that $\sim$ is an equivalence relation. Also, every equivalence class $\mathcal{C}$ is convex, i.e., if $x, y \in \mathcal{C}$ and $x < z < y$, then $z \in \mathcal{C}$. Let $[x]$ be the equivalence class of x . Since every equivalence class is convex, the order $[x] < [y]$ if and only if $x < y$ is well-defined. Hence, the quotient order $X/\sim$ is isomorphic to a subspace of X . It follows that $X/\sim$ is scattered by assumption.

Claim 2.29.1. $[x]_{\geq} := \{y \in [x] : y \geq x\}$ is a countable union of well-ordered sets for all x .

Proof of Claim. Let $\{y_\alpha : \alpha < \kappa\}$ be a cofinal subset of $[x]_{\geq}$ with $y_0 = x$. So,

$$[x]_{\geq} = \bigcup_{\alpha < \kappa} [y_\alpha, y_{\alpha+1}].$$

Since $[y_\alpha, y_{\alpha+1}]$ is a countable union of well-ordered sets, then write $[y_\alpha, y_{\alpha+1}] = \bigcup_{n < \omega} A_n^\alpha$, where A_n^α is ω^* -free. Let $A_n := \bigcup_{\alpha < \kappa} A_n^\alpha$ and notice that A_n is ω^* -free since κ is ω^* -free. Then, $[x]_\geq = \bigcup_{n < \omega} A_n$, so $[x]_\geq$ is a countable union of well-ordered sets. $\dashv$

Claim 2.29.2. $[x]_\leq := \{y \in [x] : y \leq x\}$ is a countable union of well-ordered sets for all x .

Proof of Claim. Let $\{y_\alpha : \alpha < \kappa\}$ be a coinital subset of $[x]_\leq$ with $y_0 = x$. Since X is ω_1^* -free, then κ is countable, so

$$[x]_\leq = \bigcup_{n < \omega} [y_{n+1}, y_n].$$

Since $[y_{n+1}, y_n]$ is a countable union of well-ordered sets for all $n < \omega$, then $[x]_\leq$ is a countable union of well-ordered sets. $\dashv$

The previous two claims imply that $[x]$ is a countable union of well-ordered sets for all $x \in X$. Now we show that actually $X = [x]$ for some x .

Claim 2.29.3. $(X/\sim, <)$ is dense, i.e., if $[x] < [y]$ then there is $z \in Z$ such that $[x] < [z] < [y]$.

Proof of Claim. Let $[x] < [y]$ and suppose there is no z such that $[x] < [z] < [y]$. Then, $[x, y] \subseteq [x]_\geq \cup [y]_\leq$ which implies that $[x, y]$ is a countable union of well-ordered sets by the previous two claims. Therefore, $[x] = [y]$. $\dashv$

This last claim shows that X must have a single equivalence class, since a nontrivial scattered dense order contains a copy of $\mathbb{Q}$. $\square$

If S is a countable union of well-ordered subsets of an ω_1 -free set X , then S must be countable. Also, it is clear that a subset S of $\mathbb{R}$ is countable if and only if S is a countable union of well-ordered sets. This idea, in combination with the preceding theorem, is generalized by the following.

Corollary 2.30. *If X is an ω_1 -free and ω_1^* -free linearly ordered set, then the following are equivalent for any $S \subseteq X$.*

1. S is countable.
2. S is a countable union of well-ordered sets.
3. S is σ -scattered.

2.4.3 Baker's game on linear orders

We define a variation of Baker's game in an arbitrary dense linear order $(X, <)$. Given $S \subseteq X$, denote by $BG(X, S)$ the game where two players take turns selecting elements a_n, b_n of X as in the Cantor Game for the reals. Player I wins if after ω -many turns there is an $x \in S$ such that for all $n < \omega$, $a_n < x < b_n$, and Player II wins otherwise. The density condition ensures the game never reaches a stalemate. Thus, for the rest of the section, X is a dense linearly ordered set with a fixed subset S (the winning conditions). Observe that every countable set S satisfies the subsequent proposition.

Proposition 2.31. *If S is a countable union of well-ordered sets, then Player II has a winning strategy for $BG(X, S)$.*

Proof. Let $S = \bigcup_{n < \omega} S_n$ where each S_n is ω^* -free (that is, well-founded). In the n -th turn, Player II plays $b_n = \min(S_n \cap (a_n, b_{n-1}))$ if $S_n \cap (a_n, b_{n-1}) \neq \emptyset$ and any legal move otherwise (by density, there must always be one). Now, suppose that for all $n < \omega$, $a_n < x < b_n$. This implies that $x \notin S_n$ for all $n < \omega$, and hence, $x \notin S$. In other words, this is a winning strategy for Player II. $\square$

In general, S being a countable union of well-ordered sets does not imply that S is countable (unless it contains no copies of ω_1). The previous proposition implies that if S is an (uncountable) well-ordered set, then Player II has a winning strategy for $BG(X, S)$. We now show that if S is conversely well-ordered, then Player II has a winning strategy for $BG(X, S)$. Notice that an uncountable conversely well-ordered set S has the property that S cannot be written as a countable union of well-ordered sets, so this result shows that, in general, the converse of Proposition 2.31 is false.

Theorem 2.32. *Suppose that γ is an ordinal and $\{S_\alpha : \alpha < \gamma\}$ is a collection of subsets of X on which Player II has a winning strategy. Furthermore, assume that $S_\beta < S_\alpha$ for all $\alpha < \beta < \gamma$ (i.e. $x < y$ for all $x \in S_\beta$ and $y \in S_\alpha$). Then, Player II has a winning strategy for $BG(X, \bigcup_{\alpha < \gamma} S_\alpha)$.*

Proof. Player I starts by playing $a_0 \in X$. If there is $b_0 > a_0$ such that $b_0 \leq y$ for all $y \in \bigcup_{\alpha < \gamma} S_\alpha$, then Player II wins by picking b_0 . Otherwise, Player II picks any legal move in the first turn. In this case, $a_1 > a_0$, so let $\alpha_1 < \gamma$ be the smallest such that $y_1 < a_1$ for some $y_1 \in S_{\alpha_1}$. If there is a legal move $b_1 > a_1$ such that $b_1 \leq y$ for all $y \in \bigcup_{\alpha < \alpha_1} S_\alpha$, then Player II picks such b_1 and then uses the winning strategy for S_{α_1} . Otherwise, Player II plays any legal move b_1 and then, since $a_2 > a_1$, one can pick the smallest $\alpha_2 < \alpha_1$ such that $y_2 < a_2$ for some $y_2 \in S_{\alpha_2}$.

By repeating this process, since γ is well-ordered, one cannot have a decreasing sequence $\dots < \alpha_2 < \alpha_1 < \gamma$. Hence, at some turn N , there is a legal move $b_N > a_N$ for Player II such that $b_N \leq y$ for all $y \in \bigcup_{\alpha < \alpha_N} S_\alpha$. After that, Player II continues playing with the winning strategy for $BG(X, S_{\alpha_N})$.

This describes a winning strategy for Player II. Indeed, given any $x \in X$ such that $a_n < x < b_n$ for all $n < \omega$, the following holds: Given that $y_N < a_N < x$ and the hypothesis that $S_\beta < S_\alpha$ whenever $\alpha < \beta$, $x \notin S_\alpha$ for all $\alpha > \alpha_N$; also $x \notin S_\alpha$ for all $\alpha < \alpha_N$ by the way b_N was played. Finally, $x \notin S_{\alpha_N}$, since Player II used a winning strategy for $BG(X, S_{\alpha_N})$. $\square$

Corollary 2.33. *If S is conversely well-ordered, then Player II has a winning strategy on $BG(X, S)$.*

Corollary 2.34. *For any infinite cardinal κ , there is a dense linear order $(X, <)$ of size κ on which Player II has a winning strategy on every subset of X .*

Proof. Take $X = \kappa^* \times \mathbb{Q}$ ordered lexicographically. Concretely, given $x = (\alpha, p)$ and $y = (\beta, q)$ elements of X , $x \leq y$ if either $\beta < \alpha$ or $\alpha = \beta$ and $p <_{\mathbb{Q}} q$. Since the rationals are countable and κ is infinite, X has the desired size. The fact that X is dense follows from the fact that $\mathbb{Q}$ is dense and unbounded.

Now let $S \subseteq X$ and define, for each $\alpha < \kappa$, $S_\alpha = \{\alpha\} \times \mathbb{Q}$. As each S_α is countable, Player II has a winning strategy for $BG(X, S_\alpha)$, by virtue of Proposition 2.31. Since $S = \bigcup_{\alpha < \kappa} S_\alpha$ and the S_α are ordered the way that Theorem 2.32 requires, the result follows. $\square$

Chapter 3

Deep Computations

The contents of this chapter are based on the author’s contributions to and perspectives on our article [DIMW⁺26], the second paper in a series started by the authors of [ADIW24] that connects model theory, structural Ramsey theory, and the topology of function spaces to the foundational study of “deep computations.” While the initial work in [ADIW24] established the formal framework for ultracomputations, characterizing them as accumulation points of standard computations and proving the existence of deep equilibria, this chapter focuses on the complexity and classification of this asymptotic behaviour.

Within the thesis, this chapter sits between the descriptive combinatorics of Chapter 2 and the Polish group actions of Chapter 4, and the unifying thread is descriptive set theory at successive levels of the Baire-class hierarchy: Chapter 2 works at the Borel level, while this chapter works one step higher, with compact subsets of the first Baire class. The dichotomy-and-basis paradigm developed in Chapter 2, where definability has a structural cost, and the obstructions form a small canonical library, recurs here in a new way: the obstructions are now the seven prototype Rosenthal compacta of Argyros–Dodos–Kanellopoulos, and the cost is paid in approximability. Rosenthal’s dichotomy (Theorem 1.2 of Chapter 1) is the seed of this development.

3.1 Introduction

In this chapter we study the asymptotic behavior of computations as parameters of the computation tend towards a limit, e.g., the depth of a neural network tending to infinity, or the time interval between layers of a time-series network tending toward zero. Recently, particular cases of this concept have attracted considerable attention in deep learning research (e.g., Neural Ordinary Differential Equations [CRBD], Physics-Informed Neural Networks [RPK19], and Deep Equilibrium Models [BKK]). The formal framework introduced here provides a unified setting to study these limit phenomena from a foundational viewpoint.

The main result of the chapter is the identification of a hierarchy of four distinct complexity classes for limit computations. This is obtained by invoking a major result about the topology of spaces of functions, namely, Todorćević’s trichotomy for Rosenthal compacta [Tod99] and transferring it into the realm of computation. The medium that we use to translate between topology and computation is model theory, specifically, Shelah’s classification theory [She90].

The translation between topology and computation proceeds as follows. To every computation in a given computation model, we associate a continuous real-valued function, called the *type* of the computation, that describes the logical properties of this computation with respect to the rest of the model. This association allows us to view computations in any given computational model as elements of a space of real-valued functions, which is called the *space of types* of the model. The idea of embedding models of theories into their type spaces is borrowed from model theory, where type spaces play a central role. The embedding of computations into spaces of types then allows us to utilize the vast theory of topology of function spaces, known as C_p -theory, to obtain results about complexity of topological limits of computations. As we shall indicate next, recent classification results for spaces of functions provide an elegant and powerful machinery to classify computations according to their levels of “tameness” or “wildness”, with the former corresponding roughly to polynomial approximability and the latter to exponential approximability. The model-theoretic viewpoint of spaces of types thus allows us to interconnect various classification programs: in topology, the classification of Rosenthal compacta pioneered by Todorćević [Tod99]; in logic, the classification of theories developed by Shelah [She90]; and in statistical learning, the notions of PAC learning and VC dimension pioneered by Vapnik and Chervonenkis [VC74, VC71].

In [ADIW24], the authors introduced the concept of limits of computations, which they called *ultracomputations* (given that they arise as ultrafilter limits of standard computations) and *deep computations* (following usage in machine learning [BKK]). There is a technical difference between both designations, but in this chapter, to simplify the nomenclature, we will ignore the difference and use primarily the term “deep computation”.

In [ADIW24], a new “tame vs. wild” (i.e., polynomially approximable vs. nonpolynomially approximable) dichotomy for the complexity of deep computations was proved by invoking a classical result of Grothendieck from the 1950s [Gro52]. Under our model-theoretic framework, the property of polynomial approximability of computations is identified with continuous extensibility in the sense of topology, and with the notions of *stability* and *type definability* in the sense of model theory.

In this chapter, we refine the dichotomy of [ADIW24], and obtain new complexity classes for deep computation. The “wild” part of the dichotomy from [ADIW24] is further subdivided between “tame” and “wild” classes; three new levels of tameness are identified within this new subdivision, and seven distinct minimal prototypes of tameness are identified.

The finer classification results of this chapter are obtained through more contemporary topological and model-theoretic tools than in [ADIW24]. From the topological side, instead of Grothendieck’s classical criterion, we use the modern theory of *Rosenthal compacta*, and from the model theory side, instead of stability, we use *neostability*. The main tool imported from neostability is Shelah’s *Independence Property*, or rather, its negation, which is commonly referred to as “NIP”. More on this below.

Deep computations arise as topological limits of standard (continuous) computations. In topology, the *first Baire class*, or *Baire class 1* consists of functions (also called simply *Baire-1*) arising as pointwise limits of sequences of continuous functions. Intuitively, the first Baire class consists of functions with “controlled” discontinuities, and lies just one level of topological complexity above the Baire class 0, which (by definition) consists of continuous functions.

We apply a result of Bourgain, Fremlin and Talagrand from the late 1970s on compact subsets of the first Baire class to obtain a new “tame vs. wild” dichotomy for deep computations [BFT78];

then, we apply a trichotomy of Todorćević, from the late 90s, for *Rosenthal compacta* to identify new tame classes [Tod99].

A Rosenthal compactum is a compact topological space that is homeomorphic to a subspace of the space of Baire class 1 functions on some Polish (i.e., separable, completely metrizable) space equipped with the topology of pointwise convergence. Rosenthal compacta exhibit “topological tameness,” meaning that they behave in relatively controlled ways. Since the late 1970s, they have played a crucial role in understanding the complexity of structures of functional analysis, especially Banach spaces. Todorćević’s trichotomy for Rosenthal compacta has been utilized to settle longstanding problems in topological dynamics and topological entropy [GM22]. It is noteworthy that Todorćević’s proof relies on sophisticated set-theoretic forcing and infinite Ramsey theory. At the time of writing this chapter, decades after his original argument, there are no elementary proofs of the trichotomy; in fact, Todorćević’s methods have been analyzed and exploited further [Tod23, HT19]. (Theorem 1.2 of Chapter 1 is the prototype: a pointwise-bounded sequence in $\overline{C_p(X)}$ either has a convergent subsequence or has $\beta\mathbb{N}$ as its pointwise closure: the simplest tame/wild split.)

Through our framework, Rosenthal compacta in topology correspond to the important concept of *Non-Independence Property* (known as “NIP”) in model theory, identified by Shelah [She71, She90], and to the concepts *VC-dimension* and *Probably Approximately Correct* learning (known as “PAC learnability”) in statistical learning theory [VC71, VC74, Val84].

Beyond Todorćević’s trichotomy, we invoke a more recent heptachotomy for separable Rosenthal compacta obtained by Argyros, Dodos and Kanellopoulos [ADK08]. These authors identify seven fundamental “prototypes” of separable Rosenthal compacta, and show that any non-metrizable separable Rosenthal compactum must contain a “canonical” embedding of one of these prototypes.

From the point of view of approximation theory, it is noteworthy that in the context of Grothendieck [Gro52] and Bourgain–Fremlin–Talagrand [BFT78], compactness and sequential compactness coincide; thus, the results of these papers allow us to ensure that deep computations, which in principle are abstract ultrafilter limits (over a possibly uncountable set), can always be approximated by concrete *sequences* of standard computations.

We believe that the results presented in this chapter show practitioners of computation, or topology, or descriptive set theory, or model theory, how classification invariants used in their field translate into classification invariants of other fields. In the interest of accessibility, we do not assume that the reader has previous familiarity with advanced topology, model theory, or computing. The necessary topological background beyond undergraduate topology is covered later in this section.

In this section, we also present basic topological and combinatorial preliminaries. In section 3.2, we introduce the structural/model-theoretic viewpoint (no previous exposure to model theory is needed). Section 3.4 is devoted to the classification of deep computations.

Throughout the chapter, our results pertain to classical models of computation (particularly computations involving real-valued quantities that are known and manipulated to a finite degree of precision). The final section, Section 3.5, introduces a probabilistic viewpoint, which we intend to develop in future work.

In this section we recall notions and results from general topology and function space theory. For simplicity, we assume that all spaces are Hausdorff unless explicitly stated otherwise.

A subset of a topological space is F_σ if it is a countable union of closed sets, and G_δ if it is a countable intersection of open sets. In a metric space, every open set is F_σ ; equivalently, every

closed set is G_δ .

A *Polish space* is a separable and completely metrizable topological space, i.e., one admitting a complete metric that induces its topology. The class of Polish spaces is closed under countable topological products. Classical examples of Polish spaces (besides the real line) include the Cantor space $2^\mathbb{N}$ (the set of all infinite binary sequences, endowed with the product topology), the Baire space $\mathbb{N}^\mathbb{N}$ (the set of all infinite sequences of naturals, also with the product topology), and the space $\mathbb{R}^\mathbb{N}$ of sequences of real numbers.

Fact 3.1. [Eng89, 4.3.24] *A subset of a Polish space is itself Polish in the subspace topology if and only if it is a G_δ set. In particular, closed subsets and open subsets of Polish spaces are also Polish spaces.*

If X is a topological space and A is a subset of X , then A is said to be *nowhere dense* if its closure has an empty interior, and of the *first category* if A is a countable union of closed nowhere dense sets; otherwise, A is said to be of the *second category*. Sets $A \subseteq X$ of the first category are also called *meagre*, and their complements $X \setminus A$ are called *comeagre*. Thus, the set of meagre subsets of X is the σ -ideal generated by the nowhere dense subsets of X . A topological space X is *Baire* if every comeagre subset of X is dense. Equivalently, X is Baire if any countable intersection of open dense sets remains dense. The Baire Category Theorem states that compact Hausdorff spaces and all completely metrizable spaces (hence, all Polish spaces) are Baire.¹

Given two topological spaces X and Y we denote by $C_p(X, Y)$ the set of all continuous functions $f: X \rightarrow Y$ endowed with the topology of pointwise convergence, i.e., with the relative (subspace) product topology $C_p(X, Y) \subseteq Y^X$, where the product topology on Y^X uses only Y 's topology; the topology on X enters through the requirement that elements of $C_p(X, Y)$ be continuous. The space $C_p(X, \mathbb{R})$ of continuous real-valued functions on X is denoted simply $C_p(X)$.

A function $f \in Y^X$ between topological spaces X, Y is said to be of *Baire class 1*, or a *Baire-1 function*, if there is a sequence $(f_n)_{n \in \mathbb{N}} \subseteq C_p(X, Y)$ such that $f = \lim_{n \rightarrow \infty} f_n$ (i.e., $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ for all $x \in X$). An elementary argument shows that Baire-1 functions are continuous everywhere except on a set of the first category. If X and Y are topological spaces, the space of Baire-1 functions $f: X \rightarrow Y$ endowed with the topology of pointwise convergence is denoted $B_1(X, Y)$; thus, $C_p(X, Y) \subseteq B_1(X, Y) \subseteq Y^X$ are topological inclusions. We abbreviate $B_1(X, \mathbb{R})$ as $B_1(X)$. Relatively compact subsets of $C_p(X)$ and of $B_1(X)$ (particularly for X Polish) are of interest in analysis and topological dynamics.

A topological space X is *perfectly normal* if it is normal and every closed subset of X is a G_δ set (equivalently, every open subset of X is an F_σ set). Every metrizable space (hence, every Polish space) is perfectly normal.

The following fact was proved by Baire in his groundbreaking thesis. For a proof, see [Tod97, Section 10].

Fact 3.2 (Baire). *If X is perfectly normal, then the following conditions are equivalent for a function $f: X \rightarrow \mathbb{R}$:*

- (1) *f is a Baire class 1 function, that is, f is a pointwise limit of continuous functions.*
- (2) *$f^{-1}[U]$ is an F_σ subset of X whenever $U \subseteq \mathbb{R}$ is open.*

¹More generally, any Čech-complete space is Baire [Eng89, Theorem 3.9.3].

If X is Baire, then (1) and (2) are equivalent to:

(3) For every closed $F \subseteq X$, the restriction $f|_F$ has a point of continuity.

Moreover, if X is Polish and $f \notin B_1(X)$, then there exist countable $D_0, D_1 \subseteq X$ and reals $a < b$ such that

$$D_0 \subseteq f^{-1}(-\infty, a], \quad D_1 \subseteq f^{-1}[b, \infty), \quad \overline{D_0} = \overline{D_1}.$$

3.1.1 From Rosenthal's dichotomy to the Bourgain–Fremlin–Talagrand dichotomy to Shelah's NIP

Recall the following result from the Introduction chapter.

Theorem 1.2 (Rosenthal's Dichotomy, [Ros74]). *If X is Polish and $(f_n)_{n \in \mathbb{N}} \subseteq C_p(X)$ is pointwise bounded, then (f_n) has a convergent subsequence, or a subsequence whose closure in $\mathbb{R}^X$ is homeomorphic to $\beta\mathbb{N}$.*

The genesis of the dichotomy was *Rosenthal's ℓ_1 -Theorem*, which states that a Banach space includes an isomorphic copy of ℓ_1 (the space of absolutely summable sequences), or else every bounded sequence therein is weakly Cauchy. The ℓ_1 -Theorem connects diverse areas: Banach space geometry, Ramsey theory, set theory, and topology of function spaces.

As we move from $C_p(X)$ to the larger space $B_1(X)$, a dichotomy paralleling the ℓ_1 -Theorem holds. This result is usually not phrased as a dichotomy in the above manner, but rather as an equivalence (see Theorem 3.3 below).

First, we introduce some useful notation. For any $f \in \mathbb{R}^X$ and any real a , define

$$X_{\leq a}^f := f^{-1}(-\infty, a], \quad X_{\geq a}^f := f^{-1}[a, +\infty).$$

For any set $A \subseteq \mathbb{R}^X$ and any real a , define (by a slight abuse of notation)

$$X_{\leq a}^A := \bigcap_{f \in A} X_{\leq a}^f = \{x \in X : f(x) \leq a \text{ for all } f \in A\},$$

$$X_{\geq a}^A := \bigcap_{f \in A} X_{\geq a}^f = \{x \in X : f(x) \geq a \text{ for all } f \in A\}.$$

(In case $A = \emptyset$, we define $X_{\geq a}^\emptyset = X = X_{\leq a}^\emptyset$.) For any sequence $\{f_n\} \subseteq \mathbb{R}^X$ and $I \subseteq \mathbb{N}$, define $I^c := \mathbb{N} \setminus I$ and $f_I := \{f_i : i \in I\}$.

Theorem 3.3 (“The BFT Dichotomy”. Bourgain–Fremlin–Talagrand [BFT78]). *Let X be a Polish space and $A \subseteq C_p(X)$ be pointwise bounded. The following are equivalent:*

- (i) A is relatively compact in $B_1(X)$.
- (ii) For every $(f_n)_{n \in \mathbb{N}} \subseteq A$ and every $a < b$ there is $I \subseteq \mathbb{N}$ such that

$$X_{\leq a}^{f_I} \cap X_{\geq b}^{f_{I^c}} = \emptyset.$$

(As stated above, the BFT Dichotomy is a particular case of the equivalence (ii) $\Leftrightarrow$ (v) in [BFT78, Corollary 4G].)

The sets $X_{\leq a}^{f_I}$ and $X_{\geq b}^{f_{I^c}}$ appearing in condition Theorem 3.3(ii) are defined, respectively, in terms of $|I|$ -many inequalities of the form $f_i(x) \leq a$, and $|I^c|$ -many of the form $f_j(x) \geq b$. Thus, at least one of $X_{\leq a}^{f_I}$ and $X_{\geq b}^{f_{I^c}}$ is defined by the satisfaction of infinitely (countably) many inequalities. For our purposes, it is more natural to consider only finitely many inequalities at a time, which motivates the definitions below.

Definition 3.4. We say that a collection of functions $A \subseteq \mathbb{R}^X$ has the *finitary Non-Independence Property (NIP)* if, for all sequences $(f_n)_{n \in \mathbb{N}} \subseteq A$ and reals $a < b$, there exist finite disjoint sets $E, F \subseteq \mathbb{N}$ such that $X_{\leq a}^{f_E} \cap X_{\geq b}^{f_F} = \emptyset$. We say that such E, F witness finitary NIP for (f_n) and a, b .

A set $A \subseteq \mathbb{R}^X$ has the *finitary Independence Property (IP)* if it does not have finitary NIP, i.e., if there exists a sequence $(f_n)_{n \in \mathbb{N}} \subseteq A$ and reals $a < b$ such that for every pair of finite disjoint sets $E, F \subseteq \mathbb{N}$, we have $X_{\leq a}^{f_E} \cap X_{\geq b}^{f_F} \neq \emptyset$.

If the word “finite” is omitted in the above definitions, we obtain the definitions of *countable* NIP (weaker than finitary NIP) and *countable* IP (stronger than finitary IP), respectively.

If we insist on witnesses $E, F \subseteq \mathbb{N}$ such that $F = E^c$, we call the respective properties “BFT-NIP” and “BFT-IP”. Thus, Theorem 3.3 becomes (for pointwise bounded function collections $A \subseteq C_p(X)$) that A is relatively compact in $B_1(X)$ if and only if A has BFT-NIP.

Unless otherwise unspecified, IP/NIP shall mean *finitary* IP/NIP henceforth.

Proposition 3.5. *If X is compact and $A \subseteq C_p(X)$, then A has BFT-NIP if and only if it has finitary NIP.*

(No pointwise boundedness is assumed of A .)

Proof. Trivially (as per the preceding discussion), finitary NIP implies BFT-NIP. Reciprocally, assume that X is compact and $A \subseteq C_p(X)$ has finitary IP. Fix a sequence $(f_n) \subseteq A$ and reals $a < b$ such that $X_{\leq a}^{f_E} \cap X_{\geq b}^{f_F} \neq \emptyset$ for every finite disjoint $E, F \subseteq \mathbb{N}$. Fix $I \subseteq \mathbb{N}$. Since $(f_n) \subseteq A \subseteq C_p(X)$ is a sequence of continuous functions, the collection $\mathcal{F}_I = \{X_{\leq a}^{f_n} : n \in I\} \cup \{X_{\geq a}^{f_m} : m \in I^c\}$ consists of closed (non-empty) sets. Since all finite $E \subseteq I$ and $F \subseteq I^c$ witness finitary IP for (f_n) and a, b , we see that the collection $\mathcal{F}_I$ has the finite intersection property. By compactness, $X_{\leq a}^{f_I} \cap X_{\geq b}^{f_{I^c}} = \bigcap \mathcal{F}_I \neq \emptyset$. $\square$

Theorem 3.6. *Let X be a compact metrizable (hence Polish) space. For every pointwise bounded $A \subseteq C_p(X)$, the following properties are all equivalent:*

- (i) A is relatively compact in $B_1(X)$;
- (ii) A has BFT-NIP;
- (iii) A has countable NIP;
- (iv) A has finitary NIP.

(The equivalence (ii) $\Leftrightarrow$ (iv) holds for arbitrary compact X . The equivalence (ii) $\Leftrightarrow$ (iii), for arbitrary X .)

Proof. Corollary of Theorem 3.3 and Proposition 3.5. $\square$

Theorem 3.6 may be stated as the following dichotomy (under the assumptions) for any compact metrizable X : either A is relatively compact in $B_1(X)$, or A has IP (in either sense).

The Independence Property was first isolated by Saharon Shelah in model theory as a dividing line between theories whose models are “tame” (corresponding to NIP) and theories whose models are “wild” (corresponding to IP). See [She71, Definition 4.1], [She90]. We will discuss this dividing line in more detail in the next section.

3.1.2 NIP as a universal dividing line between polynomial and exponential complexity

The particular case of the BFT dichotomy (Theorem 3.3) when A consists of $\{0, 1\}$ -valued (i.e., {Yes, No}-valued) strings was discovered independently, around 1971-1972 in many foundational contexts related to polynomial (“tame”) vs. exponential (“wild”) complexity: In model theory, by Saharon Shelah [She71], [She90], in combinatorics, by Norbert Sauer [Sau72], and Shelah [She72], [She90], and in statistical learning, by Vladimir Vapnik and Alexey Chervonenkis [VC71], [VC74].

In model theory Shelah’s classification theory is a foundational program in mathematical logic devised to categorize first-order theories based on the complexity and structure of their models. A theory T is considered classifiable in Shelah’s sense if the number of non-isomorphic models of T of a given cardinality can be described by a bounded number of numerical invariants. In contrast, a theory T is unclassifiable if the number of models of T of a given cardinality is the maximum possible number. A key fact is that the number of models of T is directly impacted by the number of *types* over sets of parameters in models of T ; a controlled number of types is a characteristic of a classifiable theory.

In Shelah’s classification program [She90], theories *without* the independence property (called NIP theories, or *dependent* theories) have a well-behaved, “tame” structure; the number of types over a set of parameters of size κ of such a theory is of polynomially or similar “slow” growth on κ .

In contrast, theories with the Independence Property (called IP theories) are considered “intractable” or “wild”. A theory with the Independence Property produces the maximum possible number of types over a set of parameters; for a set of parameters of cardinality κ , the theory will have 2^{2^κ} -many distinct types.

In combinatorics Sauer [Sau72] and Shelah [She72] proved the following independently: Let $\mathcal{F}$ be a collection of subsets of some set S . Either: for every $n \in \mathbb{N}$ there is a set $A \subseteq S$ with $|A| = n$ such that $|\{S_i \cap A : i \in \mathbb{N}\}| = 2^n$ ($\mathcal{F}$ is a collection having “exponential complexity”); or: there exists $N \in \mathbb{N}$ such that for every $A \subseteq S$ with $|A| \geq N$, one has

$$|\{S_i \cap A : i \in \mathbb{N}\}| \leq \sum_{i=0}^{N-1} \binom{|A|}{i}.$$

($\mathcal{F}$ is a collection having “polynomial complexity”). Clearly, any collection $\mathcal{F}$ of subsets of a *finite* set S has polynomial complexity. The “polynomial” name is justified: indeed, for fixed

$N > 0$, as a function of the size $|A| = m > 0$, we have

$$\sum_{i=0}^{N-1} \binom{m}{i} \leq \sum_{i=0}^{N-1} \frac{m^i}{i!} \leq \left(\sum_{i=0}^{N-1} \frac{1}{i!} \right) \cdot m^{N-1} < e \cdot m^{N-1} = O(m^N).$$

(More precisely, the order of magnitude is $O(m^{N-1})$: polynomial in m for N fixed.)

In machine learning Readers familiar with statistical learning may recognize the Sauer-Shelah lemma as the dichotomy discovered and proved slightly earlier (1971) by Vapnik and Chervonenkis [VC71, VC74] to address uniform convergence in statistics. The least integer N given by the preceding paragraph, when it exists, is called the *VC-dimension* of $\mathcal{F}$; it is a core concept in machine learning. If such an integer N does not exist, we say that the VC-dimension of $\mathcal{F}$ is infinite. The lemma provides upper bounds on the number of data points (sample size) needed to learn a concept class of known VC dimension d up to a given admissible error in the statistical sense. The Fundamental Theorem of Statistical Learning states that a hypothesis class is PAC-learnable (PAC stands for “Probably Approximately Correct”) if and only if its VC dimension is finite.

3.1.3 Rosenthal compacta

The universal classification implied by Theorem 3.3, as attested by the examples outlined in the preceding section, led to the following definition (by Gilles Godefroy [God80]):

Definition 3.7. A Rosenthal compactum is any topological space realized as a compact subset of the space $B_1(X) = B_1(X, \mathbb{R})$ (equipped with the topology of pointwise convergence) of all real functions of the first Baire class on some Polish space X .

A Rosenthal compactum K is necessarily Hausdorff since it is a topological subspace of the Hausdorff product space $\mathbb{R}^X$.

Rosenthal compacta possess significant topological and dynamical tameness properties, and play an important role in functional analysis, measure theory, dynamical systems, descriptive set theory, and model theory. In this chapter, we use them to study deep computations.

3.1.4 The special case $B_1(X, \mathbb{R}^{\mathcal{P}})$ with $\mathcal{P}$ countable

Fix an arbitrary (at most) countable set $\mathcal{P}$ whose elements $P \in \mathcal{P}$ will be called *predicate symbols* or *formal predicates*. Our present goal is to characterize relatively compact subsets of $B_1(X, \mathbb{R}^{\mathcal{P}})$, where X is always assumed to be a perfectly normal space (typically a Polish space).

The set $\mathcal{P}$ shall be considered discrete whenever regarded as a topological space. Since

$$C_p(X, \mathbb{R}^{\mathcal{P}}) \subseteq B_1(X, \mathbb{R}^{\mathcal{P}}) \subseteq (\mathbb{R}^{\mathcal{P}})^X,$$

the “ambient” space $(\mathbb{R}^{\mathcal{P}})^X$ is quite relevant. The product $X \times \mathcal{P}$ will be regarded as either a point set, or as a topological product depending on context. We have natural homeomorphic identifications

$$(\mathbb{R}^{\mathcal{P}})^X \cong \mathbb{R}^{X \times \mathcal{P}} \cong (\mathbb{R}^X)^{\mathcal{P}},$$

given by

$$\begin{aligned}\mathbb{R}^{X \times \mathcal{P}} &\rightarrow (\mathbb{R}^{\mathcal{P}})^X : \varphi \mapsto \hat{\varphi} \\ \mathbb{R}^{X \times \mathcal{P}} &\rightarrow (\mathbb{R}^X)^{\mathcal{P}} : \varphi \mapsto \varphi^\wedge,\end{aligned}$$

where

$$\hat{\varphi}(x) := \varphi(x, \cdot) \in \mathbb{R}^{\mathcal{P}}, \quad \varphi^\wedge(P) := \varphi(\cdot, P) \in \mathbb{R}^X.$$

Such identifications view X and $\mathcal{P}$ as mere point sets (the topology of X in particular plays no role).

For $x \in X$, define the “left projection” map

$$\lambda_x : \mathbb{R}^{X \times \mathcal{P}} \rightarrow \mathbb{R}^{\mathcal{P}} : \varphi \mapsto \lambda_x(\varphi) := \hat{\varphi}(x);$$

for $P \in \mathcal{P}$, the “right projection” map

$$\rho_P : \mathbb{R}^{X \times \mathcal{P}} \rightarrow \mathbb{R}^X : \varphi \mapsto \rho_P(\varphi) := \varphi^\wedge(P).$$

For fixed $x \in X$ and $P \in \mathcal{P}$, we also have canonical projection maps

$$\lambda_x : \mathbb{R}^X \rightarrow \mathbb{R} : f \mapsto f(x), \quad \rho_P : \mathbb{R}^{\mathcal{P}} \rightarrow \mathbb{R} : f \mapsto f(P).$$

When clear from context, rather than using the specific symbols (“ λ ” for left, “ ρ ” for right) to denote projections, we may use the generic symbol “ π ”; thus, π_x may mean λ_x , and π_P may mean ρ_P .

The Proposition below reduces the study of $\mathbb{R}^{\mathcal{P}}$ -valued continuous or Baire-1 functions on X to the special case of real-valued ones.

Proposition 3.8. *The identification $(\mathbb{R}^{\mathcal{P}})^X \cong \mathbb{R}^{X \times \mathcal{P}} \cong (\mathbb{R}^X)^{\mathcal{P}}$ induces identifications*

$$C_p(X, \mathbb{R}^{\mathcal{P}}) \cong C_p(X \times \mathcal{P}) \cong C_p(X)^{\mathcal{P}}, \quad B_1(X, \mathbb{R}^{\mathcal{P}}) \cong B_1(X \times \mathcal{P}) \cong B_1(X)^{\mathcal{P}}.$$

(The cardinality of $\mathcal{P}$ plays no role.)

Proof. The identification of C_p -spaces follows trivially from the definition of topological product and the fact that $\mathcal{P}$ is discrete: a continuous map $X \rightarrow \mathbb{R}^{\mathcal{P}}$ is precisely a $\mathcal{P}$ -indexed family of continuous functions $X \rightarrow \mathbb{R}$, and these correspond to continuous functions $X \times \mathcal{P} \rightarrow \mathbb{R}$. The identification of Baire-1 spaces follows immediately, since it is defined in terms of the purely topological notion of limit (in the ambient space) of sequences of continuous functions. $\square$

Given $A \subseteq Y^X$ and $K \subseteq X$ we write $A|_K := \{f|_K : f \in A\}$, i.e., the set of all restrictions of functions in A to K . The following Theorem is a slightly more general version of Theorem 3.3.

Theorem 3.9. *Assume that $\mathcal{P}$ is countable, X is a Polish space, and $A \subseteq C_p(X, \mathbb{R}^{\mathcal{P}})$ is pointwise bounded in the sense that $\pi_P \circ A (\subseteq C_p(X))$ is pointwise bounded for every $P \in \mathcal{P}$. The following are equivalent for every compact $K \subseteq X$:*

- (i) $A|_K$ is relatively compact in $B_1(K, \mathbb{R}^{\mathcal{P}})$;

(ii) $\pi_P \circ A|_K$ has NIP for every $P \in \mathcal{P}$.

Proof. Compact subsets $K \subseteq X$ are closed, hence also Polish. Therefore, the asserted equivalence follows from Theorems 3.3 and 3.5. $\square$

We conclude this section with an elementary but important observation.

Lemma 3.10. *Assume that X is Hausdorff and that $A \subseteq C_p(X)$. The following are equivalent for every $L \subseteq X$:*

(i) $A|_L$ satisfies the NIP;

(ii) $A|_{\overline{L}}$ satisfies the NIP.

Proof. The implication (i) $\Rightarrow$ (ii) is trivial. To see the opposite implication, suppose that (ii) does not hold, and choose $\{f_n\}_{n \in \mathbb{N}} \subseteq A$ and $a < b$ such that for all finite disjoint $E, F \subseteq \mathbb{N}$,

$$\overline{L} \cap \bigcap_{n \in E} f_n^{-1}(-\infty, a] \cap \bigcap_{n \in F} f_n^{-1}[b, \infty) \neq \emptyset.$$

But then, for any a', b' with $a < a' < b' < b$ and any finite disjoint $E, F \subseteq \mathbb{N}$, we can choose

$$x \in \overline{L} \cap \bigcap_{n \in E} f_n^{-1}(-\infty, a') \cap \bigcap_{n \in F} f_n^{-1}(b', \infty),$$

which, by the definition of closure, gives

$$L \cap \bigcap_{n \in E} f_n^{-1}(-\infty, a'] \cap \bigcap_{n \in F} f_n^{-1}[b', \infty) \neq \emptyset,$$

contradicting (i). $\square$

Historical remarks

The Baire hierarchy of functions was introduced by René-Louis Baire in his 1899 doctoral thesis, *Sur les fonctions de variables réelles*. The field of C_p -theory was pioneered by A. V. Arhangel'skiĭ and his students in the 1970s and 1980s [Ark92], and it has found many applications in model theory and functional analysis. For a recent survey, see [Tka11].

3.2 Compositional Computation Structures

In this section, we connect function spaces with arbitrary-precision computations. We start by summarizing some basic concepts from [ADIW24].

3.2.1 Computation States Structures

A *computation states structure* is a pair $(L, \mathcal{P})$, where L is a set whose elements we call *states* and $\mathcal{P}$ is a collection of real-valued functions on L that we call *predicates*, also called *features*. For each $P \in \mathcal{P}$, we call the value $P(v)$ the P -th (*observable*) *feature* of v .

A *transition* of a computation states structure $(L, \mathcal{P})$ is a map $f: L \rightarrow L$.

The *type* of a state $v \in L$ is the indexed family²

$$\text{tp}(v) = (P(v))_{P \in \mathcal{P}} \in \mathbb{R}^{\mathcal{P}}$$

of its features.

Intuitively, L is the set of states of a computation, and the predicates $P \in \mathcal{P}$ are primitives that are given and accepted as computable. (I.e., each symbol $P \in \mathcal{P}$ denotes a function $P(\cdot)$ taken as a computational primitive on L .)

We assume that each state $v \in L$ is uniquely characterized by its type $\text{tp}(v)$, so we may identify L with a topological subspace of $\mathbb{R}^{\mathcal{P}}$ and topologized as such (i.e., L is identified with its forward image $\text{tp}[L] \subseteq \mathbb{R}^{\mathcal{P}}$ under the type map). As a topological space, L is thus endowed with the topology of pointwise convergence induced by the product topology of $\mathbb{R}^{\mathcal{P}}$. In particular, for each $P \in \mathcal{P}$, the projection map $v \mapsto P(v)$ is continuous.

Important state spaces include $L = \mathbb{R}^{\mathbb{N}}$ (resp., $L = \mathbb{R}^n$ for some positive integer n), endowed with one predicate $P_i(v) = v_i$ (giving the i -th coordinate of v) for each $i \in \mathbb{N}$ (resp., for each $0 \leq i < n$). State subspaces $L \subseteq \mathbb{R}^{\mathbb{N}}$ and $L \subseteq \mathbb{R}^n$ are of primary interest in this paper. (Any states structure having a finite or countable family of predicates is effectively such a subspace.)

Definition 3.11. Given a computation states structure $(L, \mathcal{P})$, any type $\text{tp}(v) \in \mathbb{R}^{\mathcal{P}}$ of a state $v \in L$ will be called a *realized* (state) type. The topological closure of the set of realized types in $\mathbb{R}^{\mathcal{P}}$ (endowed with the pointwise convergence topology) will be called the *space of (state) types* of $(L, \mathcal{P})$, denoted $\mathcal{L}$; elements $\mathbf{v} \in \mathcal{L}$ are called *state types*. Elements $\mathbf{v} \in \mathcal{L} \setminus L$ are *unrealized* state types.

Intuitively, state types capture a notion of “limit state”. For each $P \in \mathcal{P}$, the P -th coordinate map (natural projection) $\mathbb{R}^{\mathcal{P}} \rightarrow \mathbb{R} : \mathbf{v} = (\mathbf{v}_Q)_{Q \in \mathcal{P}} \mapsto \mathbf{v}_P$, gives a continuous extension to $\mathcal{L}$ of the predicate $P(\cdot)$ on L ; abusing notation, the extension will still be denoted $P(\cdot)$ (or just P).

Remark 3.12. Just as L is identified with a subset of $\mathcal{L}$, but going in the opposite direction, one may regard unrealized state types as “hyper-states” (when such type is obtained as a sequential limit—or ultralimit—of realized state types, unrealized state types may be called “deep states”). A comment for readers versed in model theory: at least for the purpose of realizing state types (with parameters in the original states space L), the type space $(\mathcal{L}, \mathcal{P})$ serves as a “poor man’s” ultraproduct of $(L, \mathcal{P})$ wherein, by the very definition of $\mathcal{L}$, each type $\mathbf{v}$ is realized.

Bounds on precision and magnitude. Shards

In order for computation state structures to serve their purpose as models of real-world computing, we must keep in mind that computations involving real-valued quantities can only ever carried out approximately, to a preset degree of precision (i.e., to within acceptable error bounds). More subtly, yet equally important in practice, one must also bound the magnitude of real quantities involved

²As in §3.8 above, we regard each predicate on L as named by a specific formal label P (a *predicate symbol*) which is given the semantic meaning of a *bona fide* real function $P(\cdot): v \mapsto P(v)$ (the predicate proper) in the computation states structure. Such a formal symbol P becomes *de facto* the name of an observable feature having the real value $P(v)$ at any state $v \in L$. Whenever $\mathcal{P}$ is used as an index set (e.g., as in the topological product $\mathbb{R}^{\mathcal{P}}$ below), it is regarded as a set of formal symbols. This syntactic vs. semantic abuse of nomenclature is commonplace in mathematics, but the model-theoretic framework requires contextual awareness of the distinction. To disambiguate when necessary, we shall write $P(\cdot)$, implying the semantic meaning of a real function on L , in contrast to P , meant as a purely syntactic symbol. Nevertheless, by a standard abuse of nomenclature in appropriate contexts, we may also treat P as a function on states (i.e., take P to mean $P(\cdot)$ proper).

in all computations: one cannot possibly store—let alone compute with—arbitrarily large numbers! Thus, although the collection $\mathcal{P}$ of predicates may be infinite (possibly, uncountable even), “true” computations must imply choices, at the outset, of bounds for errors and for absolute magnitudes of quantities involved. Heuristically, different parts of a computation may require suitably matching (different) error bounds, as well as dealing with quantities of very different sizes. Focusing for now on the second of these issues, our results shall generally apply to sets of computations for which arbitrary but fixed bounds for each predicate are set in advance. This leads to the concept of *shard* first introduced in [ADIW24], as per the following definition.

Definition 3.13. Fix a computation states structure $\underline{L} = (L, \mathcal{P})$, with $L \subseteq \mathbb{R}^{\mathcal{P}}$ implicitly identified with its type space. A *sizer* is a family $r_{\bullet} = (r_P)_{P \in \mathcal{P}}$ of positive real numbers, indexed by $\mathcal{P}$. Given a sizer $r_{\bullet}$, let $\mathbb{R}[r_{\bullet}] := \prod_{P \in \mathcal{P}} [-r_P, r_P]$ (a compact space), and let the $r_{\bullet}$ -*shard* of the states space L be

$$L[r_{\bullet}] = L \cap \mathbb{R}[r_{\bullet}].$$

For a sizer $r_{\bullet}$, the $r_{\bullet}$ -*type shard* is defined as $\mathcal{L}[r_{\bullet}] = \overline{L[r_{\bullet}]}$ (a closed, hence compact subset of $\mathbb{R}[r_{\bullet}]$).

Let also $\mathcal{L}_{\text{sh}}$ (the space of *shard-supported types*) be the union of all type-shards as the sizer $r_{\bullet}$ varies.

Evidently, $\mathcal{L}_{\text{sh}} \subseteq \mathcal{L}$, although the inclusion may be proper. However, the equality $\mathcal{L}_{\text{sh}} = \mathcal{L}$ holds in the important special case when $\mathcal{P}$ is (at most) countable (see [ADIW24]).

3.2.2 Compositional Computation Structures

Definition 3.14. A *Compositional Computation Structure* (CCS) is a triple $(L, \mathcal{P}, \Gamma)$, where

- $(L, \mathcal{P})$ is a computation states structure, and
- $\Gamma \subseteq L^L$ is a semigroup under composition.

Elements of the semigroup Γ are called the *computations* of the structure $(L, \mathcal{P}, \Gamma)$. The state space L is implicitly identified with a subset of the type space $\mathcal{L} \subseteq \mathbb{R}^{\mathcal{P}}$ of $(L, \mathcal{P})$. We also assume that the identity map id on L is an element of Γ (which is thus not merely a semigroup but a monoid of transformations of L).

We topologize Γ as a subset of the topological product L^L , where (as usual) the “exponent” L serves merely as an index set, but the “base” $L \subseteq \mathbb{R}^{\mathcal{P}}$ has the relative product topology (of point-wise convergence). In this manner, the semigroup Γ is identified with a subset of the topological product $\mathcal{L}^L \subseteq (\mathbb{R}^{\mathcal{P}})^L$, which leads to an inclusion $\overline{\Gamma} \subseteq \mathcal{L}^L$. Elements $\xi \in \overline{\Gamma}$ are called (real-valued) *deep computations* or *ultracomputations*.

A collection R of sizers is *exhaustive* if $L = \bigcup_{r_{\bullet} \in R} L[r_{\bullet}]$ (shards $L[r_{\bullet}]$ exhaust L). A transformation $\gamma \in \Gamma$ is *R-confined* if γ restricts to a map $\gamma|_{L[r_{\bullet}]}: L[r_{\bullet}] \rightarrow L[r_{\bullet}]$ (into $L[r_{\bullet}]$ itself) for every $r_{\bullet} \in R$. A subset $\Delta \subseteq \Gamma$ is *R-confined* if each $\gamma \in \Delta$ is.

Unlike its subspace L^L (a topological semigroup), the space $\mathcal{L}^L$ does not have a natural semigroup structure, so ultracomputations are not generally composable. Intuitively, ultracomputations are “final” or “ultimate”.

Proposition 3.15. *If $\Delta \subseteq \Gamma$ is confined by an exhaustive sizer collection, then $\overline{\Delta}$ is a compact subset of $\mathcal{L}_{\text{sh}}^L$.*

Proof. Assume that R confines Δ . For each $v \in L$, let $r_{\bullet}^{(v)} \in R$ be a sizer such that $v \in L[r_{\bullet}^{(v)}]$. An arbitrary $\gamma \in \Delta$ restricts to a map $\gamma|_{L[r_{\bullet}^{(v)}]}: L[r_{\bullet}^{(v)}] \rightarrow L[r_{\bullet}^{(v)}]$, so $\Gamma \subseteq K := \prod_{v \in L} \mathcal{L}[r_{\bullet}^{(v)}]$. The space K is a product of compact spaces, hence compact, so $\overline{\Gamma}$ is a closed, hence compact subset thereof, and a subset of $\mathcal{L}_{\text{sh}}^L \supseteq K$ *a fortiori*. $\square$

For a CCS $(L, \mathcal{P}, \Gamma)$, we regard the elements of Γ as “standard” finitary computations, and the elements of $\overline{\Gamma}$, i.e., deep computations, as possibly infinitary limits of standard computations. The main goal of this paper is to study the computability, definability and computational complexity of deep computations. Since deep computations are defined through a combination of topological concepts (namely, topological closure) and structural and model-theoretic concepts (namely, models and types), we will import technology from both topology and model theory.

Remark 3.16. (For readers versed in model theory.)³

1. compositional computation structures as defined above are not real-valued structures in a literal sense, because there is no intrinsic structural manner to ensure that Γ is realized as a literal subset of the product space L^L . However, in a semantically transparent manner, any CSS as above may alternatively be seen (per its original definition in [ADIW24]) as a real-valued structure $(L, \mathcal{P}, \Gamma, \circ, \text{ev})$, where

- $(L, \mathcal{P})$ is a computation states structure,
- $(\Gamma, \circ)$ is a semigroup,
- $\text{ev}: \Gamma \times L \rightarrow L: (\gamma, v) \mapsto \text{ev}(\gamma, v)$ is a semigroup action of Γ on L (i.e., $\text{ev}(\gamma \circ \delta, v) = \text{ev}(\gamma, \text{ev}(\delta, v))$ for $\gamma, \delta \in \Gamma$ and $v \in L$).

Both definitions are reconciled upon identifying $\gamma \in \Gamma$ with the map $(\gamma, v) \mapsto \text{ev}(\gamma, v)$.

2. In the preceding sense of real-valued structures, the class of compositional computation structures (for a fixed predicate symbol collection $\mathcal{P}$) is first-order elementary. The limited notion of state type we have introduced (essentially, quantifier-free state types with parameters of sort L only) may be enriched to state types involving more formulas (e.g., with parameters in Γ , or including quantified formulas), and extended to “transform-types” (i.e., types of sort Γ). In particular, given a structure $\underline{C} = (L, \mathcal{P}, \Gamma, \circ, \text{ev})$, all types are realized in some ultrapower $\underline{C}^{(\mathcal{U})}$ of $\underline{C}$ (constructed in the usual manner detailed in the references above).⁴ This approach to spaces of types was introduced in Banach space theory by Krivine [Kri74, KM81, Kri72].

3.2.3 Computability of deep computations and the Extensibility Axiom

Let $(L, \mathcal{P})$ be a computation structure with *countable* predicate collection $\mathcal{P}$.

1. A *computable predicate* is any real function $\varphi: L \rightarrow \mathbb{R}$ whose restriction to an arbitrary shard is uniformly approximable by polynomials (with real coefficients) in the features $P(\cdot)$.

³Refer to [Kei23] where a general framework for real-valued structures is introduced; see also [DnI17, section 6] for a basic tutorial.

⁴By contrast (cf., Remark 3.12), the ultrapower construction is not necessary to realize state types in a computation states structures.

2. A transform $f: L \rightarrow \mathcal{L}$ is *computable* if, for each $Q \in \mathcal{P}$, the output feature $Q \circ f: L \rightarrow \mathbb{R}$ is a computable predicate.

It is shown in [ADIW24] that computable transforms $f: L \rightarrow \mathcal{L}$ are precisely those that extend to continuous maps $\tilde{f}: \mathcal{L} \rightarrow \mathcal{L}$.

To summarize, for a function $f: L \rightarrow \mathcal{L}$, the following conditions are equivalent:

- f is computable
- f has uniformly polynomially approximable features on shards
- f extends to a continuous map $\mathcal{L} \rightarrow \mathcal{L}$.

These equivalences motivate the following definition:

A computation $\gamma \in \Gamma$ is *extensible* if it has some extension $\tilde{\gamma}: \mathcal{L}_{\text{sh}} \rightarrow \mathcal{L}_{\text{sh}}$ such that, for every sizer $r_\bullet$, there is a sizer $s_\bullet$ such that $\tilde{\gamma}|_{\mathcal{L}[r_\bullet]}: \mathcal{L}[r_\bullet] \rightarrow \mathcal{L}[s_\bullet]$ is continuous.

Because $\tilde{\gamma}$ extends γ continuously on each compact type-shard, we call it the *K-extension* of γ ; such extension is evidently unique (if it exists).

Abusing notation, for any $\Delta \subseteq \Gamma$ consisting of extensible computations, the set $\{\tilde{\gamma}: \gamma \in \Delta\}$ will be denoted $\tilde{\Delta}$.

Extensibility is not automatic: a transform can be continuous, even have continuous features, without those features being *uniformly* approximable by polynomials on every shard, which is exactly what the computability equivalence above requires. Without it, a continuous but non-extensible computation would still be “uncomputable” in this precise sense, and the equivalence between computability and continuity established above would fail to hold in general. For this reason, extensibility is imposed as a standing hypothesis in what follows.

Extensibility Axiom. *Every computation $\gamma \in \Gamma$ is extensible.*

For a more detailed discussion of the Extensibility Axiom, we refer the reader to [ADIW24].

For the rest of the paper, the Extensibility Axiom is imposed on all CCSs.

3.3 Examples of Compositional Computation Structures

We begin by formalizing the Newton–Raphson root-finding method as a CCS [Gal00], [BFB16, Section 2.3]. Let $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ be the extended complex plane (the Alexandroff —one-point— compactification of $\mathbb{C}$). We regard the unit sphere $S^2 \subseteq \mathbb{R}^3$ as the Riemann sphere and endowed with the stereographic projection map $S^2 \rightarrow \hat{\mathbb{C}}$ —a topological homeomorphism whose *inverse* $\hat{\mathbb{C}} \rightarrow S^2$ is given by $z \mapsto (P_1(z), P_2(z), P_3(z))$, where $(P_1(\infty), P_2(\infty), P_3(\infty)) = (0, 0, 1)$ and, for $z \neq \infty$,

$$P_1(z) = \frac{2 \Re z}{|z|^2 + 1}, \quad P_2(z) = \frac{2 \Im z}{|z|^2 + 1}, \quad P_3(z) = \frac{|z|^2 - 1}{|z|^2 + 1}.$$

The extended complex plane $\hat{\mathbb{C}}$ is thus endowed with the real predicates P_1, P_2, P_3 with values in $[-1, 1]$; its topology is metrized by the pullback of the usual Euclidean metric on S^2 .

Newton CCSs: steps and fractals

Let $p = p(z)$ be a non-constant rational function with complex coefficients —i.e., an analytic (meromorphic) map $\hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$. The *Newton method step* (or *Newton map*) for p is the rational map

$$N_p : z \mapsto z - \frac{p(z)}{p'(z)}. \quad (3.1)$$

Successive iterations of the Newton-Raphson method (i.e., compositional powers $N_p^{(n)}$) may be regarded as computations of a CCS whose states space is $L = \hat{\mathbb{C}}$. The natural choice of predicates is $\mathcal{P} = (P_1, P_2, P_3)$, the inverse stereographic projection coordinates. (The real- and imaginary part maps $z \mapsto \Re z$, $z \mapsto \Im z$ are not defined at $z = \infty$.)

Newton CCSs Let $\Gamma_p := \{N_p^{(n)} : n \in \mathbb{N}\}$ be the semigroup of iterates of the Newton map (3.1). The CCS $(\hat{\mathbb{C}}, (P_1, P_2, P_3), \Gamma_p)$ is *Newton’s method for $p(z)$* (or the *Newton CCS for $p(z)$*). Note that the states space $\hat{\mathbb{C}}$ is compact and the Newton map N_p is continuous thereon, so the Extensibility Axiom is satisfied by Newton CCSs.

Given an initial estimate z_0 , one hopes for the sequence of Newton steps applied to z_0 (namely, $N_p^{(\bullet)}(z_0) := (z_n)_{n \in \mathbb{N}}$, where $z_n := N_p^{(n)}(z_0)$) to converge to a root r of p .

Fatou and Julia sets Recall that the *Fatou set* $\mathcal{F}_p$ consists of all points that have a compact neighbourhood on which some subsequence of the sequence $p^{(\bullet)}$ of iterates of p converges uniformly; the *Julia set* $\mathcal{J}_p := \hat{\mathbb{C}} \setminus \mathcal{F}_p$ is its complement: it is (necessarily closed and) typically a fractal. The Julia set of the Newton map N_p is called the *Newton fractal* of p .

3.3.1 Example: Newton’s CCS for the cubic $\rho(z) = z^3 - 2z + 2$

The polynomial $\rho(z) = z^3 - 2z + 2$ has Newton map

$$N_\rho : z \mapsto z - \frac{z^3 - 2z + 2}{3z^2 - 2} = \frac{2z^3 - 2}{3z^2 - 2}.$$

Denote the real root of ρ by r_0 ; the roots in the upper- and the lower half plane, by r_+ and r_- , respectively.

Successive Newton steps obtained from the initial guess $z_0 = 0$ yield the divergent sequence $N_\rho^{(\bullet)}(0) = (0, 1, 0, 1, \dots)$ having sequential limits $0, 1$ —neither a root of ρ .

Visualizing Newton iterates

Assign to each complex point $z \in \mathbb{C}$ an “RGB colour” $(R, G, B) \in [0, 1]^3$ whose entries give the corresponding (Red/Green/Blue) colour intensity; thus, $(1, 0, 0)$ is red, $(0, 1, 0)$ is green, and $(0, 0, 1)$ is blue—while, for example, $(0.5, 0, 0.5)$ is a light purple. Each of the three distinct roots of ρ will be given one of these colours (the real root r_0 , red; the roots r_{+1} on the upper- and r_{-1} on the lower half plane, respectively green and blue). The RGB colour of all other $z \in \mathbb{C}$ is a convex combination whose relative weights are $1/d_i$ where $d_i = |z - r_i|$ for $i = 0, \pm 1$; this will be called the “standard”

colour of z : the closer z is to root r_i , the closer the colours of these points are.⁵

For any $n \in \mathbb{N}$, define a colouring of $\mathbb{C}$ in the natural way: colour each $z_0 \in \mathbb{C}$ using the standard colour of $z_n := N_\rho^{(n)}(z_0)$. At any stage, particularly early ones, the colouring looks “fuzzy” (out of focus) because the colouring function is continuous; however, as n increases, the colouring becomes “sharper” (the colouring function has large regions of near-constancy, in the transition areas between which colour varies rapidly —i.e., is “nearly discontinuous”). For one thing, for many z_0 in the Fatou set, the (entire) sequence (z_n) converges to one of the three roots: the RGB colours of (z_n) tend towards (a single one of) red, green or blue correspondingly. The Fatou set also has an open region consisting of points z_0 for which $(z_n)_{n \in \mathbb{N}}$ has a subsequence converging to 0 or 1 (uniformly on some compact neighbourhood of z_0). The four open regions are immediately recognizable in Figure 3.1.

Let B_0, B_+, B_- be the respective basins of the roots r_0, r_+, r_- , and let B_{01} be the basin of the attractive cycle $0, 1, 0, 1, \dots$. It may be shown that the Fatou set of N_ρ is $F_\rho = B_{01} \cup B_0 \cup B_+ \cup B_-$, each of these four sets being nonempty open. Since B_{01} is nonempty, $N_\rho^{(\bullet)}$ does not converge pointwise; however, its subsequences of even/odd iterates are pointwise convergent on B_{01} to two distinct deep computations f_0 (constant 0 on B_{01}) and f_1 (constant 1 on B_{01}).⁶ The behavior of iterates $N_\rho^{(n)}$ on the Newton fractal of ρ (the Julia set of ρ) is chaotic.

3.3.2 Example: Newton’s CCS for 3-cyclotomic $\kappa(z) = z^3 - 1$

Consider next the 3-cyclotomic polynomial $\kappa(z) = z^3 - 1$. The roots of the 3-cyclotomic polynomial $\kappa(z) = z^3 - 1$ are the cube roots of unity $r_0 = 1$ and $r_\pm = \exp(\pm 2\pi i/3)$; the Newton map is

$$N_\kappa: z \mapsto z - \frac{z^3 - 1}{3z^2} = \frac{2z^3 + 1}{3z^2}.$$

In Figure 3.2(C), one sees a Fatou set consisting of solid-colour (disconnected) open sets, one in each of the colours red (for $r_0 = 1$), green and blue (for $r_{\pm 1}$, respectively), where the sequence of all iterates $z_n := N_\kappa^{(n)}(z_0)$ starting with an arbitrary initial point z_0 converges to the corresponding root. (Thus, each solid-colour region is the basin of attraction of the respective root).

3.3.3 Example: Square roots

When $p: \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ has real coefficients, it restricts to a map $\mathbb{RP}^1 \rightarrow \mathbb{RP}^1$ on the real-projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$. In this case, the Newton CSS for p may be regarded as having states space $\mathbb{RP}^1$. We shall explain the details in the simplest case of a polynomial $p(x) = x^2 - a$ (real $a > 0$) to illustrate iterative square-root approximations.

Let $\mathbb{R} \rightarrow S^1: x \mapsto (P_1(x), P_2(x))$ be the (inverse) stereographic projection of the real line onto $S^1 \subseteq \mathbb{R}^2$, i.e.,

$$P_1(x) = \frac{2x}{x^2 + 1}, \quad P_2(x) = \frac{x^2 - 1}{x^2 + 1},$$

extended to a projective transformation of the real projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$ by $\infty \mapsto$

⁵While ∞ is a fixed point of N_ρ , it is a repeller: if $|z|$ is large, then $N_\rho(z) = \frac{2}{3}z + O(1)$ has roughly two-thirds its magnitude. Therefore, ∞ is not the limit of any subsequence of $N_\rho^{(\bullet)}(z_0)$ with z_0 in the Fatou set of N_ρ ; in particular, ∞ is a point of the Newton fractal of $\rho(z)$.

⁶As remarked above, ∞ is a repelling fixed point of N_ρ , hence $f_0(\infty) = \infty = f_1(\infty)$.

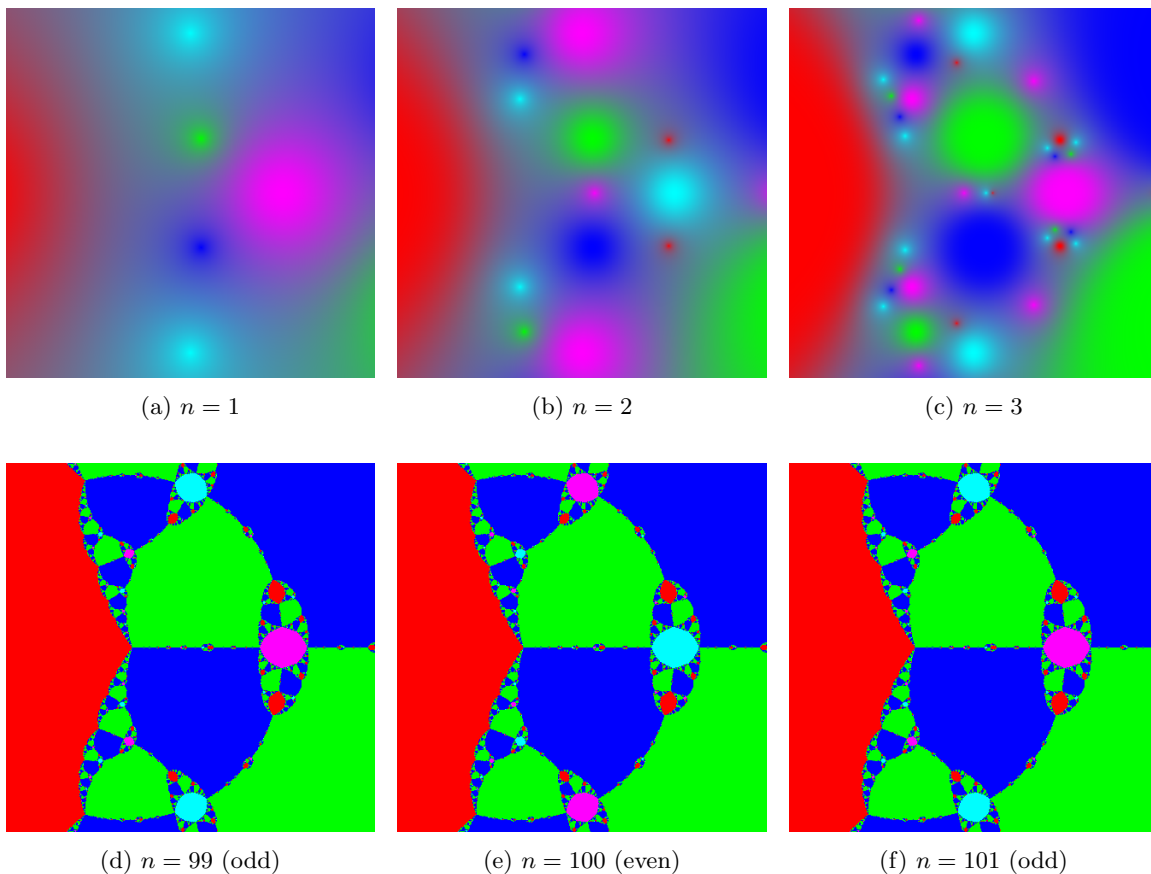


Figure 3.1: Iterates of the Newton map N_ρ of $\rho(z) = z^3 - 2z + 2$. Points in the basin of each root $r_0, r_{\pm 1}$ (respectively, red, green and blue) converge to the root. The basin of attraction of the 2-cycle $\{0, 1\}$ is the union of magenta and cyan regions; these two colours alternate like semaphore lights: points z in the cyan region are those whose subsequence $N_\rho^{(2 \cdot \bullet)}(z)$ of even Newton iterates (resp., $N_\rho^{(2 \cdot \bullet + 1)}(z)$ of odd iterates) has limit 1 (resp., limit 0). For points z in the magenta region, the limits of the even and odd iterate subsequences are exchanged. The boundaries of coloured regions constitute the Newton fractal of ρ .

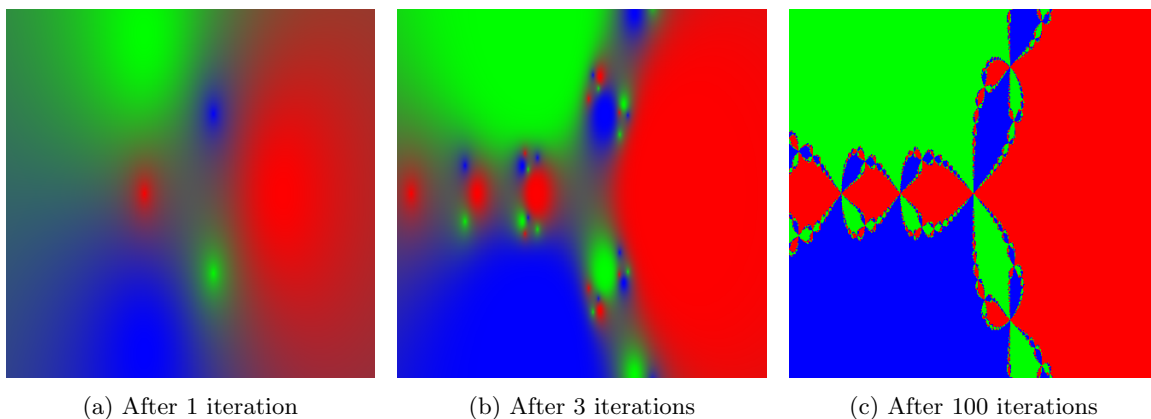


Figure 3.2: Visualization of the Newton iterates of $\kappa(z) = z^3 - 1$.

$(P_1(\infty), P_2(\infty)) = (0, 1)$; this gives a compact CCS $\mathfrak{L} = (\mathbb{RP}^1, (P_1, P_2))$. As above, consider the quadratic polynomial $p(x) = x^2 - a$ for some fixed real number $a > 0$. The Newton map N_p is

$$x \mapsto x - \frac{x^2 - a}{2x} = \begin{cases} \frac{x^2 + a}{2x}, & x \neq 0, \infty; \\ \infty, & x = 0, \infty. \end{cases}$$

These Newton steps converge to one of the square roots $\pm\sqrt{a}$ starting from any initial point $x_0 \neq 0, \infty$. More explicitly, the unique deep equilibrium $\mathfrak{f}: \mathbb{RP}^1 \rightarrow \mathbb{RP}^1$ of (the sequence of iterates of) N_p is given by

$$\mathfrak{f}(x) = \lim_{n \rightarrow \infty} N_p^n(x) = \begin{cases} \sqrt{a}, & x \in (0, \infty); \\ -\sqrt{a}, & x \in (\infty, 0); \\ \infty, & x = 0, \infty. \end{cases}$$

Thus, $\mathfrak{f}$ is discontinuous at $x = 0, \infty$.

3.3.4 Example: Threshold classifiers on Cantor's set

Consider the states space $L = 2^{\mathbb{N}} (= \{0, 1\}^{\mathbb{N}})$ consisting of all bit sequences $x = (x_n)_{n \in \mathbb{N}} \subseteq \{0, 1\}$. The product space $2^{\mathbb{N}}$ is compact⁷ and homeomorphic to the Cantor (ternary) subset of the interval $[0, 1]$. Consider the predicate collection $\mathcal{P} = (\bullet_n)_{n \in \mathbb{N}}$ of projections (coordinates)

$$\bullet_n: 2^{\mathbb{N}} \rightarrow \{0, 1\} : x \mapsto x_n.$$

Since $2^{\mathbb{N}} \subseteq \mathbb{R}^{\mathbb{N}} (\cong \mathbb{R}^{\mathcal{P}})$ is closed, the type space of $(2^{\mathbb{N}}, (\bullet_n)_{n \in \mathbb{N}})$ is effectively still $2^{\mathbb{N}}$. The constant 0 and 1 elements of $2^{\mathbb{N}}$ will be denoted $0_{\bullet}$ and $1_{\bullet}$, respectively. In this way, we obtain the *Cantor computation states structure (CSS)*

$$\mathcal{C} = (2^{\mathbb{N}}, (\bullet_i)_{i \in \mathbb{N}}).$$

The concatenation action of words

For fixed $n \in \mathbb{N}$, we call elements of $2^n = \{0, 1\}^n$ *finite bit sequences* (or *binary words*) of length n ; the finite set 2^n is topologized discretely.

The set of all finite-length words is the disjoint union $2^{<\mathbb{N}} := \bigsqcup_{n \in \mathbb{N}} 2^n$ (a discrete space); the (unambiguously defined) length n of any word $w \in 2^{<\mathbb{N}}$ is denoted $|w|$. For $m \geq n$, we also have natural continuous restriction maps (projections) $2^{\mathbb{N}} \rightarrow 2^m \rightarrow 2^n$. The restriction of x — either a sequence or a word of length $m \geq n$ — to a length- n word is denoted $x \upharpoonright_n$ (i.e., $\cdot \upharpoonright_n$ projects onto 2^n). Since the index set $\mathbb{N}$ and the value space $2 = \{0, 1\}$ are well-ordered, the spaces $2^{\mathbb{N}}$ as well as $2^{<\mathbb{N}}$ and 2^n for $n \in \mathbb{N}$ carry a natural lexicographical order henceforth denoted $\preceq$; the order topology coincides with the (product) topology of $2^{\mathbb{N}}$. All restriction maps $\cdot \upharpoonright_n$ are continuous and order-preserving.

Given lengths $n, m \in \mathbb{N}$, we have a concatenation operation $2^m \times 2^n \rightarrow 2^{m+n} : (v, w) \mapsto \widehat{vw}$; such concatenations altogether yield a semigroup structure on $2^{<\mathbb{N}}$ —more precisely, a monoid with the empty word $() \in 2^0$ as identity. The more general concatenation operation $2^{<\mathbb{N}} \times 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}} : (w, x) \mapsto \widehat{wx}$ is a continuous and order-preserving action of $2^{<\mathbb{N}}$ on $2^{\mathbb{N}}$; we shall call it the *prefixing*

⁷The topology on the space $2 = \{0, 1\}$ is discrete.

action. Thus, *prefixing by a given word* $w \in 2^{<\mathbb{N}}$ means applying the operation $\widehat{w} \cdot : x \mapsto \widehat{w}x$. In this manner, we obtain the *Cantor prefixes CCS*

$$\widehat{\mathcal{C}} = \langle 2^{\mathbb{N}}, (\bullet_i)_{i \in \mathbb{N}}, 2^{<\mathbb{N}}, \widehat{\cdot} \rangle$$

where, by an abuse of nomenclature, the operation $\widehat{\cdot}$ denotes both the concatenation $(v, w) \mapsto v\widehat{w}$ on $2^{<\mathbb{N}}$ and the prefixing $(w, x) \mapsto \widehat{w}x$ of $2^{<\mathbb{N}}$ on $2^{\mathbb{N}}$.

Shift action

An altogether different action on $2^{\mathbb{N}}$ is given by shift operations (“Bernoulli shifts”); namely, an arbitrary $m \in \mathbb{N}$ acts on $2^{\mathbb{N}}$ by the map

$$\overleftarrow{m} : x = (x_i)_{i \in \mathbb{N}} \mapsto \overleftarrow{m}x := (x_{m+i})_{i \in \mathbb{N}}.$$

Here, $\overleftarrow{0} : 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ is the identity $x \mapsto x$, and the equality $(\overleftarrow{m+n})x = \overleftarrow{m}(\overleftarrow{n}x)$ for $m, n \in \mathbb{N}$ holds identically (i.e., the additive semigroup (monoid) $(\mathbb{N}, 0, +)$ is represented as the semigroup of shifts of $2^{\mathbb{N}}$). Let $\overleftarrow{\mathbb{N}} = \{\overleftarrow{m} : m \in \mathbb{N}\}$ be the semigroup of all shifts. This gives the *Cantor shifts* (or *Cantor-Bernoulli*) CCS

$$\mathcal{B} = \langle 2^{\mathbb{N}}, (\bullet_n)_{n \in \mathbb{N}}, \mathbb{N}, (\overleftarrow{n})_{n \in \mathbb{N}} \rangle.$$

Finite-precision classifiers and definability

Notation 3.17. When useful, we will regard 0, 1 as logical constants (0 for falsehood, 1 for truth). In view of such identification, we shall call a $\{0, 1\}$ -valued function on a set X (resp., in a finite power X^n) an *(n-ary) relation on X*. The propositional connectives of negation “ $\neg$ ”, disjunction “ $\vee$ ” and conjunction “ $\wedge$ ” are realized using (linear lattice) operations on $\mathbb{R}$; explicitly, as $r \mapsto 1 - r$ (negation), binary maximum (max) and binary minimum (min), respectively.

Let $\llbracket \cdot \rrbracket$ be Iverson’s bracket;⁸ thus, for any statement $\mathbf{S}$, the value of $\llbracket \mathbf{S} \rrbracket$ is 1 if $\mathbf{S}$ is true, 0 if false.

For any fixed word $w \in 2^n$, let $\preceq_n w$ be the relation $\llbracket \cdot \upharpoonright_n \preceq w \rrbracket : x \mapsto \llbracket x \upharpoonright_n \preceq w \rrbracket$ on $2^{\mathbb{N}}$. We shall write $x \preceq_n w$ with the meaning of $x \upharpoonright_n w$ or $\llbracket x \upharpoonright_n w \rrbracket$, depending on context (thus, “ $\preceq_n w$ ” names a unary relation taking its argument x on the left). The subscript n is used only for emphasis and optional, since its value is determined uniquely by the condition $w \in 2^n$. Since both $\llbracket \cdot \preceq w \rrbracket$ and $\cdot \upharpoonright_n$ are continuous, so is $\preceq_n w$, which will be called a *precision- n threshold classifier*; any such relation $\preceq_n w$ is a *finite-precision threshold classifier*. For $n = 0$, there is only one “zeroth-precision” classifier $\preceq_0()$: the identically true relation $\mathbf{1}$ on $2^{\mathbb{N}}$. Although insubstantial for our purposes, the subsequent discussion is simpler if we regard (as we shall) the identically false relation $\mathbf{0}$ as finite-precision threshold classifier, denoted $\prec_0 \bullet$, intuitively, a threshold classifier with negative precision $n = -1$.

We remark that the statement “ $x \preceq_n w$ ” has identical semantics to “ $x \preceq \widehat{w1_\bullet}$ ” (not to “ $x \preceq \widehat{w0_\bullet}$ ”) in the usual ordering semantics of $2^{\mathbb{N}}$ which depend on all entries of x , not just those of $x \upharpoonright_n$.

A moment’s reflection shows that any finite-precision classifier can be expressed as a Boolean combination⁹ of the first n coordinates $P_i(x) = x_i$ ($i < n$) of its argument $x \in 2^{\mathbb{N}}$ (i.e., $x \preceq_n w$

⁸We choose square brackets à la Knuth, but typeset in double-strike “blackboard bold” style to preclude ambiguity.

⁹In other words, finite-precision classifiers are *definable* in the sense of real-valued logic (sans quantifiers) whose

may be obtained from entries x_i of $x \upharpoonright_n$ using the operations of maximum, minimum and negation $\neg : r \mapsto 1 - r$ in place of classical Boolean connectives).¹⁰

Infinite-precision threshold classifiers

Threshold ultra-classifiers are relations $2^{\mathbb{N}} \rightarrow \{0, 1\}$ in the closure of the set of finite-precision classifiers. Since each finite-precision classifier is a non-increasing function on $2^{\mathbb{N}}$, the same holds for any ultra-classifier. It is easily shown that such ultra-classifiers have, for some sequence $z \in 2^{\mathbb{N}}$, one of the forms

- $\llbracket \cdot \preceq z \rrbracket$, denoted simply “ $\preceq z$ ”, or
- $\llbracket \cdot \prec z \rrbracket$, denoted “ $\prec z$ ”.

Any finite-precision classifier is an ultra-classifier. Indeed, $\prec 0_{\bullet}$ is the false relation $\mathbf{0}$ (in agreement with our earlier convention); all other finite-precision classifiers are of the form $\prec_n w$ and hence identical to the ultra-classifier $\preceq \widehat{w1_{\bullet}}$ as remarked above. Since (countably many) sequences of the form $\widehat{w1_{\bullet}}$ for words $w \in 2^{<\mathbb{N}}$ are dense in $2^{\mathbb{N}}$, one sees that ultra-classifiers are *sequential* limits of finite classifiers, hence are Baire-1 functions on $2^{\mathbb{N}}$.

An ultra-classifier not of finite precision is called *of infinite-precision*; such infinite-precision classifiers are *discontinuous* Baire-1 functions on $2^{\mathbb{N}}$: the sole point of discontinuity of an infinite-precision classifier $\prec x$ or $\prec x$ is x . Unlike finite-precision classifiers, those of infinite precision are not definable, i.e., admit no (finite) Boolean expression in the coordinates x_i .

Split Cantor When threshold classifiers are regarded as elements of $B_1(2^{\mathbb{N}})$, we will denote $\preceq x$ by $\bar{x}$, and $\prec x$ by $\hat{x}$. Per the preceding discussion, such classifiers constitute a set

$$\mathcal{C} = \hat{\mathcal{C}} \cup \bar{\mathcal{C}} \subseteq B_1(2^{\mathbb{N}})$$

consisting of two parts

$$\hat{\mathcal{C}} = \{\hat{x} : x \in 2^{\mathbb{N}}\}, \quad \bar{\mathcal{C}} = \{\bar{x} : x \in 2^{\mathbb{N}}\}.$$

In the topology of pointwise convergence, $\mathcal{C}$ is homeomorphically identified with the (topologized by the lexicographic order)¹¹ *split Cantor space* $2^{\mathbb{N}} \times \{0, 1\}$ by

$$\hat{\mathcal{C}} \cong 2^{\mathbb{N}} \times \{0\} : \hat{x} \mapsto (x, 0), \quad \bar{\mathcal{C}} \cong 2^{\mathbb{N}} \times \{1\} : \bar{x} \mapsto (x, 1).$$

Each of these two parts is homeomorphic to the Cantor set $2^{\mathbb{N}}$, but they are *not* open; indeed, each is essentially dense:¹² $\text{Cl}(\hat{\mathcal{C}}) = \mathcal{C} \setminus \{\mathbf{1}\}$ and $\text{Cl}(\bar{\mathcal{C}}) = \mathcal{C} \setminus \{\mathbf{0}\}$.

connectives are vector-lattice operations (maximum, minimum, and linear combinations), plus logical constants (i.e., real constants) as nullary connectives.

¹⁰The constant relations $\mathbf{0}$, $\mathbf{1}$ are obtained per the usual convention: the conjunction (resp., disjunction) of an empty set of propositions is $\mathbf{1}$ (resp., $\mathbf{0}$).

¹¹The primary key of a pair $(x, b) \in 2^{\mathbb{N}} \times \{0, 1\}$ is the sequence $x \in 2^{\mathbb{N}}$; the secondary key is the bit $b \in \{0, 1\}$. Thus, $\mathcal{C}$ is *not* homeomorphic to the product $\{0, 1\} \times 2^{\mathbb{N}}$ (i.e., to the “disjoint union” $2^{\mathbb{N}} \sqcup 2^{\mathbb{N}}$ of two open subspaces).

¹²The constant relations $\mathbf{0}$, $\mathbf{1}$ are isolated points of $\mathcal{C}$.

The upshot is that, starting from a countable set of quite simple finite-precision classifiers (non-increasing Boolean functions) on Cantor space, ultraclassifiers form a much more complicated split Cantor $\mathcal{C} \subseteq B_1(2^{\mathbb{N}})$.

3.3.5 Example: Prefix tests

Let $\mathcal{C} = (2^{\mathbb{N}}, (\bullet_i)_{i \in \mathbb{N}})$ be the Cantor CSS of Section §3.3.4. For each $w \in 2^{<\mathbb{N}}$, let $\equiv_n w$ be the unary $\{0, 1\}$ -valued relation $\llbracket \cdot \upharpoonright_n = w \rrbracket : x \mapsto \llbracket x \upharpoonright_n = w \rrbracket$ which holds if and only if w is the n -prefix of x (i.e., the initial n -word of x); any relation $\equiv_n w$ will be called an n -*prefix test*, or simply a *finite-prefix test*, when n is otherwise unspecified. As for finite-precision classifiers in §3.3.4 above, the subscript n in $\equiv_n w$ is determined as $n = |w|$ and thus optional. One sees that all finite-prefix tests are continuous maps $2^{\mathbb{N}} \rightarrow \{0, 1\}$; moreover, given w (of length n), the formula $x \equiv_n w$ is explicitly definable as suitable Boolean combination of the entries of $x \upharpoonright_n$. The test $\equiv w$ will also be denoted $\tilde{w}$ when thought of as a $\{0, 1\}$ valued function on $2^{\mathbb{N}}$.

Ultra-prefix tests and equality

Ultra-prefix tests are relations $2^{\mathbb{N}} \rightarrow \{0, 1\}$ that belong to the closure of prefix tests in the topology of pointwise convergence.

Proposition 3.18. *The identically false relation $\mathbf{0}$ is an ultra-prefix test.*

Proof. It suffices to show that the sets

$$W_x^{\neq} := \{w \in 2^{<\mathbb{N}} : x \neq w\} \quad (x \in 2^{\mathbb{N}})$$

are a filter sub-base on $2^{<\mathbb{N}}$, i.e., that for any finite collection $(x_i)_{i < m} \subseteq 2^{\mathbb{N}}$, the corresponding family $(W_i^{\neq})_{i < m} := (W_{x_i}^{\neq})_{i < m}$ has nonempty intersection.¹³ Let $k \in \mathbb{N}$ satisfy¹⁴ $2^k > m$; thus, at most m -many distinct k -words $w \in 2^k$ are of the form $w = x_i$ for some $i < m$; some k -word $w^* \in 2^k$ is not of that form; the intersection $\bigcap_{i < m} W_i^{\neq}$ contains w^* , hence is nonempty. $\square$

Proposition 3.19. *Any ultra-prefix test is either the constant $\mathbf{0}_\bullet$, a finite-prefix test, or an equality test*

$$\bar{x} : 2^{\mathbb{N}} \rightarrow \{0, 1\} : y \mapsto \llbracket y = x \rrbracket$$

for some fixed $x \in 2^{\mathbb{N}}$. Such equality tests are Baire-1 on $2^{\mathbb{N}}$.

Proof. From the proof of Proposition 3.18 and the remark in footnote¹³, one sees that any non-zero ultra-prefix φ is necessarily an accumulation point of a finite union of sets of the form $\widetilde{W}_x := \{\tilde{w} : x \equiv w \in 2^{\mathbb{N}}\}$, and therefore must be an accumulation point of some (single) such set $\widetilde{W}_{x^*}$ for some $x^* \in 2^{\mathbb{N}}$.

One readily sees that $\widetilde{W}_{x^*} = \{\tilde{w}_n : n \in \mathbb{N}\}$ where $w_n := x^* \upharpoonright_n$. It remains to show that $\bar{x}^*$ is the sole limit point of $(\tilde{w}_n)_{n \in \mathbb{N}}$, and sequentially so in the pointwise topology. Indeed, for all $y \in 2^{\mathbb{N}}$, we have

$$\lim_{n \rightarrow \infty} \tilde{w}_n(y) = \lim_{n \rightarrow \infty} \llbracket y \equiv_n x^* \upharpoonright_n \rrbracket = \llbracket y = x^* \rrbracket = \bar{x}^*(y). \quad \square$$

¹³From the definition of the sets $W_x^{\neq}$, it should be evident that $\mathbf{0}$ is an ultra-prefix if and only if they altogether form a filter sub-base.

¹⁴One may trivially take $k = m$, but the flow of the proof is obscured.

Unlike finite-prefix tests, equality tests are discontinuous.

Extended Alexandroff compactification

The set $\mathcal{D}_c := \{\bar{x} : x \in 2^{\mathbb{N}}\} \subseteq B_1(2^{\mathbb{N}})$ is *discrete* of size $\mathfrak{c}$ (continuum); its Alexandroff (one-point) compactification is $\mathcal{A} := \mathcal{D}_c \cup \{\mathbf{0}\}$. Let $\mathcal{A}^{\mathbb{C}} := \{\tilde{w} : w \in 2^{<\mathbb{N}}\}$. The compact space $\mathcal{A}_E = \mathcal{A} \cup \mathcal{A}^{\mathbb{C}} \subseteq B_1(2^{\mathbb{N}})$ extends the Alexandroff compactification $\mathcal{A}$ by the countable set $\mathcal{A}^{\mathbb{C}}$. The point $\mathbf{0} \in \mathcal{A}_E$ is not a G_δ point, i.e., $\{\mathbf{0}\}$ is not a countable intersection of open sets; $\mathbf{0}$ is the only such non- G_δ point. This key topological property implies that the space of ultra-prefix tests is not first-countable (in a first-countable Hausdorff space all points are G_δ).

3.4 Classifying deep computations

3.4.1 NIP, Rosenthal compacta, and deep computations

This is the main section of the chapter. We use topology of function spaces to obtain a new hierarchy of complexity classes for deep computations. The main topological tool is Todorćević's trichotomy for Rosenthal compacta [Tod99]. Our inspiration is the work of Glasner and Megrelishvili on tame semigroups in topological dynamics [GM22].

Theorem 3.20 below is the cornerstone of the section; it establishes a powerful link between deep computations and Rosenthal compacta. It extends the results of [ADIW24] from the setting of continuous deep computations to the more general (and realistic) setting of deep computations of the first Baire class. Informally, Theorem 3.20 states that if $\mathcal{P}$ and $\Delta \subseteq \Gamma$ are countable, then the space of deep computations generated by Δ is a Rosenthal compactum (when restricted to shards) if and only if Δ satisfies the NIP feature by feature. Furthermore, Theorem 3.20 shows that every deep computation generated by Δ is a sequential limit (rather than an abstract ultrafilter limit) of computations in Δ .

Theorem 3.20. *Let $(L, \mathcal{P}, \Gamma)$ be a compositional computation structure (Definition 3.14) satisfying the Extensibility Axiom with $\mathcal{P}$ countable. Let R be an exhaustive collection of sizers, and suppose that $\Delta \subseteq \Gamma$ is R -confined. Then the following conditions are equivalent:*

- (i) $\overline{\tilde{\Delta} \upharpoonright_{\mathcal{L}[r_\bullet]}} \subseteq B_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ for all $r_\bullet \in R$;
- (ii) $\pi_P \circ \Delta \upharpoonright_{L[r_\bullet]}$ satisfies the NIP for all $P \in \mathcal{P}$ and $r_\bullet \in R$; that is, for all $P \in \mathcal{P}$, $r_\bullet \in R$, $a < b$, $\{\gamma_n\}_{n \in \mathbb{N}} \subseteq \Delta$ there are finite disjoint $E, F \subseteq \mathbb{N}$ such that

$$L[r_\bullet] \cap \bigcap_{n \in E} (\pi_P \circ \gamma_n)^{-1}(-\infty, a] \cap \bigcap_{n \in F} (\pi_P \circ \gamma_n)^{-1}[b, \infty) = \emptyset.$$

Furthermore, if any (hence all) of the preceding conditions hold, then every deep computation $f \in \overline{\Delta}$ can be extended to a Baire-1 function on shards: there is $\tilde{f} : \mathcal{L}_{sh} \rightarrow \mathcal{L}_{sh}$ such that $\tilde{f} \upharpoonright_{\mathcal{L}[r_\bullet]} \in B_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ for all $r_\bullet \in R$. In particular, on each shard, every deep computation is the pointwise limit of a countable sequence of computations.

Proof. Since $\mathcal{P}$ is countable, $\mathcal{L}[r_\bullet] \subseteq \mathbb{R}^{\mathcal{P}}$ is Polish. The Extensibility Axiom implies that $\pi_P \circ \tilde{\Delta} \upharpoonright_{\mathcal{L}[r_\bullet]}$ is a pointwise bounded set of continuous functions for all $P \in \mathcal{P}$. Hence, Theorem 3.9 and Lemma

3.10 prove the equivalence of (i) and (ii). If (i) holds and $f \in \overline{\Delta}$, then we can write f as an ultralimit of computations in Δ , say $f = \mathcal{U}\lim_i \gamma_i$. Define $\tilde{f} := \mathcal{U}\lim_i \tilde{\gamma}_i$. Then, for all $r_\bullet \in R$ we have $\tilde{f} \upharpoonright_{\mathcal{L}[r_\bullet]} \in \overline{\tilde{\Delta} \upharpoonright_{\mathcal{L}[r_\bullet]}} \subseteq B_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$. The assertion that every deep computation is a pointwise limit of a countable sequence of computations follows from the fact that for a Polish space X , every compact subset of $B_1(X)$ is Fréchet-Urysohn (that is, a space where any point in the closure of a set A is the limit of a sequence of points in A ; see Theorem 3F in [BFT78] or Theorem 4.1 in [Deb13]). $\square$

3.4.2 The Todorćević trichotomy, and levels of NIP and PAC learnability

Recall that a topological space X is *hereditarily separable* if every subspace is separable, and that X is *first countable* if every point in X has a countable local basis. Every separable metrizable space is hereditarily separable, and R. Pol proved that every hereditarily separable Rosenthal compactum is first countable (see section 10 of [Deb13]). This suggests the following definition:

Definition 3.21. Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extensibility Axiom and R be an exhaustive collection of sizers. Let $\Delta \subseteq \Gamma$ be an R -confined countable set of computations satisfying the NIP on shards and features (condition (ii) in Theorem 3.20). We say that Δ is:

- (i) NIP_1 if $\overline{\tilde{\Delta} \upharpoonright_{\mathcal{L}[r_\bullet]}}$ is first countable for every $r_\bullet \in R$;
- (ii) NIP_2 if $\overline{\tilde{\Delta} \upharpoonright_{\mathcal{L}[r_\bullet]}}$ is hereditarily separable for every $r_\bullet \in R$;
- (iii) NIP_3 if $\overline{\tilde{\Delta} \upharpoonright_{\mathcal{L}[r_\bullet]}}$ is metrizable for every $r_\bullet \in R$.

Observe that $\text{NIP}_3 \Rightarrow \text{NIP}_2 \Rightarrow \text{NIP}_1 \Rightarrow \text{NIP}$. We now present examples of separable and non-separable examples Rosenthal compacta that show that implications are strict. These examples are due to Todorćević [Tod99].

Example 3.22.

1. *Alexandroff compactification of a discrete space of size continuum.* The subset $\mathcal{A} = \mathcal{D}_c \cup \{0\} \subseteq B_1(2^\mathbb{N})$ in §3.3.5 is the Alexandroff (one-point) compactification of the continuum-size discrete set $\mathcal{D}_c = \{\bar{x} : x \in 2^\mathbb{N}\}$. Hence, $\mathcal{A}$ is a Rosenthal compactum that is not hereditarily separable (and therefore not first countable). In particular, $\mathcal{A}$ is not separable.
2. *Extended Alexandroff compactification.* The Alexandroff compactification $\mathcal{A}$ of $\mathcal{D}_c$ above can be embedded into a separable compactum of $B_1(2^\mathbb{N})$ by adding a countable set $\mathcal{A}^c$ of new points that “soften” neighbourhoods in $\mathcal{A}$, yielding the extended compactification $\mathcal{A}_E = \mathcal{A} \cup \mathcal{A}^c$ of §3.3.5. However, the separable Rosenthal compactum $\mathcal{A}_E$ is not first countable. It is an example of a CCS that is NIP but not NIP_1 .
3. *Split Cantor.* The split Cantor set $\mathcal{C}$ is a separable Rosenthal compactum that is NIP_2 but not NIP_3 ; it is hereditarily separable, but it is not metrizable. The set of all finite-precision classifiers is countable dense in $\mathcal{C}$.
4. *Alexandroff Duplicate.* Let K be any compact metric space. Construct the space $X = K \sqcup C(K)$, i.e., the disjoint union of K and the space $C(K)$ of real-valued continuous functions on

K with the Banach topology of uniform convergence (supremum-norm topology). Clearly, X is Polish. For each $a \in K$, define $g_a^0, g_a^1: X \rightarrow \mathbb{R}$ as follows:

$$g_a^0(x) = \begin{cases} 0, & \text{if } x \in K \\ x(a), & \text{if } x \in C(K); \end{cases}$$

$$g_a^1(x) = \begin{cases} \llbracket a = x \rrbracket, & \text{if } x \in K; \\ x(a), & \text{if } x \in C(K). \end{cases}$$

Let $D(K) = \{g_a^0 : a \in K\} \cup \{g_a^1 : a \in K\}$. Observe that all points g_a^1 are isolated and that open neighbourhoods of $g_{a_0}^0$ are of the form $\{g_a^i : a \in U, i \in \{0, 1\}\} \setminus \{g_a^1 : a \in F\}$ where $U \subseteq K$ is an open neighbourhood of a_0 and $F \subseteq K$ is a finite set.

Another way to present this space is as the space $K \times \{0, 1\}$ with the identifications $(a, 0) \mapsto g_a^0$ and $(a, 1) \mapsto g_a^1$ and the basic open neighbourhoods are given as above. We can also embed $D(K)$ into the product $A(K) \times K$ by identifying $(a, 0) \mapsto (\mathbf{0}, a)$ and $(a, 1) \mapsto (\delta_a, a)$.

Notice that $D(K)$ is a first countable Rosenthal compactum. It is not separable if K is uncountable; thus, we typically study the interesting case when $K = 2^{\mathbb{N}}$.

As with the Alexandroff compactification $A(2^{\mathbb{N}})$, we can make the space $D(2^{\mathbb{N}})$ separable by “softening” the neighbourhoods of each point through the addition of a countable set. For example, the closure of the set $\{(v_s, s \smallfrown 0^\infty) : s \in 2^{<\mathbb{N}}\} \subseteq \hat{A}(2^{\mathbb{N}}) \times 2^{\mathbb{N}}$ is $\{(\mathbf{0}, a) : a \in 2^{\mathbb{N}}\} \cup \{(\delta_a, a) : a \in 2^{\mathbb{N}}\} \cup \{(v_s, s \smallfrown 0^\infty) : s \in 2^{<\mathbb{N}}\}$, where $\{(\mathbf{0}, a) : a \in 2^{\mathbb{N}}\} \cup \{(\delta_a, a) : a \in 2^{\mathbb{N}}\}$ is homeomorphic to $D(2^{\mathbb{N}})$.

5. *Extended Alexandroff Duplicate of the Split Cantor.* For each finite binary sequence $t \in 2^{<\mathbb{N}}$, let $a_t \in 2^{\mathbb{N}}$ be the sequence starting with t and ending with 0’s and let $b_t \in 2^{\mathbb{N}}$ be the sequence starting with t and ending with 1’s. Define $h_t: 2^{\mathbb{N}} \rightarrow \mathbb{R}$ by

$$h_t(x) = \begin{cases} 0, & \text{if } x < a_t; \\ 1/2, & \text{if } a_t \leq x \leq b_t; \\ 1, & \text{if } b_t < x. \end{cases}$$

Let $\hat{D}(S(2^{\mathbb{N}}))$ be the pointwise closure of the set $\{h_t : t \in 2^{<\mathbb{N}}\}$. The identification $h_t \mapsto (v_t, f_{t \smallfrown 0^\infty}^+)$ lifts to a homeomorphism between $\hat{D}(S(2^{\mathbb{N}}))$ and the subspace of $\hat{A}(2^{\mathbb{N}}) \times S(2^{\mathbb{N}})$ consisting of $(\mathbf{0}, f_a^+)$, $(\mathbf{0}, f_a^-)$, (δ_a, f_a^+) and $(v_t, f_{t \smallfrown 0^\infty}^+)$ for $a \in 2^{\mathbb{N}}$ and $t \in 2^{<\mathbb{N}}$ (see 4.3.7 in [ADK08]). Hence, $\hat{D}(S(2^{\mathbb{N}}))$ is a separable first countable Rosenthal compactum that is not hereditarily separable. In fact, it contains an uncountable discrete subspace.

Theorem 3.23 (Todorćević’s Trichotomy, [Tod99], [ADK08, Theorem 3]). *Let K be a separable Rosenthal Compactum.*

- (i) *If K is hereditarily separable but non-metrizable, then $S(2^{\mathbb{N}})$ embeds into K .*
- (ii) *If K is first countable but not hereditarily separable, then either $D(2^{\mathbb{N}})$ or $\hat{D}(S(2^{\mathbb{N}}))$ embeds into K .*

(iii) If K is not first countable, then $A(2^{\mathbb{N}})$ embeds into K .

We thus have the following classification for separable Rosenthal compacta:

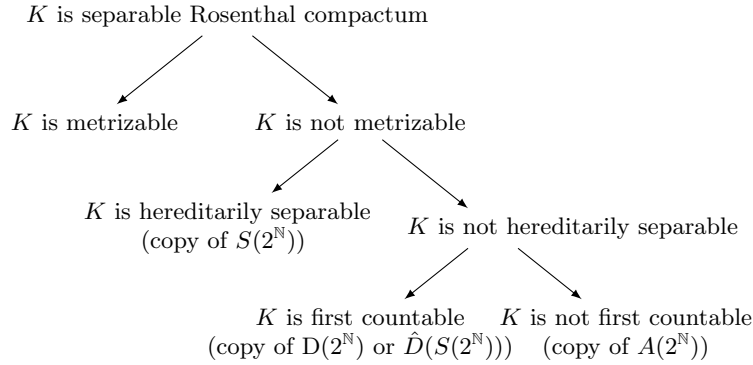


Figure 3.3: Classification of separable Rosenthal compacta.

Todorćević's Trichotomy suggests that in order to distinguish the classes NIP_i , the examples in 3.22 are essential. The following examples show that the levels NIP_i ($i = 1, 2, 3$) may be distinguished by the topological complexity of deep computations.

Example 3.24.

1. Let $(L, \mathcal{P}, \Gamma)$ be the computation of square root (example 3.3.3 with $\Delta = \Gamma$). We saw that $\overline{\Delta} = \tilde{\Delta} \cup \{\tilde{f}\} \subseteq B_1(\mathcal{L}, \mathcal{L})$. This corresponds to the Alexandroff compactification of a countable discrete set, which is metrizable. Hence, Δ is NIP_3 but it is not stable, in the sense that $\overline{\Delta} \not\subseteq C(\mathcal{L}, \mathcal{L})$.
2. Let $(L, \mathcal{P}, \Gamma)$ be the finite precision threshold classifiers (Section 3.3.4) with $\Delta = \{\phi_w : w \in 2^{<\mathbb{N}}\} \cup \{0^\infty, 1^\infty\}$. We saw that $\overline{\Delta}$ is homeomorphic to the Split Cantor space $S(2^{\mathbb{N}})$ (Example 3.22-(3)), which is hereditarily separable but not metrizable. Hence, Δ is NIP_2 but not NIP_3 .
3. Let $(L, \mathcal{P}, \Gamma)$ be the CCS given by $L = 2^{\mathbb{N}}$, $\mathcal{P} = \{P_n : n \in \mathbb{N}\}$ and Γ is the semigroup generated by $\Delta = \{\gamma_t : t \in 2^{<\mathbb{N}}\}$, where $P_n : 2^{\mathbb{N}} \rightarrow \{0, 1\}$ is the projection map $P_n(x) = x(n)$ and $\gamma_t : L \rightarrow L$ is given by

$$\gamma_t(x) = \begin{cases} 0^\infty, & \text{if } x <_{\text{lex}} t \frown 0^\infty; \\ (01)^\infty, & \text{if } t \frown 0^\infty \leq_{\text{lex}} x \leq_{\text{lex}} t \frown 1^\infty; \\ 1^\infty, & \text{if } t \frown 1^\infty <_{\text{lex}} x. \end{cases}$$

where $(01)^\infty$ denotes the sequence of alternating bits: $010101 \dots$. As in the other examples, it is not difficult to see that $(L, \mathcal{P}, \Gamma)$ satisfies the Extensibility Axiom. For example, the condition $t \frown 0^\infty \leq_{\text{lex}} x \leq_{\text{lex}} t \frown 1^\infty$ is equivalent to x extending t . Observe that the set of deep computations is homeomorphic to $\hat{D}(S(2^{\mathbb{N}}))$ (see Example 3.22-(5)). Hence, Δ is NIP_1 but not NIP_2 .

4. Let $(L, \mathcal{P}, \Gamma)$ be the finite precision prefix test (Section 3.3.5) with $\Delta = \{\psi_w : w \in 2^{<\mathbb{N}}\}$. We saw that $\overline{\Delta}$ is homeomorphic to the Extended Alexandroff compactification $\hat{A}(2^{\mathbb{N}})$ (Example 3.22-(3)), which is separable but not first countable. Hence, Δ is NIP but not NIP_1 .

The definitions provided here for NIP_i ($i = 1, 2, 3$) are topological. This raises the following question:

Question 3.1. Is there a non-topological characterization for NIP_i , $i = 1, 2, 3$?

3.4.3 The Argyros-Dodos-Kanellopoulos heptachotomy, and approximability of deep computation by minimal classes

In the three separable cases given in 3.22, namely, $\hat{A}(2^{\mathbb{N}})$, $S(2^{\mathbb{N}})$ and $\hat{D}(S(2^{\mathbb{N}}))$, the countable dense subsets are indexed by the binary tree $2^{<\mathbb{N}}$. This choice of index is useful because our emphasis is computational. Real numbers can be represented as infinite binary sequences, i.e., infinite branches of the binary tree $2^{\mathbb{N}}$, and we approximate real numbers with elements in $2^{<\mathbb{N}}$, i.e., finite bitstrings.

Definition 3.25. Let X be a Polish space.

1. If I is countable and $\{f_i : i \in I\} \subseteq \mathbb{R}^X$, $\{g_i : i \in I\} \subseteq \mathbb{R}^X$ are two pointwise families by I , we say that $\{f_i : i \in I\}$ and $\{g_i : i \in I\}$ are *equivalent* if and only if the map $f_i \mapsto g_i$ is extended to a homeomorphism from $\overline{\{f_i : i \in I\}}$ to $\overline{\{g_i : i \in I\}}$.
2. If $\{f_t : t \in 2^{<\mathbb{N}}\}$ is a pointwise bounded family, we say that $\{f_t : t \in 2^{<\mathbb{N}}\}$ is *minimal* if and only if for every dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$, $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to $\{f_t : t \in 2^{<\mathbb{N}}\}$.

One of the main results in [ADK08] is that, up to equivalence, there are seven minimal families of Rosenthal compacta and that for every relatively compact $\{f_t : t \in 2^{<\mathbb{N}}\} \subseteq B_1(X)$ there is a dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ such that $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to one of the minimal families. We shall describe the seven minimal families next. We follow the same notation as in [ADK08]. For any node $t \in 2^{<\mathbb{N}}$, we denote by $t \frown 0^\infty$ ($t \frown 1^\infty$) the infinite binary sequence starting with t and continuing with all 0's (respectively, all 1's). Fix a regular dyadic subtree $R = \{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$ (i.e., a dyadic subtree such that every level of R is contained in a level of $2^{<\mathbb{N}}$) with the property that for all $s, s' \in R$, we have $s \frown 0^\infty \neq s' \frown 0^\infty$ and $s \frown 1^\infty \neq s' \frown 1^\infty$. Given $t \in 2^{<\mathbb{N}}$, let v_t be the characteristic function of the set $\{x \in 2^{\mathbb{N}} : x \text{ extends } t\}$. Let $<$ be the lexicographic order in $2^{\mathbb{N}}$. Given $a \in 2^{\mathbb{N}}$, let $f_a^+ : 2^{\mathbb{N}} \rightarrow \{0, 1\}$ be the characteristic function of $\{x \in 2^{\mathbb{N}} : a \leq x\}$ and let $f_a^- : 2^{\mathbb{N}} \rightarrow \{0, 1\}$ be the characteristic function of $\{x \in 2^{\mathbb{N}} : a > x\}$. Given two maps $f, g : 2^{\mathbb{N}} \rightarrow \mathbb{R}$ we denote by $(f, g) : 2^{\mathbb{N}} \sqcup 2^{\mathbb{N}} \rightarrow \mathbb{R}$ the function which is f on the first copy of $2^{\mathbb{N}}$ and g on the second copy of $2^{\mathbb{N}}$.

1. $D_1 = \{\frac{1}{|t|+1}v_t : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_1} = A(2^{\mathbb{N}})$.
2. $D_2 = \{s_t \frown 0^\infty : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_2} = 2^{\leq \mathbb{N}}$.
3. $D_3 = \{f_{s_t \frown 0^\infty}^+ : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_3} = S(2^{\mathbb{N}})$.
4. $D_4 = \{f_{s_t \frown 1^\infty}^- : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_4} = S(2^{\mathbb{N}})$.
5. $D_5 = \{v_t : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_5} = \hat{A}(2^{\mathbb{N}})$.
6. $D_6 = \{(v_{s_t}, s_t \frown 0^\infty) : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_6} = \hat{D}(2^{\mathbb{N}})$.

7. $D_7 = \{(v_{s_t}, x_{s_t^-}^+ : 0^\infty) : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_7} = \hat{D}(S(2^\mathbb{N}))$.

Theorem 3.26 (Heptachotomy of minimal families, Theorem 2 in [ADK08]). *Let X be Polish. For every relatively compact $\{f_t : t \in 2^{<\mathbb{N}}\} \subseteq B_1(X)$, there exists $i = 1, 2, \dots, 7$ and a regular dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$ such that $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to D_i . Moreover, the families D_i are minimal and mutually non-equivalent.*

The implication of this result for deep computations is the following: for any countable set of computations Δ satisfying the NIP (for some CCS $(L, \mathcal{P}, \Gamma)$), we can always find a countable discrete set of deep computations that approximates all the other deep computations. For example: in the finite precision prefix test example (subsection 3.3.5), the prefix test computations (family D_5) approximate all other deep computations. However, note that this discrete set D_i may not consist of continuous functions (i.e., they will not be computable in general). For example, functions in D_3 are not continuous.

3.5 Monte Carlo computability

In this section, we replace deterministic computability by probabilistic (‘Monte Carlo’) computability. We do not assume that $\mathcal{P}$ is countable. (The countability assumption on $\mathcal{P}$ played a crucial role in the proof of Theorem 3.9, as it makes $\mathbb{R}^\mathcal{P}$ a Polish space.) The main results of the section are Theorem 3.33, which connects NIP and Monte Carlo computability, and 3.40, which connects Talagrand stability and Monte Carlo computability.

Fundamental in this section is a measure-theoretic version of Theorem 3.9, namely, Theorem 3.31.

3.5.1 NIP and Monte Carlo computability of deep computations

By Fact 3.2, for perfectly normal X , a function f is in $B_1(X, Y)$ if and only if $f^{-1}[U]$ is an F_σ subset of X for every open $U \subseteq Y$. This motivates the following definition.

Definition 3.27. Given a Hausdorff space X and a measurable space (Y, Σ) , we say that $f : X \rightarrow Y$ is *universally measurable* (with respect to Σ) if $f^{-1}(E)$ is μ -measurable for every Radon measure μ on X and every $E \in \Sigma$. When $Y = \mathbb{R}$, we will always take $\Sigma = \mathcal{B}(\mathbb{R})$, the Borel σ -algebra of $\mathbb{R}$.

If X is a compact (Hausdorff) space, then Radon measures on X are finite, and hence, through renormalization, we can regard them as probability measures. We thus have the following remark:

Remark 3.28. If X is compact, then a function $f : X \rightarrow \mathbb{R}$ is universally measurable if and only if $f^{-1}(U)$ is μ -measurable for every Radon probability measure μ on X and every open set $U \subseteq \mathbb{R}$.

Intuitively, a function is universally measurable if it is “measurable no matter which reasonable way you try to measure things on its domain”. The concept of universal measurability emerged from work of Kallianpur and Sazonov, in the late 1950’s and 1960s — with later developments by Blackwell, Darst and others — building on earlier ideas of Gnedenko and Kolmogorov from the 1950s. See [Pap02, Chapters 1 and 2].

Notation 3.29. Following [BFT78], the collection of all universally measurable real-valued functions on X will be denoted by $M_r(X)$. Given a fixed Radon measure μ on X , the collection of all μ -measurable real-valued functions on X will be denoted by $\mathcal{M}^0(X, \mu)$.

In the context of deep computations, we are interested in transition maps of a state space $L \subseteq \mathbb{R}^{\mathcal{P}}$ into itself. In the product space $\mathbb{R}^{\mathcal{P}}$, we have two natural σ -algebras: the Borel σ -algebra, i.e., the σ -algebra generated by open sets in $\mathbb{R}^{\mathcal{P}}$, and the cylinder σ -algebra (i.e., the σ -algebra generated by the sub-basic open sets in $\mathbb{R}^{\mathcal{P}}$, that is, the set $\pi_i^{-1}(U)$ with $U \subseteq \mathbb{R}$ open). In general, the cylinder σ -algebra is strictly smaller; however, when $\mathcal{P}$ is countable, both σ -algebras coincide. We will use the cylinder σ -algebra to define universally measurable maps $f: \mathbb{R}^{\mathcal{P}} \rightarrow \mathbb{R}^{\mathcal{P}}$. The reason for this choice is the following characterization:

Proposition 3.30. *Let X be a Hausdorff space and $Y = \prod_{i \in I} Y_i$ be any product of measurable spaces (Y_i, Σ_i) for $i \in I$. Let Σ_Y be the cylinder σ -algebra generated by the measurable spaces (Y_i, Σ_i) . The following are equivalent for $f: X \rightarrow Y$:*

- (i) $f: X \rightarrow Y$ is universally measurable (with respect to Σ_Y).
- (ii) $\pi_i \circ f: X \rightarrow Y_i$ is universally measurable (with respect to Σ_i) for all $i \in I$.

Proof. (i) $\Rightarrow$ (ii) is clear since the projection maps π_i are measurable and the composition of measurable functions is measurable. To prove (ii) $\Rightarrow$ (i), note that if $C = \prod_{i \in I} C_i$ is a measurable cylinder and J is the finite subset of I such that $C_i \neq Y_i$ for $i \in J$, then, $f^{-1}(C) = \bigcap_{i \in J} (\pi_i \circ f)^{-1}(C_i)$ is universally measurable by assumption. $\square$

The preceding proposition says that we can check measurability of a transition just by checking measurability feature by feature. This is analogous to the Baire class 1 setting (see Proposition 3.8).

We saw in section 2.4.2 that, when the set $\mathcal{P}$ of predicates is countable, PAC-learning (or NIP) corresponds to relative compactness in the space of Baire class 1 functions (see Theorems 3.3, 3.6, 3.9). The following result (which does not assume countability of $\mathcal{P}$) gives an analogous characterization of the NIP in terms of universal measurability:

Theorem 3.31 (Bourgain-Fremlin-Talagrand, Theorem 2F in [BFT78]). *Let X be a Hausdorff space and $A \subseteq C(X)$ be pointwise bounded. The following are equivalent:*

- (i) $\overline{A} \subseteq M_r(X)$.
- (ii) For every compact $K \subseteq X$, $A|_K$ satisfies the NIP.
- (iii) For every Radon measure μ on X , A is relatively countably compact in $\mathcal{M}^0(X, \mu)$, i.e., every countable subset of A has a limit point in $\mathcal{M}^0(X, \mu)$.

This result allows us to formalize the concept of a deep computation being *Monte Carlo computable*, which we define below (Definition 3.32). To motivate the definition, let us first recall two facts:

1. Littlewood's second principle states that every Lebesgue measurable function is "nearly continuous." The formal statement of this, which is Luzin's theorem, is that if (X, Σ, μ) a Radon measure space and Y is a second-countable topological space (e.g., $Y = \mathbb{R}^{\mathcal{P}}$ with $\mathcal{P}$ countable) equipped with the Borel σ -algebra, then any given $f: X \rightarrow Y$ is measurable if and only if for every $E \in \Sigma$ and every $\varepsilon > 0$ there exists a closed $F \subseteq E$ such that the restriction $f|_F$ is continuous and $\mu(E \setminus F) < \varepsilon$.

2. Computability of deep computations is characterized in terms of continuous extensibility of computations. This is at the core of [ADIW24].

These two facts motivate the following definition:

Definition 3.32. Let $(L, \mathcal{P}, \Gamma)$ be a CCS. We say that a transition $f: L \rightarrow L$ is *universally Monte Carlo computable* if and only if there exists an extension $\tilde{f}: \mathcal{L}_{\text{sh}} \rightarrow \mathcal{L}_{\text{sh}}$ of f satisfying the following property: for every sizer $r_\bullet$ there is a sizer $s_\bullet$ such that the restriction $\tilde{f}|_{\mathcal{L}[r_\bullet]}: \mathcal{L}[r_\bullet] \rightarrow \mathcal{L}[s_\bullet]$ is universally measurable, i.e., $\pi_P \circ \tilde{f}|_{\mathcal{L}[r_\bullet]}: \mathcal{L}[r_\bullet] \rightarrow [-s_P, s_P]$ is μ -measurable for every Radon probability measure μ on $\mathcal{L}[r_\bullet]$ and $P \in \mathcal{P}$.

Note that, since $\mathcal{L}_{\text{sh}}$ is compact, by remark 3.28, in the context of Monte Carlo computability, we restrict our attention to Radon probability measures instead of general Radon measures.

Theorem 3.33. Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extensibility Axiom. Let R be an exhaustive collection of sizers. Let $\Delta \subseteq \Gamma$ be R -confined. If $\pi_P \circ \Delta|_{L[r_\bullet]}$ satisfies the NIP for all $P \in \mathcal{P}$ and all $r_\bullet \in R$, then every deep computation in Δ is universally Monte Carlo computable.

Proof. Fix $P \in \mathcal{P}$ and $r_\bullet \in R$. By the Extensibility Axiom, $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ is a set of pointwise bounded continuous functions on the compact set $\mathcal{L}[r_\bullet]$. Since $\pi_P \circ \Delta|_{L[r_\bullet]}$ has the NIP, so does $\pi_P \circ \tilde{\Delta}|_{L[r_\bullet]} = \pi_P \circ \Delta|_{\mathcal{L}[r_\bullet]}$, by Lemma 3.10. Let $f \in \overline{\Delta}$ be a deep computation; write f as an ultralimit of computations in Δ , say $f = \mathcal{U} \lim_i \gamma_i$, and define an extension $\tilde{f}: \mathcal{L}_{\text{sh}} \rightarrow \mathcal{L}_{\text{sh}}$ of f naturally by $\tilde{f} := \mathcal{U} \lim_i \tilde{\gamma}_i$. Since, by hypothesis, Δ is R -confined, we have $f: L[r_\bullet] \rightarrow L[r_\bullet]$ and hence, $\tilde{f}: \mathcal{L}[r_\bullet] \rightarrow \mathcal{L}[r_\bullet]$. Now, by Theorem 3.31, we have $\overline{\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq M_r(\mathcal{L}[r_\bullet])$; therefore, $\pi_P \circ \tilde{f}|_{\mathcal{L}[r_\bullet]} \in \overline{\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq M_r(\mathcal{L}[r_\bullet])$. $\square$

Question 3.2. Under the same assumptions of the preceding theorem, suppose that every deep computation of Δ is universally Monte Carlo computable. Must $\pi_P \circ \Delta|_{L[r_\bullet]}$ have the NIP for all $P \in \mathcal{P}$ and all $r_\bullet \in R$?

3.5.2 Talagrand stability and Monte Carlo computability of deep computations

There is another notion closely related to NIP, introduced by Talagrand in [Tal84] while studying Pettis integration. Suppose that X is a compact Hausdorff space and $A \subseteq \mathbb{R}^X$. Let μ be a Radon probability measure on X . Given a μ -measurable set $E \subseteq X$, a positive integer k and real numbers $a < b$, let

$$D_k(A, E, a, b) = \bigcup_{f \in A} \{x \in E^{2k} : f(x_{2i}) \leq a, f(x_{2i+1}) \geq b \text{ for all } i < k\}.$$

Definition 3.34. We say that A is *Talagrand μ -stable* if and only if for every μ -measurable set $E \subseteq X$ of positive measure and for every $a < b$ there is $k \geq 1$ such that

$$(\mu^{2k})^*(D_k(A, E, a, b)) < (\mu(E))^{2k},$$

where μ^* denotes the outer measure.

Note that the reference to outer measure in Definition 3.34 is necessary since $D_k(A, E, a, b)$ need not be measurable. The inequality certainly holds when A is a countable set of continuous (or μ -measurable) functions.

Remark 3.35. Let $X, \mu, A \subseteq \mathbb{R}^X$, and $\subseteq X$ be as above. Then,

1. If $A \subseteq B \subseteq \mathbb{R}^X$, then

$$D_k(A, E, a, b) \subseteq D_k(B, E, a, b).$$

2. If $a < a' < b' < b$, then

$$D_k(\bar{A}, E, a, b) \subseteq D_k(A, E, a', b').$$

[*Proof:* If $x \in D_k(\bar{A}, E, a, b)$, then there is $f \in \bar{A}$ such that $f(x_{2i}) \leq a < a'$ and $f(x_{2i+1}) \geq b > b'$ for all $i < k$. By definition of pointwise convergence topology, there exists $g \in A$ such that $g(x_{2i}) < a'$ and $g(x_{2i+1}) > b'$ for all $i < k$. Hence, $x \in D_k(A, E, a', b')$.]

3. Any subset of a μ -stable set is μ -stable. (This is immediate from (1).)

4. If A is stable, then $\bar{A}$ is μ -stable. (This is immediate from (2).)

The main result of this section is that deep computations that are limit points of a Talagrand stable set of computations are Monte Carlo computable; this is Corollary 3.40 below. To prove it, we first show that the closure of a Talagrand μ -stable set is μ -measurable (lemma 3.37); but first, we need the following useful characterization of measurable functions (compare with Fact 3.2):

Fact 3.36 (Lemma 413G in [Fre03]). *Suppose that (X, Σ, μ) is a measure space and let $\mathcal{K} \subseteq \Sigma$ be a collection of measurable sets satisfying the following conditions:*

- (X, Σ, μ) is complete, i.e., for all $E \in \Sigma$ with $\mu(E) = 0$ and $F \subseteq E$ we have $F \in \Sigma$.
- (X, Σ, μ) is semi-finite, i.e., for all $E \in \Sigma$ with $\mu(E) = \infty$ there exists $F \subseteq E$ such that $F \in \Sigma$ and $0 < \mu(F) < \infty$.
- (X, Σ, μ) is saturated, i.e., $E \in \Sigma$ if and only if $E \cap F \in \Sigma$ for all $F \in \Sigma$ with $\mu(F) < \infty$.
- (X, Σ, μ) is inner regular with respect to $\mathcal{K}$, i.e., for all $E \in \Sigma$

$$\mu(E) = \sup\{\mu(K) : K \in \mathcal{K} \text{ and } K \subseteq E\}.$$

(In particular, if X is compact Hausdorff, $\mathcal{K}$ is the collection of compact subsets of X , μ is a Radon probability measure on X , Σ is the completion of the Borel σ -algebra by μ , and all these conditions hold). Then, $f : X \rightarrow \mathbb{R}$ is measurable if and only if for every $K \in \mathcal{K}$ with $0 < \mu(K) < \infty$ and $a < b$, either $\mu^*(P) < \mu(K)$ or $\mu^*(Q) < \mu(K)$, where $P = \{x \in K : f(x) \leq a\}$ and $Q = \{x \in K : f(x) \geq b\}$.

The following technical lemma will be instrumental for proving Proposition 3.39, which, in turn, will yield the main result of the subsection, namely Corollary 3.40

Lemma 3.37. *If A is Talagrand μ -stable, then $\bar{A}$ is also Talagrand μ -stable and $\bar{A} \subseteq \mathcal{M}^0(X, \mu)$.*

Proof. If A is Talagrand μ -stable, so is $\overline{A}$, by Remarks 3.35. Suppose that A is Talagrand μ -stable to show that $\overline{A} \subseteq \mathcal{M}^0(X, \mu)$. Assume, for the sake of contradiction that there exists $f \in \overline{A}$ such that $f \notin \mathcal{M}^0(X, \mu)$. By Fact 3.36, there exists a μ -measurable set E of positive measure and $a < b$ such that $\mu^*(P) = \mu^*(Q) = \mu(E)$ where $P = \{x \in E : f(x) \leq a\}$ and $Q = \{x \in E : f(x) \geq b\}$. Then, for any $k \geq 1$: $(P \times Q)^k \subseteq D_k(\{f\}, E, a, b)$, so $(\mu^{2k})^*(D_k(\{f\}, E, a, b)) = (\mu^*(P)\mu^*(Q))^k = (\mu(E))^{2k}$. Thus, the set $\{f\}$ is not μ -stable. However, $\{f\} \subseteq \overline{A}$ and we know (Remarks 3.35) that a subset of a μ -stable set must be μ -stable, so we have a contradiction. $\square$

Definition 3.38. We say that A is *universally Talagrand stable* if A is Talagrand μ -stable for every Radon probability measure μ on X .

We first observe that universal Talagrand stability implies NIP:

Proposition 3.39. *Let X be a compact Hausdorff space and $A \subseteq C(X)$ be pointwise bounded. If A is universally Talagrand stable, then A satisfies the NIP.*

Proof. Suppose that A is universally Talagrand stable. By Theorem 3.31, it suffices to show that A is relatively countably compact in $\mathcal{M}^0(X, \mu)$ for every Radon probability measure μ on X . Since, for any such μ , A is Talagrand μ -stable, Lemma 3.37 gives us $\overline{A} \subseteq \mathcal{M}^0(X, \mu)$. In particular, A is relatively countably compact in $\mathcal{M}^0(X, \mu)$. $\square$

Corollary 3.40. *Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extensibility Axiom. If $\pi_P \circ \Delta \upharpoonright_{L[r_\bullet]}$ is universally Talagrand stable for all $P \in \mathcal{P}$ and all sizers $r_\bullet$, then every deep computation is universally Monte Carlo computable.*

Proof. This follows directly from Proposition 3.39 and Theorem 3.33. $\square$

In the context of deep computations, we have identified two ways to obtain Monte Carlo computability, namely, NIP/PAC and Talagrand stability. It is natural to ask whether these two notions are equivalent. The following results show that, even in the simple case of countably many computations, this question is sensitive to the set-theoretic axioms. On the one hand, it is consistent (with respect to the standard ZFC axioms of set theory) that these two classes are the same:

Theorem 3.41 (Talagrand, Theorem 9-3-1(a) in [Tal84]). *Let X be a compact Hausdorff space and $A \subseteq M_r(X)$ be countable and pointwise bounded. Assume that $[0, 1]$ is not the union of $< \mathfrak{c}$ closed measure zero sets. If A satisfies the NIP, then A is universally Talagrand stable.*

(The assumption that $[0, 1]$ is not the union of $< \mathfrak{c}$ closed measure zero sets is a consequence of, for example, the Continuum Hypothesis.)

On the other hand, the opposite is also consistent for the particular case of the Lebesgue measure:

Theorem 3.42 (Fremlin, Shelah, [FS93]). *It is consistent with the usual axioms of set theory that there exists a countable pointwise bounded set of Lebesgue measurable functions with the NIP which is not Talagrand stable with respect to Lebesgue measure.*

Notice that the preceding two results apply to sets of measurable functions, a class of functions larger than the class of continuous functions. However, by the Extensibility Axiom, finitary computations are continuous, i.e., if A is a set of computations, then $A \subseteq C_p(X)$.

Chapter 4

Group Actions on Graphs

4.1 Introduction

This chapter deals with contributions achieved within various areas of graph theory, and Section 4.5 develops a descriptive-set-theoretic theory of the **Fer** group acting on infinite graphs, completing the thesis-wide thread on Polish group actions.

Section 4.2 reports a peripheral algorithmic contribution; the remaining sections develop the chapter's main theme, the **Fer** group on finite and infinite graphs.

The subsequent sections deal with a specific group action defined on finite graphs called the *group of feasible edge replacements* or the **Fer** group, for short. Section 4.3 presents the historic motivation for studying this action by defining the *balancing number* of a graph. This number is not defined for an arbitrary graph, but only for graphs which are *balanceable*, which is a particular extremal or Ramsey-theoretic property. The same section then presents a family of graphs which admit a rich **Fer** group structure, called *global amoeba* graphs. These graphs are always balanceable and bounding their balancing numbers is an important question in this research field. More specifically, this section studies an infinite family of finite trees, obtained through recursive constructions, with rich **Fer** groups and computes their balancing numbers, thus providing the answer to several open questions. Section 4.4 presents a general study of structural properties of trees which have rich **Fer** groups. While the existing literature only deals with the case when the graph is finite, the remaining sections of the chapter aim to extend the study of the **Fer** group to the infinite case, which is the main topic of discussion of Section 4.5.

4.2 Border tracing in oriented adjacency graphs of polygonal tilings

This section briefly reports a side project, which arose within a specific application of graph theory to the analysis of digital images modelled on polygonal plane tilings, where objects of interest appear as connected sets of tiles. Full details of this project, including proofs of all stated results, appear in [WMW26a].

4.2.1 Introduction

Traditionally, the support of a two-dimensional digital image is modelled as $\mathbb{Z}^2$, the set of points of the plane with integer coordinates, called grid points or pixels. $\mathbb{Z}^2$ can be identified with a set of square tiles of the plane. Connectivity in the digital plane $\mathbb{Z}^2$ most commonly relies on the so-called 4-adjacency or 8-adjacency: points $p_1 = (x_1, y_1), p_2 = (x_2, y_2) \in \mathbb{Z}^2$ are called 4-adjacent if $|x_1 - x_2| + |y_1 - y_2| = 1$, and 8-adjacent if $\max\{|x_1 - x_2|, |y_1 - y_2|\} = 1$. The set $\mathbb{Z}^2$ forms a graph together with the k -adjacency relation, $k \in \{4, 8\}$. The usual concept of k -connectivity for a subset $S \subseteq \mathbb{Z}^2$, based on k -paths lying in S , hence applies [GW18, KR04]. In a binary digital image, the pixels belonging to an object of interest are identified and labelled, for example, having the value 1, whereas all other pixels are given the value of 0. Thus there is a partition of $\mathbb{Z}^2$ into object pixels and so-called background pixels. For a k -connected object $S \subseteq \mathbb{Z}^2$, an important task in digital image analysis consists in finding some type of border or boundary of S , with the purpose of separating it from the background. To achieve this, the k -adjacency graph is equipped with an additional structure, specifically, a cyclic ordering of the k -neighbours of each pixel. Then a so-called *border-tracing* algorithm is applied, which finds a closed k -path of object pixels that, intuitively, circumscribes the object or some hole in the object. Such techniques for square pixels and the 4- or 8-adjacency, are described in textbooks on digital image analysis [GW18, KR04].

A natural generalization replaces square pixels by the tiles of some edge-to-edge polygonal tiling (typically called a *tessellation* or *mosaic* in computational geometry) of the Euclidean plane $\mathbb{R}^2$, for example, a rectangular, triangular or hexagonal tiling. This is motivated by theoretical considerations within mathematical disciplines such as digital topology and digital geometry, but also by practical applications. Triangular and hexagonal pixels have been used for decades, as alternative to square pixels, for the design of sensors and digitization hardware and in techniques of digital image processing and pattern recognition, computer graphics and data visualization, for example, in thinning of objects and geometric modelling of straight lines and curves. All this is reported by numerous references in [WMW26a]. Additionally, the paper [Mag25], published in 2025, summarizes modern tendencies of modelling digital objects within polygonal tessellations and provides outlooks to applications in the future.

In [WMW26a], we develop a border-tracing algorithm that works for objects modelled as connected sets of tiles in certain types of polygonal tiling, where connectivity is provided by an adjacency relation, and the adjacency graph is enriched by a so-called orientation. Our algorithm adapts and corrects a classical *Border Mesh Determination Algorithm* due to Voss [Vos93], which originally was proposed only for planar adjacency graphs¹.

4.2.2 Oriented adjacency graphs and border meshes

The notions of adjacency, orientation, and mesh defined below originate with Voss [Vos93] and are kept here consistent with their standard use throughout the border-tracing literature.

Let $\mathcal{T}$ be an edge-to-edge polygonal tiling of $\mathbb{R}^2$ consisting of convex polygons, where each tile intersects only a finite number of other tiles. Then $\mathcal{T}$ is an infinite set and any two distinct intersecting tiles share a common side or point belonging to at least three tiles. Any non-empty finite subset $\mathcal{C} \subseteq \mathcal{T}$ is called an *object*.

¹Planar graphs are graphs that are embeddable in the plane such that graph edges intersect only at nodes.

Definition 4.1 (Adjacency; Voss [Vos93]). An *adjacency* on $\mathcal{T}$ is an irreflexive, symmetric binary relation A on $\mathcal{T}$ such that: any two A -adjacent tiles intersect; the graph $G = (\mathcal{T}, A)$ is connected; and every finite subset of $\mathcal{T}$ has exactly one infinite A -connected complementary component. A -adjacent tiles are also called *A -neighbours*, and the number $v(t)$ of A -neighbours of a tile t is its *degree*. The graph $G = (\mathcal{T}, A)$ is called an *adjacency graph*.

Remark 4.2. The third property ensures that an object has only finitely many holes, each of them finite (Section 4.2.3). The two most natural choices of A are *edge-adjacency* (tiles share a whole side) and *vertex-adjacency* (tiles share at least a vertex); since A -adjacent tiles always intersect, every adjacency is contained in the vertex-adjacency.

Definition 4.3 (Orientation; Voss [Vos93]). An *orientation* of G is a set $Z = \{z(t) : t \in \mathcal{T}\}$, where $z(t) = (q_1, \dots, q_{v(t)})$ is a cyclic ordering of the A -neighbours of t , i.e., $z(t)$ is considered only up to a shifting of the sequence. Then $G = (\mathcal{T}, A, Z)$ is called an *oriented adjacency graph*.

Remark 4.4. This notion is purely combinatorial: Z need not bear any relation to a planar embedding of $\mathcal{T}$, and in particular need not be counterclockwise or clockwise in any geometric sense. The counterclockwise and clockwise orientations (Definition 4.12 below) are two specific, geometrically motivated choices of Z , not the only ones this definition allows.

Definition 4.5 (Mesh; Voss [Vos93]). In an oriented adjacency graph $G = (\mathcal{T}, A, Z)$, or in a subgraph equipped with the cyclic orders induced by Z , define the successor of a directed graph edge (t_i, t_{i+1}) by

$$\varphi(t_i, t_{i+1}) := (t_{i+1}, t_{i+2}),$$

where t_{i+2} is the A -neighbour of t_{i+1} immediately following t_i in the cyclic order $z(t_{i+1})$. Since each $z(t)$ is a cyclic ordering, φ is a bijection on the set of directed graph edges of G ; in particular, every directed graph edge e has a unique successor $\varphi(e)$ and a unique predecessor $\varphi^{-1}(e)$, and $\varphi(e) \neq e$, since a tile is never its own A -neighbour.

The *mesh* generated by a directed graph edge e_0 is obtained by iterating φ from e_0 : setting $e_{i+1} := \varphi(e_i)$ for $i \geq 0$, either $e_r = e_0$ for some smallest $r \geq 1$, in which case the mesh is *closed* and equals the finite cyclic sequence $(e_0, \dots, e_{r-1})$ of length r ; or $e_n \neq e_0$ for every $n \geq 1$, in which case the mesh is the infinite sequence $(e_0, e_1, e_2, \dots)$. Cyclic shifts of the sequence of a closed mesh represent the same mesh, and two infinite sequences represent the same mesh if they have a common directed graph edge. In particular, $(e_0, e_1, e_2, \dots)$, $(e_1, e_2, e_3, \dots)$, $(e_2, e_3, e_4, \dots)$ and so on, represent the same mesh. Denote the set of meshes of G by M , and write the oriented adjacency graph as $G = (\mathcal{T}, A, Z, M)$ when this set is included in the notation.

Lemma 4.6 (No repetition before closure). *Let $(e_0, e_1, e_2, \dots)$ be the sequence of directed graph edges obtained by iterating the successor bijection φ from e_0 . If $e_i = e_j$ for some $0 \leq i < j$, then $e_{j-i} = e_0$. Consequently, no directed graph edge repeats before the sequence first returns to e_0 : the edges of a closed mesh are distinct up to that return, whereas no edge of an infinite mesh is ever repeated.*

Proof. Since φ is a bijection, so is φ^i . The equality $e_i = e_j$ is the equality $\varphi^i(e_0) = \varphi^i(e_{j-i})$, and injectivity of φ^i therefore gives $e_0 = e_{j-i}$. The two conclusions now follow from the minimality of the return time r in the closed case and the absence of any return to e_0 in the infinite case. $\square$

Remark 4.7. By Lemma 4.6, every directed graph edge determines a unique mesh: starting the procedure of iterating φ from another directed graph edge of that mesh generates the same mesh. If $e_i = (t_i, t_{i+1})$ for indices modulo r , a closed mesh may also be written in the shorthand tile notation $(t_0, \dots, t_{r-1})$, of length r (the tiles need not be distinct, but every mesh contains at least one directed graph edge and therefore at least two distinct tiles).

Definition 4.8 (Border mesh; Voss [Vos93]). Let $G = (\mathcal{T}, A, Z, M)$ be an oriented adjacency graph. Every object $\mathcal{C} \subseteq \mathcal{T}$ induces an oriented subgraph $G' = (\mathcal{C}, A', Z', M')$: its underlying graph is the subgraph of $(\mathcal{T}, A)$ induced by $\mathcal{C}$, and Z' arises from Z by restricting each cyclic order $z(t)$, $t \in \mathcal{C}$, to the A -neighbours of t that lie in $\mathcal{C}$; the set M' is then determined by Definition 4.5. A *border mesh* of $\mathcal{C}$ is any mesh $m \in M' \setminus M$, i.e., any mesh of G' that is not a mesh of G .

Remark 4.9. Figure 4.1 shows examples of border meshes that trace (part of) the boundary of the object within the tiling. Since any object $\mathcal{C}$ is finite, $\mathcal{C} \neq \mathcal{T}$; since G is connected (Definition 4.1), $\mathcal{C}$ cannot be closed off from the rest of $\mathcal{T}$, so some tile $t \in \mathcal{C}$ has an A -neighbour outside $\mathcal{C}$, hence $z'(t)$ is strictly shorter than $z(t)$; consequently, the successor rule can act differently in G' than in G , which allows $M' \setminus M$ to be nonempty.

Lemma 4.10. Let $G = (\mathcal{T}, A, Z, M)$ be an oriented adjacency graph on a polygonal tiling $\mathcal{T}$, and let $\mathcal{T}'$ be a finite subset of $\mathcal{T}$. Denote by $G' = (\mathcal{T}', A', Z', M')$ the oriented adjacency subgraph induced by $\mathcal{T}'$, as in Definition 4.8. Then every mesh of G' is closed, and the meshes of G' partition its set of directed graph edges.

Proof. Let φ' be the successor bijection of G' . Since $\mathcal{T}'$ is finite, G' has finitely many directed graph edges, so φ' is a bijection of a finite set and hence decomposes into disjoint finite cycles. Thus iterating φ' from any directed graph edge of G' returns to it after finitely many steps, which is exactly the closed case of Definition 4.5, and the resulting cycles partition the set of directed graph edges. $\square$

Remark 4.11. On the infinite tiling $\mathcal{T}$ itself, some orientations do produce infinite meshes. Nevertheless, by Lemma 4.10 (applied with $\mathcal{T}' = \mathcal{C}$), every mesh of G' is closed and M' partitions the directed graph edges of G' ; in particular, every border mesh, being a mesh of G' , is closed. A border mesh may trace the whole of G' , i.e., pass through every directed graph edge of G' . Border meshes can be determined for arbitrary adjacency A , orientation Z , and A -connected object $\mathcal{C}$ with at least two tiles, by the algorithm described in [WMW26a].

The counterclockwise and clockwise orientations are of special interest for algorithmic aspects of border tracing.

Definition 4.12 (Counterclockwise and clockwise orientation). For a given adjacency graph $G = (\mathcal{T}, A)$, $Z = \{z(t) : t \in \mathcal{T}\}$ is the *counterclockwise orientation* if, for each tile $t \in \mathcal{T}$, $z(t)$ orders the A -neighbours of t in counterclockwise sense around t .² The *clockwise orientation* is defined analogously.

²Observe that the topological frontier of the convex polygon t is a Jordan curve γ , which can be traced counterclockwise as a concatenation $(s_1, s_2, \dots, s_m)$ of the sides of t . Here *frontier* denotes the usual topological notion (the closure minus the interior of a point set); we reserve *boundary* for the tile-theoretic notion $\mathcal{B}(\mathcal{C})$ introduced later, in the “Boundary coincidence” property of Section 4.2.3.

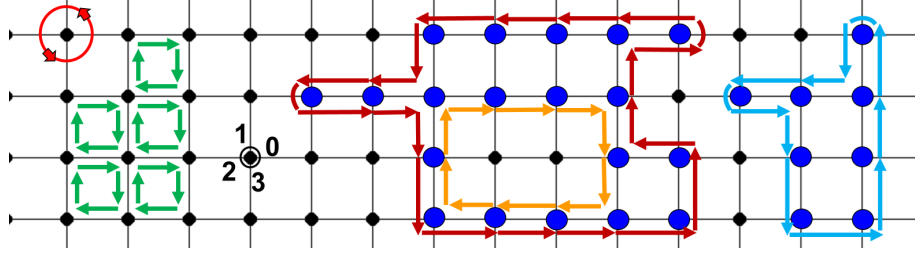


Figure 4.1: The nodes of the edge-adjacency graph G of a square tiling are drawn as dots, the graph edges as black straight line segments, and the underlying tiling is not shown. The counterclockwise orientation (illustrated in the top left corner with light red arrows) generates meshes, illustrated by green arrows, which circumscribe all graph faces (each mesh has length four, since every face of this square-tiling graph is a 4-cycle). The nodes drawn as big blue dots form an object $\mathcal{C}$ with two components and a hole, which induces a subgraph and three border meshes of $\mathcal{C}$, depicted by blue, dark red and orange arrows. Reproduced from [WMW26a, Figure 11].

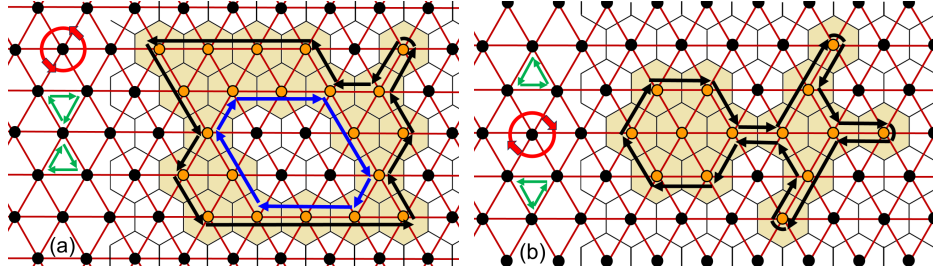


Figure 4.2: The nodes of the edge-adjacency graph G of a hexagonal tiling are drawn as dots, the graph edges as brown straight line segments, the underlying tiling is only shown in part. The counterclockwise orientation in (a) generates meshes that circumscribe all graph faces, depicted by green arrows. The clockwise orientation in (b) causes a distinct tracing sense of the meshes. In each graph, the nodes drawn as orange dots form a connected object whose tiles are shaded in, and its border meshes are shown by black and blue arrows. Reproduced from [WMW26a, Figure 13].

Remark 4.13. Let Z be the counterclockwise orientation on an adjacency graph $(\mathcal{T}, A)$, let $t \in \mathcal{T}$, and list the sides $s_1, s_2, \dots, s_m$ of the polygon t in the order encountered while tracing the Jordan curve $\gamma = \text{fr}(t)$ counterclockwise. If A is the edge-adjacency, $z(t)$ coincides, up to a shifting of the cyclic sequence, with $(p_1, \dots, p_m)$, where each tile p_i shares the side s_i with t . If A is the vertex-adjacency, $z(t)$ lists all polygons intersecting t in counterclockwise order around γ : for each vertex a of t , the polygons $h_1, \dots, h_r$ sharing a with t (where h_1, h_r are edge-neighbours of t) appear successively in $z(t)$, with consecutive ones sharing a side containing a .

Figures 4.1 and 4.2 show examples of border meshes for the edge-adjacency on square and hexagonal tilings.

For applications in digital image modelling and analysis, tilings with certain homogeneity properties are highly relevant, for example, where all tiles are congruent polygons. Triangular and hexagonal tilings with this property were considered in detail in [WMW26a]. A particularly simple adjacency would satisfy that each tile $t \in \mathcal{T}$ has the same number $v(t) = k$ of neighbours, which defines a k -adjacency on $\mathcal{T}$. When A is a k -adjacency, A -neighbours also are called k -neighbours, and the concepts of k -path, k -connected, k -component, etc., are employed. The commonly used 4- and 8-adjacencies for square tilings in [GW18, KR04] are examples of k -adjacencies.

4.2.3 The algorithm BorderTracer

The general border-tracing procedure [WMW26a, Algorithm 1] works for any oriented adjacency graph $G = (\mathcal{T}, A, Z)$ and any A -connected object $\mathcal{C} \subset \mathcal{T}$ that has at least two tiles. In particular, the algorithm itself places no restriction on the orientation Z used; the counterclockwise or clockwise orientations become relevant only when relating the resulting border mesh to the geometric boundary of $\mathcal{C}$, as in the properties discussed at the end of this subsection. The algorithm determines a border mesh of $\mathcal{C}$. It starts with A -neighbours (t_0, q) with $t_0 \in \mathcal{C}$ and $q \in (\mathcal{T} \setminus \mathcal{C})$; any pair satisfying these conditions may be chosen. To find such t_0, q , a finite set $\mathcal{N} \subset \mathcal{T}$ surrounding the object $\mathcal{C} \subset \mathcal{N}$ is used, i.e. such that all (including non-object) A -neighbours of every object tile belong to $\mathcal{N}$.

The algorithm locates q in $z(t_0)$ and analyses, for each one of the tiles subsequent to q in $z(t_0)$, whether it belongs to $\mathcal{C}$. If t_1 is the first found object tile, the directed graph edge (t_0, t_1) initialises the border mesh. At a fixed point of the iteration, if t_m is the last tile found so far, then t_{m-1} is located in $z(t_m)$, so the algorithm examines whether the tiles successive to t_{m-1} in $z(t_m)$ belong to $\mathcal{C}$, the first found object tile gives the next border tile t_{m+1} . When $t_{m+1} = t_0$, the algorithm stops if also $t_{m+2} = t_1$, otherwise the tracing is continued. This is justified because the border mesh can only be closed if the starting directed graph edge (t_0, t_1) is found again. Our End Condition Test constitutes a correction over the original method of Voss [Vos93] as it guarantees a correct performance even in case that a border mesh may pass through the starting tile t_0 multiple times before closing.

The algorithm returns an ordered cyclic list `Cont` of tiles, which represents the border mesh of $\mathcal{C}$ uniquely determined by the directed graph edge (t_0, t_1) .

When A is a k -adjacency, the cyclic orderings $z(t)$, $t \in \mathcal{T}$, can be encoded uniformly, which is important since it turns the cyclic order into a simple integer lookup so that the algorithm can efficiently move from one border tile to the next. For k -adjacent tiles t and t' with $z(t) = (q_1, q_2, \dots, q_k)$ and $t' = q_i$, the *Freeman code* of the ordered pair (t, t') is defined as the integer $f(t, t') = i - 1$. Thus $z(t)$ can be represented by the sequence $(0, 1, 2, \dots, k - 1)$, then a k -path $(t_1, t_2, \dots, t_n)$ can be encoded by the *Freeman chain code* $(f(t_1, t_2), f(t_2, t_3), \dots, f(t_{n-1}, t_n))$ (adding $f(t_n, t_1)$ if the path is closed). This encoding is the natural extension of the classical Freeman chain code, which has been applied since the beginning of digital image analysis for 4- and 8-neighbours on the square grid [GW18]. Figure 4.3 illustrates Freeman codes for the standard triangular tiling.

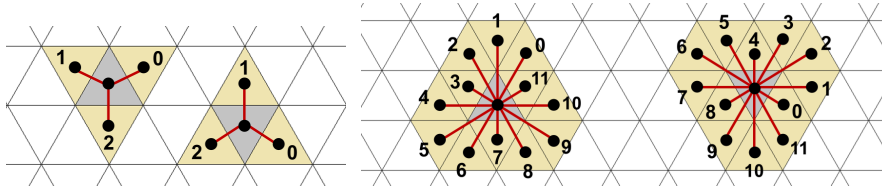


Figure 4.3: Freeman codes for the edge- and vertex-neighbours in the standard triangular tiling with the counterclockwise orientation. Reproduced from [WMW26a, Figure 12].

For the case of a k -adjacency, the border-tracing algorithm simplifies to the version presented below as Algorithm 1, which uses the Freeman code to drive the neighbour search. The algorithm has linear time complexity in the length of the output list `Cont`: at each step, locating the previous tile in the cyclic order and scanning for the next object tile each require at most k comparisons.

For example, repeated applications of Algorithm 1 to the appropriate A -connected components

Algorithm 1: Border tracing for a k -adjacency graph $G = (\mathcal{T}, A, Z, M)$

Input : k -adjacency graph $G = (\mathcal{T}, A, Z, M)$, where Z is represented by the Freeman code sequence $(0, 1, \dots, k-1)$;
 object $\mathcal{C} \subseteq \mathcal{T}$;
 a finite set $\mathcal{N}$ with $\mathcal{C} \subseteq \mathcal{N} \subseteq \mathcal{T}$ surrounding the object.
Output: Ordered cyclic list Cont of tiles representing a border mesh of $\mathcal{C}$ in G .

```

1 Step 1.
2 Scan  $\mathcal{N}$  to find a tile  $t = t_0 \in \mathcal{C}$  having an  $A$ -neighbour  $q \in \mathcal{N} \setminus \mathcal{C}$ .
3 Set  $\text{Cont} \leftarrow (t_0)$ .
4 Step 2.
5 Set  $b \leftarrow f(t, q)$ .
6 repeat
7    $b \leftarrow (b + 1) \bmod k$ 
8   Let  $t'$  be the tile with  $f(t, t') = b$ 
9 until a tile  $t' \in \mathcal{C}$  is found (at most  $k$  iterations)
10 if no object tile was found then return error: “ $\mathcal{C}$  is an isolated tile”
11 Append  $t'$  to  $\text{Cont}$ , so  $\text{Cont} = (t_0, t_1)$ .
12 Step 3.
13 Let  $\text{Cont} = (t_0, t_1, \dots, t_n)$ .
14 Set  $b \leftarrow f(t_n, t_{n-1})$ .
15 repeat
16    $b \leftarrow (b + 1) \bmod k$ 
17   Let  $t'$  be the tile with  $f(t_n, t') = b$ 
18 until a tile  $t' \in \mathcal{C}$  is found
19 End Condition Test:
20 if  $t' \neq t_0$  then
21   Append  $t'$  to  $\text{Cont}$ ; go to Step 3.
22 else
23   Append  $t'$  to  $\text{Cont}$ . Run Step 3 once more to obtain the next tile  $t' \in \mathcal{C}$ .
24   if  $t' = t_1$  then
25     Remove the last element of  $\text{Cont}$ .
26     return  $\text{Cont}$ .
27   else
28     Append  $t'$  to  $\text{Cont}$ ; go to Step 3.

```

and starting pairs determine the border meshes shown in Figures 4.1 and 4.2.

The border-tracing algorithm finds a border mesh of an object $\mathcal{C}$. We now record the geometric guarantees proved in [WMW26a] for the hole-free setting. Suppose that A is the edge- or vertex-adjacency, Z is the counterclockwise (respectively, clockwise) orientation, and $\mathcal{C}$ is an A -connected object with at least two tiles whose complement $\mathcal{C}^c := \mathcal{T} \setminus \mathcal{C}$ is edge-connected (and hence has no finite component, or *hole*). For every choice of edge-neighbours t_0 and q with $t_0 \in \mathcal{C}$ and $q \in \mathcal{C}^c$, the algorithm returns a border mesh Cont uniquely determined by that starting pair. Every such output circumscribes $\mathcal{C}$ and is called a *contour* of $\mathcal{C}$. Thus the choice of (t_0, q) is arbitrary within the stated conditions: different choices may change the starting point or tile representation, but each resulting contour traces the same geometric frontier in the prescribed direction. More precisely, a contour satisfies the following properties:

1. **Circumscription** [WMW26a, Theorem 1]. Write $\text{Cont} = (t_0, \dots, t_n)$ and let $\gamma := \text{fr}(|\mathcal{C}|)$, where $|\mathcal{C}| \subseteq \mathbb{R}^2$ is the point-set union of the tiles in $\mathcal{C}$. For each $i \in \{0, \dots, n\}$, the intersection

$t_i \cap \gamma$ contains a connected set F_i that is either a vertex of t_i or a union of consecutive sides of t_i . The ordered sequence $(F_0, \dots, F_n)$ traces $\gamma = F_0 \cup \dots \cup F_n$ counterclockwise (respectively, clockwise).

2. **Separation** [WMW26a, Theorem 2] (under edge-adjacency). If A is the edge-adjacency, the graph $(\mathcal{T}, A)$ is planar. Then, deleting all graph edges between the contour and the background $\mathcal{C}^c$ disconnects $\mathcal{C}$ from $\mathcal{C}^c$, i.e. there is no edge-path in the resulting graph from the object to the background.
3. **Boundary coincidence** [WMW26a, Lemma 5] (under edge-adjacency). The set of tiles appearing in the contour equals the boundary $\mathcal{B}(\mathcal{C})$, defined as the set of all object tiles that intersect $\text{fr}(|\mathcal{C}|)$.

Remark 4.14 (Objects with holes). A single run of Algorithm 1 follows the successor orbit determined by its starting pair and therefore returns one border mesh, or contour. If an object has holes, its frontier may have several components (an outer component and components surrounding the holes) which can correspond to distinct border meshes determined by different starting pairs. Repeated runs can extract these meshes, as in the implementation described below, but a complete geometric characterization of their relation to objects with holes lies beyond the scope of the present algorithmic guarantees and is left for future work.

A complete **implementation** of Algorithm 1, designed, implemented, and documented by the author, is available in a public repository [WMW26b]. It supports the edge- and vertex-adjacencies on rectangular, triangular, and hexagonal tilings, and includes routines for generating the Freeman codes, detecting starting pairs (t_0, q) , and extracting all border meshes via repeated application of the algorithm with appropriate marking of already-traced borders.

Future work within this project also aims to implement tools for diverse polygonal tilings and to consider applications of modelling digital objects in such tilings.

4.3 Recursive constructions

4.3.1 Balanceable graphs

Ramsey’s classical theorem for finite graphs states that, for a fixed graph G , there is a large enough complete graph K_n such that no matter how the edges of K_n are coloured (in say two colours), one can find a monochromatic copy of G inside of K_n . Suppose one wishes to find a copy of G inside the coloured graph K_n , but instead of monochromatic, with half of the edges (rounded down or up when G has an odd number of edges) in one colour, and the other half in the remaining colour. Clearly, one needs to impose additional conditions on the colourings of K_n , namely, that each colour class is “well represented” (since otherwise one could colour K_n in a single colour, making it impossible to find a non-trivial graph with the desired property) and one needs additional conditions on the structure of G . We do not go into the details of what those properties are, beyond what is needed to state and expand the results from my own research, as well as tie these concepts to the other ideas in this section. The interested reader will find a thorough discussion and complete characterization of these graphs in [CHM21, Section 2].

Definition 4.15. Suppose that a finite graph G has an edge-colouring $c: E(G) \rightarrow 2$ in two colours. This coloured graph is called *balanced* if the number of edges of each colour differs by at most one; that is, if G has an even number of edges, there are exactly as many edges of each colour.

A finite graph G is called *balanceable* if in any 2-colouring of the complete graph K_n , with sufficiently many edges of each colour and n sufficiently large, one can find a balanced copy of G .

The *balancing number* $\text{bal}(n, G)$ of a balanceable graph G is the maximum number k for which there is a 2-colouring of $c: E(K_n) \rightarrow 2$ with $\min\{|c^{-1}\{0\}|, |c^{-1}\{1\}|\} = k$ and no balanced copy of G .

In other words, any 2-colouring of K_n with more than $\text{bal}(n, G)$ edges in each colour contains a balanced copy of G .

Theorem 4.16 ([CHM21, Corollary 2.5]). *Let G be a finite graph with k edges. Then G is balanceable if and only if:*

1. *there are disjoint $X, Y \subseteq V(G)$ such that $V(G) = X \cup Y$ and there are either $\lfloor \frac{1}{2}k \rfloor$ or $\lceil \frac{1}{2}k \rceil$ edges between the vertices of X and Y ; and*
2. *there is $W \subseteq V(G)$ such that there are either $\lfloor \frac{1}{2}k \rfloor$ or $\lceil \frac{1}{2}k \rceil$ edges with both endpoints in W .*

We can study more specific aspects of graphs than whether they are balanceable or not. Interestingly, the balancing number may or may not depend on n , as the following examples demonstrate. We include a detailed proof of the first example for the convenience of the reader, as it often used without proof in the literature.

Proposition 4.17. *The 4-cycle, C_4 , has balancing number 1. More specifically, $\text{bal}(n, C_4) = 1$ whenever $n \geq 4$.*

Proof. Indeed, it is obvious that the balancing number cannot be less than 1, as C_4 has four edges, and hence a balanced copy needs two edges of each colour.

As for the upper bound, let $n \geq 4$ and fix an arbitrary edge colouring $c: E(K_n) \rightarrow \{\text{red}, \text{blue}\}$ with at least two edges of each colour. First, suppose there is a vertex $x \in V(K_n)$ all of whose incident edges are same colour (say red). Then, if there are two blue edges in K_n that intersect, they form a blue path of vertices p_0, p_1, p_2 , none of which can be x by assumption. The edges $xp_0, p_0p_1, p_1p_2, p_2x$ are red, blue, blue, red, respectively, and thus form a balanced copy of C_4 . If no two blue edges intersect, then the edges joining the endpoints of two particular blue edges (of which there must be at least two) must be all red, which we use to form a balanced copy of C_4 .

Now, consider the case when every vertex has red and blue incident edges. Continue by fixing a red edge xy and partitioning the vertices $w \in V(K_n) \setminus \{x, y\}$ into four classes depending on the value of the pair $(c(wx), c(wy))$. There are two colours, hence four possible values for this pair; let us abbreviate the classes as RR for those vertices w with pair (red, red) and similarly, RB, BR and BB . By assumption, x and y have blue incident edges and thus $V(K_n) \setminus \{x, y\} \notin \{RR, RB, BR\}$. If $V(K_n) \setminus \{x, y\} = BB$, then any red edge different from xy must be disjoint from xy and forms a balanced copy of C_4 by connecting said edge to xy (via blue edges). We can therefore assume that at least two of the four classes are nonempty, giving six sub-cases to analyze:

If there are $w \in RR$ and $z \in BB$, or if there are $w \in RB$ and $z \in BR$, then (x, w, y, z) is a balanced 4-cycle; if $w \in RB$ and $z \in BB$, then either (x, w, z, y) (whenever wz is blue) or (x, y, w, z)

(whenever wz is red) is a balanced 4-cycle; the case $w \in BR$ and $z \in BB$ is symmetric; if $w \in RR$, $z \in RB \cup BR$ and wz is blue, then either (x, y, z, w) (whenever $z \in RB$) or (x, y, w, z) (whenever $z \in BR$) is a balanced 4-cycle.

The only case that is not yet covered so far is when $w \in RR$, $z \in RB$ (or, respectively, when $z \in BR$) and wz is a red edge. Invoking our assumption one last time, there must be a blue edge incident in w , say ww' . Crucially, we may assume that $w' \in RR$ since any other possibility is covered by the preceding discussions. Finally, the cycle (w, w', y, z) (or, respectively (w, w', x, z) , when $z \in BR$) is balanced. $\square$

Example 4.18.

1. The path on three edges, P_4 , has balancing number 0.

Suppose that the edges of K_n are coloured in red and blue with at least one edge of each colour. In particular, a red edge and a blue edge must intersect, forming a path of two edges of different colour. Taking any third edge that completes the copy of P_4 , successfully builds a balanced copy regardless of the colour of the third edge.

2. For $n \geq 10$, $\text{bal}(n, 4K_2) = n - 1$.³ This is [CLZ20, Theorem 2.3].
3. When $n \geq 5$, $\text{bal}(n, K_{1,4}) = n - 1$. This is an immediate consequence of [CHM21, Theorem 3.2].
4. The complete graph K_5 is not balanceable.

Consider a colouring of K_{2n} where we split the vertex set into two disjoint sets A and B of size n . We colour every edge connecting vertices in A red, and edges connecting vertices in B blue; edges connecting A with B are also red. Suppose a balanced copy C of K_5 sits in K_{2n} with vertex set $V(C) \subseteq V(K_{2n})$; then five of its ten edges must be blue. The vertex set $V(C) \cap B$ induces a complete graph (a clique in K_{2n}) such that an edge $xy \in E(C)$ is blue if and only if $\{x, y\} \subseteq B$. Since $\binom{k}{2} = 5$ has no integer solution, this is a contradiction.

This example illustrates the characterization provided by Theorem 4.16. The preceding argument shows that K_5 fails the second condition of the theorem; it is straightforward (albeit unnecessary) to show that it also fails the first.

Let us elucidate the case when the value of $\text{bal}(n, G)$ is independent of the parameter n . So, suppose that G is a graph on k edges (for simplicity, let us say k is always even), such that $\text{bal}(n, G) \leq C(G)$. In such a case, we say that G has *constant balancing number*, such as in Example 4.18(1). The graph $\frac{k}{2}K_2$ is a set of $k/2$ disjoint edges (a *matching*), and $K_{1,k/2}$ is a *star graph*, that is, $k/2$ edges incident on a single vertex.

In [CGH⁺23], we gave a concrete structural characterization of graphs of constant balancing number independently of, and concurrently with, the characterization in [CHM26, Section 3.2], where the found values of $C(G)$ involve Ramsey numbers, which grow exponentially. In our paper, the characterization offers a deeper understanding of the structure of the graphs with constant balancing number, and furthermore shows that $\text{bal}(n, G)$ is quadratic in k (and not just bounded by a function exponential in k). Moreover, we extend the characterization to graphs with an odd number of edges.

³Here $4K_2$ is the graph consisting of four disjoint edges.

Theorem 4.19. *Let $k > 0$ be an even integer and G a graph with k edges and no isolated vertices. Then, the following are equivalent.*

1. G has constant balancing number.
2. There are injective graph homomorphisms $\frac{k}{2}K_2 \rightarrow G$ and $K_{1,k/2} \rightarrow G$.
- 3.

$$\text{bal}(n, G) \leq \frac{(|V(G)| - 3)(|V(G)| + k)}{2}.$$

Moreover, if G has either k or $k + 1$ edges, then

$$\frac{1}{18}k^2 - \frac{11}{18}k + \frac{14}{9} \leq \text{bal}(n, G) \leq \frac{15}{8}k^2 - \frac{7}{4}k - 1.$$

Proof sketch of (1) $\implies$ (2). Fix the complete graph K_n . One can always pick an arbitrary vertex v_0 and colour all edges incident in v_0 red and every other edge of K_n blue, as done in the proof of Proposition 4.17. Provided one makes n large enough, one can guarantee that K_n has more than $C(G)$ red edges. By definition, there is a balanced copy of G inside K_n . In particular, all of its red edges (which account for exactly $k/2$ edges of G) intersect at some vertex. This vertex in G can be then chosen as the image of the centre of the star graph $K_{1,k/2}$ under an injective graph homomorphism $K_{1,k/2} \rightarrow G$.

Consider a different colouring of K_n . This time, pick a maximal matching in K_n and colour those edges red, and everything else blue. The size of the matching increases with n , and so once again, for large enough n , we can guarantee that G satisfies Definition 4.15. Consequently, (at least) half of the edges of G must be pairwise disjoint. In other words, one can construct an injective homomorphism $\frac{k}{2}K_2 \rightarrow G$. $\square$

The implication (2) $\implies$ (1) appears as [CHM26, Theorem 3.6], where it is proved using Ramsey numbers, and also as [CGH⁺23, Theorem 1.1], where we prove it by a structural induction. The implication (2) $\implies$ (3) is [CGH⁺23, Theorem 3.7] and is the only part that requires that G has no isolated vertices. The moreover part is [CGH⁺23, Theorem 5.1]. The implication (3) $\implies$ (1) is immediate, as the bound in (3) does not depend on n .

If one wishes to study an analogous notion for infinite graphs, one has to define what it means for a large complete graph to have “sufficiently many edges of each colour.” In [BLM20, Section 4], the authors study a somewhat similar notion of balanceability and suggest an approach to generalizing these ideas to the infinite case. Drawing inspiration from this approach, we adapt some of the ideas to this theory. Namely, that “sufficiently many edges of each colour” corresponds to non-meagre colour classes in a Polish space.

Section 4.5 includes an analogue of Theorem 4.19 that says that if the colours are well represented in a complete graph on a perfect Polish space, then one can find a star graph (in this context: a fixed centre connected to a Cantor set through edges) in one colour, joined with perfectly-many pairwise disjoint edges in the other colour.

4.3.2 The group of feasible edge replacements

In this subsection, we introduce the group $\text{Fer}(G)$, one of the central objects studied in the rest of the chapter. It was first defined in [CHM23] to study extremal graph properties, including the balancing

number, and has since appeared in several works [CHM21, CHM26, CLZ20, EHMWV25, GL22]. We work only with finite graphs here and follow the conventions of this literature. The definitions are extended to infinite graphs in Section 4.5.

Applying label permutations.

Definition 4.20. Given a set X , the collection of all bijections $\sigma: X \rightarrow X$ forms a group under composition, called the *symmetric group on X* and denoted by S_X . When $X = \{0, 1, \dots, n-1\}$, we write $S_n = S_X$. Products of permutations are evaluated from right to left.

For the remainder of this subsection, fix a finite graph G of order n and enumerate its vertices as

$$V(G) = \{v_0, v_1, \dots, v_{n-1}\}.$$

We call $X := \{0, 1, \dots, n-1\}$ the *label set* of G , and refer to $i \in X$ as the *label* of the vertex v_i . An element $\sigma \in S_n$ is called a *label permutation*. The group $\text{Fer}(G)$ will be a subgroup of S_n .

Definition 4.21 (Applying a label permutation). For any graph H on the vertex set $V(G)$ and any $\sigma \in S_n$, define H_σ on the same vertex set by

$$E(H_\sigma) := \{v_i v_j : v_{\sigma(i)} v_{\sigma(j)} \in E(H)\}.$$

We mostly apply this to $H = G$ itself, writing G_σ , but the operation makes sense for any graph on this vertex set, including graphs obtained from G by feasible edge replacements.

A step-by-step example helps visualize the definition. Figure 4.4 starts with a path on four vertices and applies $\sigma = (1\ 3\ 2)$. The construction proceeds as follows.

- (a) Draw G with vertices $v_0, v_1, \dots, v_{n-1}$.
- (b) Remove its edges and annotate each vertex v_i with $\sigma(i)$, displayed as $v_{i \mapsto \sigma(i)}$.
- (c) Join two positions precisely when their displayed labels are adjacent in G .
- (d) Remove the annotations; the resulting graph is G_σ on the original vertex set.

For example, $v_0 v_2$ is an edge of $G_{(1\ 3\ 2)}$ because $v_{\sigma(0)} v_{\sigma(2)} = v_0 v_1 \in E(G)$. In contrast, $v_0 v_1$ is an edge of G but not of $G_{(1\ 3\ 2)}$.

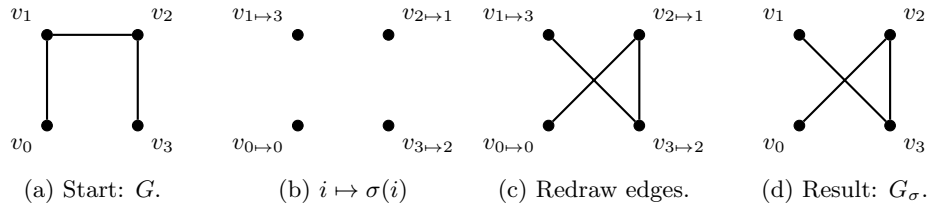


Figure 4.4: Applying the label permutation $\sigma = (1\ 3\ 2)$ to the graph G .

Remark 4.22. The map $\varphi_\sigma: V(G_\sigma) \rightarrow V(G)$ defined by $\varphi_\sigma(v_i) = v_{\sigma(i)}$ is an isomorphism, since

$$v_i v_j \in E(G_\sigma) \iff v_{\sigma(i)} v_{\sigma(j)} \in E(G).$$

This convention makes σ , rather than σ^{-1} , describe the isomorphism $G_\sigma \rightarrow G$ and agrees with the established literature.

Definition 4.23 (Automorphism label group). Define

$$A(G) := \{\sigma \in S_n : G_\sigma = G\}.$$

The correspondence $\sigma \mapsto \varphi_\sigma$ gives an isomorphism $A(G) \simeq \text{Aut}(G)$.

Every isomorphism $\psi: H \rightarrow G$ from a graph H on the same vertex set uniquely determines the permutation σ satisfying $\psi(v_i) = v_{\sigma(i)}$; then $H = G_\sigma$. The permutation is unique for the chosen isomorphism, but two fixed isomorphic graphs generally admit several isomorphisms.

Remark 4.24. If G and H are isomorphic graphs on the same vertex set, then exactly $|\text{Aut}(G)|$ label permutations σ satisfy $H = G_\sigma$.

Figure 4.5 uses the same graph G in a different drawing and applies the permutation $\sigma = (0\ 3\ 1\ 2)$.

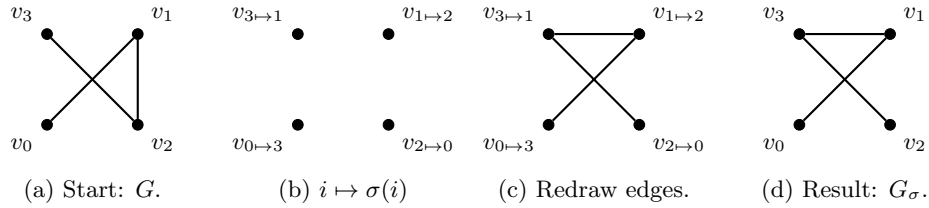


Figure 4.5: Applying the label permutation $\sigma = (0\ 3\ 1\ 2)$ to the graph G .

Feasible edge replacements. If $e = \{v_i, v_j\}$ is a pair of vertices and $\sigma \in S_n$, write $\sigma(e) := \{v_{\sigma(i)}, v_{\sigma(j)}\}$.

Definition 4.25 (Feasible edge replacement). Let e and f be 2-element subsets of $V(G)$. The pair (e, f) , denoted by $(e \rightarrow f)$, is a *feasible edge replacement* of G if

$$e \in E(G), \quad f \notin E(G), \quad G - e + f \simeq G,$$

where $G - e + f$ has edge set $(E(G) \setminus \{e\}) \cup \{f\}$.

Definition 4.26 (Witness permutations). If $(e \rightarrow f)$ is a feasible edge replacement for G , define

$$\text{Fer}_G(e \rightarrow f) := \{\sigma \in S_n : G_\sigma = G - e + f\}.$$

Its elements are the label permutations that witness the feasibility of $(e \rightarrow f)$.

The literature uses the notation $\text{Fer}_G(\emptyset \rightarrow \emptyset) := A(G)$ for the permutations induced by automorphisms. They move no edge, but they are retained because $\text{Fer}(G)$ records label permutations rather than merely the graphs they produce: two permutations producing the same graph differ by an element of $A(G)$. Their inclusion also retains the label symmetries of graphs with no nontrivial feasible edge replacements, such as complete graphs. This group structure is exactly what Subsection 4.3.4 relies on to compute the balancing number of G , for instance by establishing $\text{Fer}(G) = S_X$ through a comparison of group orders.

Figure 4.6 illustrates a feasible edge replacement. Both $\sigma = (2\ 3)$ and $\sigma = (0\ 3\ 1\ 2)$ witness it, since

$$G_{(2\ 3)} = G_{(0\ 3\ 1\ 2)} = G - v_1v_2 + v_1v_3.$$

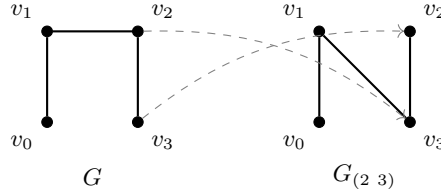


Figure 4.6: A feasible edge replacement with more than one witness permutation.

Figure 4.4 gives a two-step example. Starting from G , the sequence of feasible edge replacements

$$(v_1v_2 \rightarrow v_1v_3), \quad (v_0v_1 \rightarrow v_0v_2)$$

produces G_α , where $\alpha = (1\ 3\ 2)$.

To make use of the group structure, it is important to understand how feasible edge replacements are applied sequentially. The following puzzle makes the issue visible. Figure 4.7 shows G and $G_{(0\ 3)}$. Informally, applying the transposition $(0\ 3)$ to G twists the two bottom endpoints of the path. Can one pass from G to $G_{(0\ 3)}$ by a sequence of feasible edge replacements, keeping every intermediate graph isomorphic to G ?



Figure 4.7: Two paths on the same vertex set.

The first two feasible edge replacements already produce the two diagonal edges required in the final graph. Now let $\beta = (0\ 3\ 1\ 2)$, which also witnesses the feasible edge replacement $(v_1v_2 \rightarrow v_1v_3)$ for the original graph G . Since $\beta \circ \alpha = (0\ 3)$, it is tempting to write the sequence

$$(v_1v_2 \rightarrow v_1v_3), \quad (v_0v_1 \rightarrow v_0v_2), \quad (v_1v_2 \rightarrow v_1v_3).$$

This sequence cannot be executed as written: the last instruction tries to remove the edge v_1v_2 , but that edge was removed by the first feasible edge replacement and was never added back. The final instruction must therefore be translated into the current labelling.

Proposition 4.27 (Relabelling feasible edge replacements). *Let $\sigma \in S_n$ and let e, f be 2-element subsets of $V(G)$. Then $(e \rightarrow f)$ is a feasible edge replacement for G_σ if and only if $(\sigma(e) \rightarrow \sigma(f))$ is a feasible edge replacement for G . When these conditions hold,*

$$\text{Fer}_{G_\sigma}(e \rightarrow f) = \sigma^{-1} \circ \text{Fer}_G(\sigma(e) \rightarrow \sigma(f)) \circ \sigma.$$

Proof. The isomorphism $\varphi_\sigma: G_\sigma \rightarrow G$ sends e to $\sigma(e)$ and f to $\sigma(f)$. It therefore sends $G_\sigma - e + f$ to $G - \sigma(e) + \sigma(f)$, proving the equivalence.

Directly from Definition 4.21,

$$(G_\sigma)_\rho = G_{\sigma \circ \rho} \quad \text{and} \quad (G - \sigma(e) + \sigma(f))_\sigma = G_\sigma - e + f.$$

Hence ρ witnesses $(e \rightarrow f)$ for G_σ exactly when $\sigma \circ \rho \circ \sigma^{-1}$ witnesses $(\sigma(e) \rightarrow \sigma(f))$ for G . $\square$

Apply the proposition with $\sigma = \alpha$. The instruction associated with β becomes

$$\alpha^{-1}(v_1 v_2) \rightarrow \alpha^{-1}(v_1 v_3), \quad \text{that is,} \quad v_2 v_3 \rightarrow v_1 v_2.$$

Thus the puzzle has the executable solution

$$(v_1 v_2 \rightarrow v_1 v_3), \quad (v_0 v_1 \rightarrow v_0 v_2), \quad (v_2 v_3 \rightarrow v_1 v_2),$$

whose total label permutation is $\beta \circ \alpha = (0 \ 3)$.

Definition 4.28 (The group of feasible edge replacements). Define the set of *canonical generators*

$$\mathcal{E}_G := A(G) \cup \bigcup \{\text{Fer}_G(e \rightarrow f) : (e \rightarrow f) \text{ is a feasible edge replacement for } G\}.$$

The *group of feasible edge replacements* of G is

$$\text{Fer}(G) := \langle \mathcal{E}_G \rangle \leq S_n.$$

Proposition 4.27 explains how a product of canonical generators is executed as a sequence of feasible edge replacements: after each step, the next instruction is translated into the current labelling. In particular,

$$1 \leq A(G) \leq \text{Fer}(G) \leq S_n.$$

Amoebas.

Definition 4.29 (Local amoeba). A finite graph G on the label set X is a *local amoeba* if

$$\text{Fer}(G) = S_X.$$

Thus every label permutation, and hence every isomorphic copy of a local amoeba on the same vertex set, can be realized by a finite product of canonical generators. This notion first appeared in [CHM21] and has strong balanceability properties.⁴

If C_4 and P_4 denote the 4-cycle and the path on four vertices, respectively, then $\text{Fer}(C_4) = A(C_4) \simeq D_4$, whereas $\text{Fer}(P_4) = S_4$. Therefore, C_4 is not a local amoeba, while P_4 is. Every finite complete graph K_n is a local amoeba because $A(K_n) = S_n$.

Definition 4.30 (Rigid graph). A finite graph G is *rigid* if it has no nontrivial feasible edge replacement, equivalently, if

$$\text{Fer}(G) = A(G).$$

⁴The word *balanceable* is defined in Subsection 4.3.1.

Constructing local amoeba trees with arbitrarily large maximum degree was the main precursor to [EHMWV25]; the construction is presented in Subsection 4.3.3.

The groups $\text{Fer}(G)$ range over all finite groups, as the following proposition shows.

Proposition 4.31. *For every finite group Γ , there is a finite graph G such that $\text{Fer}(G) \simeq \Gamma$.*

The main ingredient is Frucht's theorem.

Lemma 4.32 (Frucht's theorem [Fru49]). *For every finite group Γ , there exists a finite cubic graph G such that $\text{Aut}(G) \simeq \Gamma$.*

Proof of Proposition 4.31. Let G be a cubic graph given by Frucht's theorem. If $e \in E(G)$ and $f \notin E(G)$, then $G - e + f$ is not cubic and hence is not isomorphic to G . Thus G is rigid, and

$$\text{Fer}(G) = A(G) \simeq \text{Aut}(G) \simeq \Gamma.$$

□

4.3.3 Local amoeba trees with large maximum degree

Intuitively, if one is trying to transform one graph into another by means of feasible edge replacements, it may come in handy to have isolated vertices that allow new positions for the graph. We previously mentioned that complete graphs are rigid graphs, meaning they have no non-trivial feasible edge replacements. Adding isolated vertices to a complete graph destroys its rigidity, since now new movements are possible.

Definition 4.33. A finite graph G is called a *global amoeba* if there is a $t \geq 1$ such that $G \cup tK_1$ (that is, the graph obtained as the disjoint union of G and t many isolated vertices) is a local amoeba.

Perhaps surprisingly, there is no general logical implication between a graph G being a local amoeba and being a global amoeba.⁵ One case where such an implication does exist is for trees (see Corollary 4.35).

The main result of [CGH⁺23] is the following classification of global amoebas. Notably, if a value of t exists such that Definition 4.33 holds, then $t = 1$ is always the least such value, as the following theorem states.

Theorem 4.34 ([CGH⁺23, Theorem 4.1]). *For any non-empty graph G with vertex set $V(G) = \{v_0, v_1, \dots, v_{n-1}\}$, the following are equivalent.*

1. G is a global amoeba.
2. $G \cup K_1$ is a local amoeba.
3. Every orbit of $\text{Fer}(G)$ contains some label $i < n$ such that v_i has degree 1 in G .

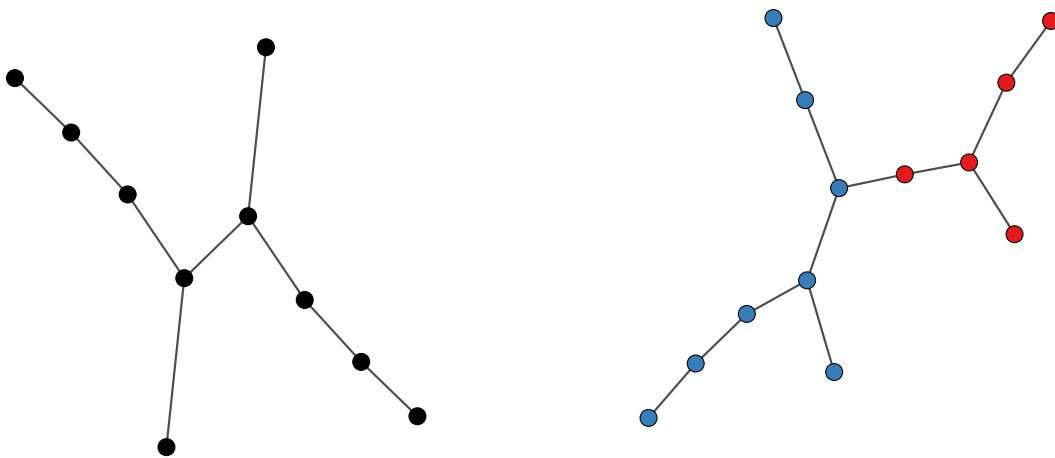
A well-known fact about a finite tree T with at least one edge is that it must have at least two vertices of degree 1. Recalling that if T is a local amoeba, then $\text{Fer}(T)$ consists of a single orbit (it is in fact, the whole symmetric group), it is clear that condition (3) of Theorem 4.34 is satisfied for any tree T that happens to be a local amoeba.

⁵With this in mind, one might try to suggest more descriptive names for these graphs that do not hint at this nonexistent causal link, but the author chose to respect the terminology from the well-established literature.

Corollary 4.35. *If T is a finite tree for which $\text{Fer}(T)$ acts transitively on (the vertex labels of) T (for example if T is a local amoeba), then T is a global amoeba.*

The acyclicity of T is indispensable: a complete graph with at least three vertices is a local amoeba but not a global amoeba.

Figure 4.8a depicts the smallest tree T that is a global amoeba but not a local amoeba. Interestingly, the Fer group acts transitively on it (that is, it consists of a single orbit). There are also global amoeba trees whose Fer group is not transitive, and the smallest such example has two orbits, depicted in Figure 4.8b with vertices coloured by the orbit of their labels under the action of the Fer group.⁶ The main conjecture of Section 4.4 is that the Fer groups of all global amoeba trees have at most two orbits.



(a) The smallest non-local global amoeba tree.

(b) The smallest global amoeba tree on which Fer does not act transitively.

Figure 4.8: The smallest global amoeba trees with these properties.

The construction we describe now (see Figure 4.9), loosely inspired by the Fibonacci sequence, illustrates the main results of [EHMWV25]. Start by defining T_1 and T_2 to each be the tree consisting of a single edge, in other words, $T_1 \simeq T_2 \simeq K_2$. Now, suppose that T_{k-1} and T_{k-2} have already been constructed. In each of these trees, select a vertex of maximum degree. The tree T_k is the result of joining T_{k-1} and T_{k-2} by the edge connecting these two vertices. The choice of the vertex of maximum degree is irrelevant, as one can easily verify: either such vertex is unique, or there are

⁶Recall from elementary group theory that the set of orbits of a group action forms a partition of the set. Thus, in this example, two vertices have the same colour if and only if some element of $\text{Fer}(T)$ sends the label of one to the label of the other.

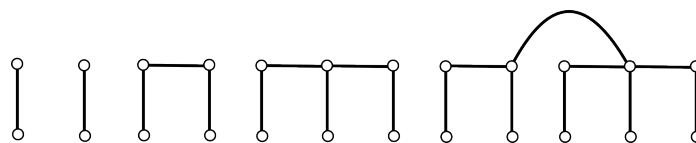


Figure 4.9: Fibonacci-type trees T_k , for $1 \leq k \leq 5$.

exactly two such vertices that are similar (that is, they can be interchanged via an automorphism), and thus the resulting sequence of graphs is unique up to isomorphisms.

The sequence of graphs $\{T_k : k \in \mathbb{N}\}$ is a particular case of a more general scheme of recursive constructions studied in [EHMWV25, Section 3].

Theorem 4.36 ([EHMWV25, Theorem 9]). *For every $k \in \mathbb{N}$, T_k is a local amoeba.*

The proof that a finite graph G of order n is a local amoeba often involves a counting argument: one shows that $\text{Fer}(G)$ has $n!$ elements and then, since $\text{Fer}(G) \leq S_n$, concludes that $\text{Fer}(G) = S_n$. There are two disadvantages of this method.

- Given an arbitrary permutation $\sigma \in S_n$ (that is, an arbitrary copy of G on the same vertex set), the proof generally offers no method of finding the sequence of feasible edge replacements. In other words, this proof technique shows the existence of a solution to the puzzle, without giving an algorithm to find it. The algorithm published in [EHMWV25, Section 5] resolves this for an infinite family of trees.
- Counting methods do not generalize to infinite graphs. This is addressed in Section 4.5.

The following notion will also be useful in Section 4.5.

Definition 4.37. Let G be a finite graph whose vertices are indexed by a label set X , so that $V(G) = \{v_j : j \in X\}$, and let $i \in X$. The subgroup of $\text{Fer}(G)$ generated by the canonical generators that fix i is denoted by

$$\text{Fer}^i(G) := \langle \mathcal{E}_G \cap \text{Stab}_{S_X}(i) \rangle.$$

We say a graph G is *stem-symmetric with respect to the label i* (equivalently, *stem-symmetric with respect to the vertex v_i*) if $\text{Fer}^i(G) = \text{Stab}_{S_X}(i)$, that is, if $\text{Fer}^i(G)$ realizes every label permutation of X that fixes i .

In other words, a finite graph G is stem-symmetric with respect to a label i if every permutation fixing i is a product of canonical generators that each fix i .

Lemma 4.38. *For every $n \in \mathbb{N}$, the path whose vertices, in order, are $v_0, v_1, \dots, v_n$ is stem-symmetric with respect to the endpoint label n .*

Proof. Denote this path by P_{n+1} . The result is immediate for $n \leq 1$, so assume that $n \geq 2$. Let $1 \leq \ell < n$ and notice that the graph $P_{n+1} - v_\ell v_{\ell+1} + v_0 v_{\ell+1}$ is a path on $n+1$ vertices and thus isomorphic to P_{n+1} . The label permutation

$$\sigma_\ell := \prod_{j=0}^{\lfloor (\ell-1)/2 \rfloor} (j \ \ell - j)$$

belongs to $\text{Fer}_{P_{n+1}}(v_\ell v_{\ell+1} \rightarrow v_0 v_{\ell+1})$ and fixes the label n .

Set $\sigma_0 := \text{id}_{n+1}$ and $c_\ell := \sigma_\ell \sigma_{\ell-1}$ for $1 \leq \ell < n$. Then $c_\ell = (0 \ 1 \ \dots \ \ell)$, so $(0 \ 1) = c_1$ and $(\ell - 1 \ \ell) = c_{\ell-1}^{-1} c_\ell$ for $2 \leq \ell < n$. Thus $\text{Fer}^n(P_{n+1})$ contains every adjacent transposition of $\{0, 1, \dots, n-1\}$ and is therefore equal to $\text{Stab}_{S_{n+1}}(n)$. $\square$

4.3.4 Balancing numbers of Fibonacci-type trees

By Theorem 4.36 together with Corollary 4.35, every Fibonacci-type tree T_k is a global amoeba and is, in particular, balanceable in the sense of Definition 4.15, as is any global amoeba (see the proof of Theorem 4.41 below). The aim of this subsection is to give bounds, often sharp, on $\text{bal}(n, T_k)$ as a function of n and k . The treatment follows [EHMWV25, Section 4]; the proofs of the results concerning T_k are reproduced below, and the reader is referred to that work for the parallel arguments concerning the amoeba families $\mathcal{A}$ and $\mathcal{B}$ studied there.

The principal result of the subsection is the following.

Theorem 4.39 ([EHMWV25, Theorem 15, partial]). *For n sufficiently large,*

$$n - 1 \leq \text{bal}(n, T_6) \leq 2n - 2,$$

and for $k \geq 7$,

$$F_{k-4}(n - F_{k-4}) \leq \text{bal}(n, T_k) \leq (F_{k-2} - 2)n - \binom{F_{k-2} - 1}{2} + \delta_{k=7},$$

where $\delta_{k=7} = 1$ if $k = 7$ and is zero otherwise.

For $1 \leq k \leq 5$, where the balancing number is constant in n , the values are recorded in Table 4.1; the case $T_3 \simeq P_4$ was already handled in Example 4.18(1), and the cases T_4 and T_5 are addressed in Lemma 4.48.

Remark 4.40. The proof of [EHMWV25, Lemma 24] handles the cases T_1, T_2, T_3 with the remark that “the first few elements of $\mathcal{T}, \mathcal{A}, \mathcal{B}$ are paths and their balancing numbers are straightforward to obtain,” and reports $\text{bal}(n, T_3) = 1$, for sufficiently large n , in the corresponding table. Since $T_3 \simeq P_4$, by Example 4.18(1) we have $\text{bal}(n, T_3) = 0$, for sufficiently large n , not 1. Indeed, applying Theorem 4.41 gives $\text{bal}(n, T_3) = \text{ex}(n, \mathcal{H}_{T_3}) = \text{ex}(n, K_2) = 0$, since T_3 has 3 edges and so $\mathcal{H}_{T_3} = \{K_2\}$. The values reported in Table 4.1 and the corresponding entries of Lemma 4.48 reflect this correction.

G	$e(G)$	$\text{bal}(n, G)$
$T_1 \simeq T_2 \simeq K_2$	1	0
$T_3 \simeq P_4$	3	0
T_4	5	1
T_5	9	6

Table 4.1: Balancing numbers of the smallest Fibonacci-type trees.

The proof of Theorem 4.39 occupies the remainder of the subsection. The strategy combines three ingredients:

1. an exact identity expressing the balancing number of any global amoeba as a Turán-type extremal number for an associated class of subgraphs (Theorem 4.41);
 2. a lower bound on this extremal number coming from the degree sequence of G (Theorem 4.43);
- and

3. an explicit family of star forests embedded in T_k on half its edges, together with the Turán number of star forests due to Lidický, Liu, and Palmer (Proposition 4.47).

The next three paragraphs implement these three ingredients in order.

The balancing number as a Turán-type extremal number. For a graph G with $m = e(G)$ edges, define

$$\mathcal{H}_G := \{H \subseteq G : e(H) = \lfloor m/2 \rfloor \text{ and } H \text{ has no isolated vertices}\}. \quad (4.1)$$

For a class $\mathcal{F}$ of graphs, recall that $\text{ex}(n, \mathcal{F})$ denotes the maximum number of edges in an n -vertex graph that contains no member of $\mathcal{F}$ as a subgraph. Since containing a given subgraph is a property invariant under isomorphism, $\text{ex}(n, \mathcal{F})$ depends only on the isomorphism types occurring in $\mathcal{F}$. Accordingly, we read (4.1) up to isomorphism: isomorphic subgraphs of G determine the same member of $\mathcal{H}_G$. For instance, $\mathcal{H}_{P_4} = \{K_2\}$, even though the three distinct edges of P_4 give three distinct one-edge subgraphs.

Theorem 4.41 ([EHMWV25, Theorem 11]). *Let G be a global amoeba. Then for n sufficiently large, $\text{bal}(n, G) = \text{ex}(n, \mathcal{H}_G)$.*

Proof. Let $m = e(G)$. By a classical averaging argument of Erdős, every graph contains a bipartite subgraph with at least half of its edges. Retaining exactly $\lfloor m/2 \rfloor$ of these edges and deleting isolated vertices gives a bipartite member of $\mathcal{H}_G$; consequently, $\text{ex}(n, \mathcal{H}_G) = o(n^2)$ by standard bounds on Turán numbers of bipartite graphs (cf. [EHMWV25, Section 4]).

Lower bound on $\text{bal}(n, G)$. Pick a 2-edge-colouring $E(K_n) = R \sqcup B$ with $|B| = \text{ex}(n, \mathcal{H}_G)$ chosen so that the graph induced by B contains no member of $\mathcal{H}_G$. For n large, $|R| = \binom{n}{2} - |B| \geq |B|$, so this colouring has at least $|B|$ edges of each colour. No balanced copy of G exists, since any balanced copy of G would contribute exactly $\lfloor m/2 \rfloor$ blue edges that, after deleting isolated vertices from the subgraph formed by those edges, would give a member of $\mathcal{H}_G$ inside B . Hence $\text{bal}(n, G) \geq \text{ex}(n, \mathcal{H}_G)$.

Upper bound on $\text{bal}(n, G)$. Let $E(K_n) = R \sqcup B$ be a 2-edge-colouring with $\min\{|R|, |B|\} > \text{ex}(n, \mathcal{H}_G)$. Both colour classes contain a member of $\mathcal{H}_G$; extending each to a copy of G in K_n yields copies G_R and G_B with at least $\lfloor m/2 \rfloor$ red edges and at least $\lfloor m/2 \rfloor$ blue edges, respectively. If either G_R or G_B is itself balanced we are done; otherwise the number of red edges ℓ_R in G_R satisfies $\ell_R > \lceil m/2 \rceil$, while the number of red edges in G_B is strictly less than $\lfloor m/2 \rfloor$.

Since G is a global amoeba, there is a chain $G_R = G_0, G_1, \dots, G_t = G_B$ of copies of G in K_n in which each G_i is obtained from G_{i-1} by a feasible edge replacement. Let $\ell_{R,i}$ denote the number of red edges in G_i . Each replacement changes $\ell_{R,i}$ by at most one, so $|\ell_{R,i} - \ell_{R,i-1}| \leq 1$. Since $\ell_{R,0} > \lceil m/2 \rceil$ and $\ell_{R,t} < \lfloor m/2 \rfloor$, the discrete intermediate value theorem produces some G_i with $\ell_{R,i} \in \{\lfloor m/2 \rfloor, \lceil m/2 \rceil\}$, which is therefore a balanced copy of G in the given colouring. Hence $\text{bal}(n, G) \leq \text{ex}(n, \mathcal{H}_G)$. $\square$

A general lower bound from the degree sequence. The lower bound $\text{bal}(n, G) \geq \text{ex}(n, \mathcal{H}_G)$ from Theorem 4.41 can be exploited to extract a quantitative lower bound on $\text{bal}(n, G)$ directly from the degree sequence of G . For a graph $G = (V, E)$ and a subset $S \subseteq V$, denote by $e_G(S, V \setminus S)$ the

number of edges of G with one endpoint in S and the other in $V \setminus S$. For $1 \leq \ell \leq |V|$, set

$$\text{MaxCut}(G, \ell) := \max\{e_G(S, V \setminus S) : S \subseteq V, |S| \leq \ell\}, \quad (4.2)$$

$$\ell_G := \max\{\ell \geq 0 : \text{MaxCut}(G, \ell) < \lfloor e(G)/2 \rfloor\}. \quad (4.3)$$

The quantity ℓ_G is well-defined: $\text{MaxCut}(G, \ell)$ is non-decreasing in ℓ , and the same averaging argument of Erdős used in the proof of Theorem 4.41 gives $\text{MaxCut}(G, \lfloor |V|/2 \rfloor) \geq \lfloor e(G)/2 \rfloor$, so $\ell_G < |V|/2$.

Lemma 4.42 ([EHMWV25, Lemma 13]). *For any graph G and any $\ell \leq \ell_G$,*

$$\text{ex}(n, \mathcal{H}_G) \geq \ell_G(n - \ell_G) \geq \ell(n - \ell).$$

Proof. The function $\ell \mapsto \ell(n - \ell)$ is non-decreasing on $[0, n/2]$, and $\ell_G < |V|/2 \leq n/2$, which gives the second inequality. For the first, we claim that the complete bipartite graph $K_{\ell_G, n-\ell_G}$ contains no member of $\mathcal{H}_G$ as a subgraph; granting this, $\text{ex}(n, \mathcal{H}_G) \geq e(K_{\ell_G, n-\ell_G}) = \ell_G(n - \ell_G)$.

Suppose, for contradiction, that some member of $\mathcal{H}_G$ embeds into $K_{\ell_G, n-\ell_G}$. Choose a representative $H \subseteq G$ of this isomorphism type and transfer to it the bipartition induced by the embedding. Write its parts as $S \sqcup R$, choosing their order so that $|S| \leq \ell_G$. Denoting $m = e(G)$,

$$e_G(S, V \setminus S) \geq e_H(S, R) = \lfloor m/2 \rfloor,$$

while by the definition of ℓ_G ,

$$e_G(S, V \setminus S) \leq \text{MaxCut}(G, \ell_G) < \lfloor m/2 \rfloor,$$

a contradiction. □

Theorem 4.43 ([EHMWV25, Theorem 12]). *Let G be a global amoeba on k vertices and m edges, with degree sequence $d_1 \geq d_2 \geq \dots \geq d_k$. For any integer $\ell \geq 1$ satisfying $\sum_{i=1}^{\ell} d_i < \lfloor m/2 \rfloor$, $\text{bal}(n, G) \geq \ell(n - \ell)$ for n sufficiently large.*

Proof. For any $S \subseteq V(G)$,

$$e_G(S, V \setminus S) \leq \sum_{v \in S} \deg_G(v) \leq \sum_{i=1}^{|S|} d_i,$$

where the second inequality uses that $d_1, \dots, d_k$ is in non-increasing order. Hence, if $\sum_{i=1}^{\ell} d_i < \lfloor m/2 \rfloor$, then $\text{MaxCut}(G, \ell) < \lfloor m/2 \rfloor$, so $\ell \leq \ell_G$. Combining Theorem 4.41 and Lemma 4.42,

$$\text{bal}(n, G) = \text{ex}(n, \mathcal{H}_G) \geq \ell(n - \ell).$$

□

Profile of T_k and the lower bound. For a graph G and $i \geq 0$, let $V_i(G)$ denote the set of vertices of G of degree i , and call the sequence $(|V_i(G)|)_{i \geq 0}$ the *degree profile* of G . By induction on

the recursive construction of T_k one obtains an explicit description of its degree profile in terms of Fibonacci numbers.

Lemma 4.44 ([EHMWV25, Lemma 17]). *The number of vertices of T_k is $2F_k$. For $k \geq 4$,*

$$|V_i(T_k)| = \begin{cases} F_{k+1-i} & 1 \leq i \leq 2, \\ F_{k-1-i} & 3 \leq i \leq k-2, \\ 1 & i = k-1, \\ 0 & i \geq k. \end{cases}$$

Proposition 4.45 ([EHMWV25, Proposition 19]). *For every $k \geq 6$ and n sufficiently large, $\text{bal}(n, T_k) \geq F_{k-4}(n - F_{k-4})$.*

Proof. The tree T_k has $m = 2F_k - 1$ edges, hence $\lfloor m/2 \rfloor = F_k - 1$. Letting $d_1 \geq d_2 \geq \dots \geq d_{2F_k}$ be its degree sequence, by Theorem 4.43 it suffices to show that

$$\sum_{i=1}^{F_{k-4}} d_i \leq F_k - 2 < \lfloor m/2 \rfloor. \quad (4.4)$$

The case $k = 6$ is immediate: $F_{k-4} = F_2 = 1$, so the partial sum is equal to $d_1 = \Delta(T_6) = k - 1 = 5$, while $F_6 - 2 = 6$.

For $k \geq 7$, denote by $V_{>4}(T_k)$ the set of vertices of T_k of degree exceeding 4. By Lemma 4.44,

$$|V_{>4}(T_k)| = \sum_{i=5}^{k-1} |V_i(T_k)| = 1 + \sum_{i=5}^{k-2} F_{k-i-1} = 1 + \sum_{j=1}^{k-6} F_j.$$

The well-known identity $\sum_{j=1}^r F_j = F_{r+2} - 1$ gives $|V_{>4}(T_k)| = F_{k-4}$. Hence the F_{k-4} largest entries in the degree sequence are exactly the degrees of vertices of degree at least 5, and so

$$\sum_{i=1}^{F_{k-4}} d_i = \sum_{v \in V_{>4}(T_k)} \deg_{T_k}(v) = \sum_{i=5}^{k-1} i |V_i(T_k)|.$$

Since T_k is a tree on $2F_k$ vertices, it has $2F_k - 1$ edges. The handshake lemma therefore gives $\sum_{i=1}^{k-1} i |V_i(T_k)| = 2(2F_k - 1) = 4F_k - 2$, so

$$\sum_{i=1}^{F_{k-4}} d_i = 4F_k - 2 - \sum_{i=1}^4 i |V_i(T_k)|. \quad (4.5)$$

Lemma 4.44 and the Fibonacci recursion give

$$\sum_{i=1}^4 i |V_i(T_k)| = F_k + 2F_{k-1} + 3F_{k-4} + 4F_{k-5}.$$

The Fibonacci recursion together with $F_{k-4} \leq 2F_{k-5}$ (valid for $k \geq 7$, since $F_{k-4} = F_{k-5} + F_{k-6} \leq 2F_{k-5}$) implies

$$2F_{k-2} = 4F_{k-4} + 2F_{k-5} \leq 3F_{k-4} + 4F_{k-5},$$

and therefore

$$\sum_{i=1}^4 i |V_i(T_k)| \geq F_k + 2F_{k-1} + 2F_{k-2} = F_k + 2(F_{k-1} + F_{k-2}) = 3F_k.$$

Substituting back into (4.5) yields $\sum_{i=1}^{F_k-4} d_i \leq 4F_k - 2 - 3F_k = F_k - 2$, establishing (4.4). $\square$

The upper bound via star forests. To produce upper bounds on $\text{bal}(n, T_k) = \text{ex}(n, \mathcal{H}_{T_k})$, it suffices, by definition of $\mathcal{H}_{T_k}$, to exhibit a single member $H \in \mathcal{H}_{T_k}$ with a tractable Turán number, since then $\text{ex}(n, \mathcal{H}_{T_k}) \leq \text{ex}(n, H)$. Such a member is provided by the following recursive star forest construction, which mirrors the recursion of the trees T_k themselves.

For $k \geq 5$, define recursively two star forests $S_k \subseteq S_k^+ \subseteq T_k$:

- S_5 is a single star of degree 4, and S_5^+ is obtained from S_5 by adding an independent edge;
- S_6 is the disjoint union of one star of degree 4 and one star of degree 3, and S_6^+ is obtained from S_6 by adding an independent edge;
- for $k \geq 7$, the natural decomposition of T_k into a copy of T_{k-1} and a copy of T_{k-2} allows us to define, on the T_{k-1} summand, a copy of S_{k-1} for S_k (and a copy of S_{k-1}^+ for S_k^+); on the T_{k-2} summand, a copy of S_{k-2}^+ in both cases. That is,

$$S_k := S_{k-1} \sqcup S_{k-2}^+ \quad \text{and} \quad S_k^+ := S_{k-1}^+ \sqcup S_{k-2}^+.$$

A direct induction on $k \geq 5$ establishes the following properties:

1. $S_k \subseteq S_k^+ \subseteq T_k$;
2. $e(S_k^+) = e(S_k) + 1 = F_k$;
3. S_k has $F_{k-2} - 1$ connected components, of which F_{k-4} are stars of (maximum) degree 4, F_{k-5} are stars of degree 3, and $F_{k-4} - 1$ are independent edges (i.e., stars of degree 1).

In particular, $e(S_k) = F_k - 1 = \lfloor (2F_k - 1)/2 \rfloor = \lfloor e(T_k)/2 \rfloor$, so $S_k \in \mathcal{H}_{T_k}$ for every $k \geq 5$.

The Turán number of any star forest is given by the following theorem of Lidický, Liu, and Palmer.

Theorem 4.46 (Lidický, Liu, Palmer [LLP13]; cf. [EHMWV25, Theorem 21]). *Let $H = \bigsqcup_{j=1}^p S^j$ be a star forest, where S^j has maximum degree d_j , and $d_1 \geq d_2 \geq \dots \geq d_p$. For n sufficiently large,*

$$\text{ex}(n, H) = \max_{1 \leq j \leq p} \left\{ (j-1)(n-j+1) + \binom{j-1}{2} + \left\lfloor \frac{d_j-1}{2} \right\rfloor (n-j+1) \right\}.$$

Proposition 4.47 ([EHMWV25, Proposition 22]). *For n sufficiently large, $\text{bal}(n, T_6) \leq 2n - 2$, and for every $k \geq 7$,*

$$\text{bal}(n, T_k) \leq (F_{k-2} - 2)n - \binom{F_{k-2} - 1}{2} + \delta_{k=7}.$$

Proof. Since $S_k \in \mathcal{H}_{T_k}$ and T_k is a global amoeba, by Theorem 4.41,

$$\text{bal}(n, T_k) = \text{ex}(n, \mathcal{H}_{T_k}) \leq \text{ex}(n, S_k).$$

Order the connected components of S_k in non-increasing order of maximum degree, and let $s_{k,j}$ denote the maximum degree of the j -th component. The component count from item (3) of the construction above gives

$$s_{k,j} = \begin{cases} 4 & 1 \leq j \leq F_{k-4}, \\ 3 & F_{k-4} < j \leq F_{k-3}, \\ 1 & F_{k-3} < j \leq F_{k-2} - 1, \end{cases}$$

using the identity $F_{k-4} + F_{k-5} = F_{k-3}$ and the total component count $F_{k-3} + (F_{k-4} - 1) = F_{k-2} - 1$.

For $1 \leq j \leq F_{k-2} - 1$, set

$$h_k(j, n) := (j-1)(n-j+1) + \binom{j-1}{2} + \left\lfloor \frac{s_{k,j}-1}{2} \right\rfloor (n-j+1).$$

By Theorem 4.46, $\text{ex}(n, S_k) = \max_j h_k(j, n)$ for n sufficiently large. Within each of the three intervals on which $s_{k,j}$ is constant, $h_k(\cdot, n)$ is non-decreasing for $n \geq 2(F_{k-2} - 1)$ (recall that we assume that n is sufficiently large), so the maximum is attained at one of the right endpoints. Substituting $\lfloor (4-1)/2 \rfloor = 1$, $\lfloor (3-1)/2 \rfloor = 1$, and $\lfloor (1-1)/2 \rfloor = 0$, and simplifying, one obtains

$$h_k(F_{k-4}, n) = F_{k-4}(n - F_{k-4} + 1) + \binom{F_{k-4}-1}{2}, \quad (4.6)$$

$$h_k(F_{k-3}, n) = F_{k-3}(n - F_{k-3} + 1) + \binom{F_{k-3}-1}{2}, \quad (4.7)$$

$$h_k(F_{k-2} - 1, n) = (F_{k-2} - 2)(n - F_{k-2} + 2) + \binom{F_{k-2}-2}{2}. \quad (4.8)$$

A short calculation rearranges (4.8) as

$$h_k(F_{k-2} - 1, n) = (F_{k-2} - 2)n - \binom{F_{k-2}-1}{2}. \quad (4.9)$$

For $k \geq 8$, the leading-in- n coefficients in (4.6), (4.7), and (4.9) are F_{k-4} , F_{k-3} , and $F_{k-2} - 2$, respectively, and one has $F_{k-4} < F_{k-3} < F_{k-2} - 2$. Hence, for n sufficiently large, $\text{ex}(n, S_k) = h_k(F_{k-2} - 1, n) = (F_{k-2} - 2)n - \binom{F_{k-2}-1}{2}$.

The cases $k \in \{6, 7\}$ are computed directly:

- For $k = 6$, S_6 has $F_4 - 1 = 2$ components, and $h_6(1, n) = n$, $h_6(2, n) = 2(n-1) = 2n-2$, so $\text{ex}(n, S_6) = 2n-2$.
- For $k = 7$, $F_3 = 2$, $F_4 = 3$, and $F_5 - 1 = 4$, giving $h_7(2, n) = 2n-2$, $h_7(3, n) = 3n-5$, and $h_7(4, n) = 3n-6$. The maximum is $h_7(F_{k-3}, n) = 3n-5 = h_7(F_{k-2} - 1, n) + 1$, which accounts for the correction $\delta_{k=7} = 1$.

Combining all cases, for n sufficiently large,

$$\text{bal}(n, T_k) \leq \text{ex}(n, S_k) = (F_{k-2} - 2)n - \binom{F_{k-2}-1}{2} + \delta_{k=7} \quad \text{for } k \geq 7,$$

and $\text{bal}(n, T_6) \leq 2n-2$. □

Constant cases: T_3 , T_4 , T_5 .

Lemma 4.48 (cf. [EHMWV25, Lemma 24]). *For n sufficiently large, $\text{bal}(n, T_4) = 1$ and $\text{bal}(n, T_5) = 6$.*

Proof. By Theorem 4.41, it suffices to compute $\text{ex}(n, \mathcal{H}_{T_4})$ and $\text{ex}(n, \mathcal{H}_{T_5})$.

Case T_4 . Since T_4 has 5 edges, $\mathcal{H}_{T_4}$ consists of the 2-edge subgraphs of T_4 with no isolated vertices. Both the matching $2K_2$ and the path P_3 embed into T_4 , and these are the only two isomorphism types available, so $\mathcal{H}_{T_4} = \{2K_2, P_3\}$. A graph that contains neither P_3 nor $2K_2$ as a subgraph must be a matching of size at most 1, hence has at most one edge, so $\text{ex}(n, \mathcal{H}_{T_4}) = 1$.

Case T_5 . Since T_5 has 9 edges, $\mathcal{H}_{T_5}$ is the family of 4-edge subgraphs of T_5 with no isolated vertices. The lower bound $\text{ex}(n, \mathcal{H}_{T_5}) \geq 6$ is witnessed by K_4 , which has 6 edges and contains no forest on 4 edges as a subgraph (since K_4 has only 4 vertices, every subgraph of K_4 on 4 edges contains a cycle). For the matching upper bound, a case analysis on the components of an arbitrary graph with 7 edges shows that it must contain a member of $\mathcal{H}_{T_5}$; the details are routine and given in [EHMWV25, Lemma 24]. Hence $\text{ex}(n, \mathcal{H}_{T_5}) = 6$. $\square$

The proof of Theorem 4.39 is now complete: the lower bound is Proposition 4.45, the upper bound is Proposition 4.47, and the constant cases recorded in Table 4.1 are addressed in Example 4.18(1) and Lemma 4.48. Analogous bounds for the recursive amoeba families $\mathcal{A} = \{A_k\}$ and $\mathcal{B} = \{B_k\}$ studied in [EHMWV25] can be obtained by the same methods, exhibiting suitable star forests adapted to the recursions of A_k and B_k ; the details, together with the corresponding analogues of Theorem 4.39, are given in [EHMWV25, Theorem 15 and Propositions 20–22].

4.4 Amoeba trees

Section 4.3 concentrated on recursive constructions of amoebas with prescribed quantitative features, such as Fibonacci-type trees with high maximum degree. This section turns to the structural side of the same subject and studies global amoeba trees through the interaction between automorphisms and feasible edge replacements. The material is largely drawn from joint work in [GGHMW], which is in preparation as a paper at the time of writing; within this collaboration, the author proposed the section's central conjecture, discussed in Subsection 4.3.3, that the **Fer** group of every global amoeba tree has at most two orbits.

While Section 4.3 produced concrete amoebas by recursive construction, the present section seeks general structural information about the relationship between $\text{Fer}(T)$ and $\text{Aut}(T)$. Section 4.4.1 develops the necessary background on automorphisms of trees, forests, and unicyclic graphs, the latter arising naturally as $T + f$ for $f \in E(T^c)^7$, i.e., as the unicyclic extensions whose automorphism groups generate elements of $\text{Fer}(T)$. Section 4.4.2 then turns to the central question of when $A(T)$ is a normal subgroup of $\text{Fer}(T)$, decomposing the canonical generators $\mathcal{E}_T$ in terms of automorphisms of T , of $T - e$, and of $T + f$, and showing that for trees the contribution from automorphisms of unicyclic extensions $T + f$ subsumes the others (Proposition 4.62).

Throughout this section, all graphs are finite. The infinitary counterparts of several of the results presented here are discussed in Section 4.5.

⁷Recall that T^c denotes the *complement* of T , that is the graph on the same vertex set obtained from flipping all of the adjacencies of T .

4.4.1 Automorphisms of trees, forests and unicyclic graphs

For any graph G , we say $\varphi \in \text{Aut}(G)$ *fixes* the vertex $x \in V(G)$ if $\varphi(x) = x$, and that $\text{Aut}(G)$ *fixes* x if every automorphism does. A set $S \subseteq V(G)$ is *fixed pointwise* (by either a single automorphism or by the entire group, respectively) if every element of it is fixed, and is *fixed set-wise* by φ if $\varphi[S] = S$. Observe that S is fixed set-wise if and only if $V(G) \setminus S$ is fixed set-wise. An edge xy is *fixed* if the set $\{x, y\}$ is fixed set-wise. Now let xy be an edge of G . If $\varphi \in \text{Aut}(G)$ satisfies $\varphi(x) = y$ and $\varphi(y) = x$, then φ is called an *edge-inversion*. In other words, an edge-inversion fixes the edge xy but does not fix either of its endpoints.

The next lemma is a cornerstone result that we use throughout the section: automorphisms permute connected components.

Lemma 4.49. *Let φ be an automorphism of G and C a connected component of G . Then $\varphi|_C$ is an isomorphism between C and a connected component (possibly C itself) of G .*

Proof. If $\varphi[V(C)] = V(C)$, then the result is clear. Thus suppose that there is $v \in V(C)$ such that $\varphi(v) \notin V(C)$. Let D be the connected component containing $\varphi(v)$. We claim that $\varphi[V(C)] = V(D)$. Indeed, given any $c \in V(C)$, a cv -path is preserved by φ , so there is a $\varphi(c)\varphi(v)$ -path in G . Hence, $\varphi(c) \in V(D)$. The reverse containment holds because φ^{-1} is also an isomorphism. $\square$

Trees and forests. We begin by showing that any automorphism of a tree either fixes at least one vertex or inverts exactly one edge.

Lemma 4.50. *Every tree T has a vertex or an edge that is fixed by every $\varphi \in \text{Aut}(T)$.*

Proof. Let T be a tree and $L(T) \subseteq V(T)$ the set of *leaves*, i.e., the vertices of degree 1. Denote by $C(T)$ the *centre* of T : the set of vertices c for which $\varepsilon(c) := \max\{d(c, x) : x \in L(T)\}$ is minimal among all vertices of T . Recall that every isomorphism is an isometry on the usual graph distance. Consequently, $C(T)$ is fixed set-wise by every $\varphi \in \text{Aut}(T)$.

It remains to argue that $C(T)$ consists of either a single vertex or two adjacent vertices, since then the conclusion follows from this and the preceding paragraph. Routine arguments show that whenever T has more than two vertices, $T - L(T)$ is a tree with the same centre as T . For any tree T , $L(T)$ has at least two elements; thus, one can inductively remove all leaves from T until there remains a tree of either one or two vertices, at which point the centre is the whole tree. But as argued, the centre is invariant under this operation. In other words, $C(T)$ consists of at most two vertices. $\square$

For convenience, refer to an automorphism of a tree that fixes a vertex as a *branch permutation* and use the symbol ψ for it; for an automorphism that exchanges two adjacent vertices, use the letter φ and call it an *edge-inversion*, as defined above. By Lemma 4.50, every tree with non-trivial automorphism group has either a branch permutation or an edge-inversion. The next corollary shows that the two conditions are mutually exclusive.

Corollary 4.51. *Every automorphism of a tree is either an edge-inversion or a branch permutation, and no automorphism is both.*

Proof. By Lemma 4.50, it remains to show the exclusiveness. Since trees are acyclic, if φ is an edge-inversion of a tree T that inverts the edge xy , then every vertex in the connected component of

$T - xy$ containing x gets mapped to the connected component of $T - xy$ containing y . In particular, no such φ has a fixed vertex. $\square$

The procedure of iteratively removing leaves used in the proof of Lemma 4.50 also provides an algorithm for finding $C(T)$, and thus the vertex or edge fixed by the entire automorphism group, without computing any automorphisms of T .

Definition 4.52 (Relative automorphisms). Suppose that H is an induced subgraph of G . The group of automorphisms of H relative to G , denoted $\text{Aut}_G(H)$, is the set of automorphisms of G that fix $V(G) \setminus V(H)$ pointwise. In symbols,

$$\text{Aut}_G(H) := \{\varphi \in \text{Aut}(G) : \varphi|_{G-H} = \text{id}_{G-H}\},$$

where id_{G-H} denotes the identity map on $V(G) \setminus V(H)$.

If H is an induced subgraph of G , then $\text{Aut}_G(H)$ is a subgroup of $\text{Aut}(G)$ under composition.

Lemma 4.53. *The map $\text{Aut}_G(H) \rightarrow \text{Aut}(H)$ given by $\varphi \mapsto \varphi|_H$ is a group monomorphism. Furthermore, if $V(H)$ is fixed set-wise by $\text{Aut}(G)$, then $\text{Aut}_G(H)$ is a normal subgroup of $\text{Aut}(G)$.*

The graph consisting of two disjoint triangles, taking H to be one of them, gives an example where neither condition holds.

Proof. Given $\varphi, \psi \in \text{Aut}_G(H)$, by definition they agree on vertices not in $V(H)$. If they have the same restriction to $V(H)$, they agree on every vertex of G and are therefore equal. This proves injectivity.

Now assume that $V(H)$ is fixed set-wise by $\text{Aut}(G)$, let $\varphi \in \text{Aut}(G)$, and let $\psi \in \text{Aut}_G(H)$. To prove normality, we show that $\varphi \circ \psi \circ \varphi^{-1}$ fixes every vertex of $V(G) \setminus V(H)$. Given $x \in V(G) \setminus V(H)$, the assumption implies that $V(G) \setminus V(H)$ is also fixed set-wise, so $\varphi^{-1}(x) \in V(G) \setminus V(H)$; the choice of ψ then yields $\psi(\varphi^{-1}(x)) = \varphi^{-1}(x)$. Hence $\varphi \circ \psi \circ \varphi^{-1}(x) = \varphi(\varphi^{-1}(x)) = x$. $\square$

The well-known result that the alternating group A_n has no proper non-trivial normal subgroups for $n \geq 5$ [DF04, Section 3.5], combined with a check of small cases by hand, yields the following.

Corollary 4.54 (Folklore). *If T is a tree of order n , then $\text{Aut}(T)$ is not isomorphic to A_n .*

Now turn to the automorphisms of forests.

Remark 4.55. Let F be a forest with connected components $T_1, \dots, T_k$, and let $\sigma \in \text{Aut}(F)$. By Lemma 4.49, for each $1 \leq i \leq k$ there exists $1 \leq j \leq k$ such that $\sigma|_{T_i}$ is an isomorphism from T_i to T_j . Therefore, in addition to the automorphisms specific to each tree T_i , a forest also has automorphisms that exchange isomorphic components.

Let T be a tree and assume that $\psi \in \text{Aut}(T)$ is a branch permutation that fixes a vertex x . Let $T_1, \dots, T_k$ be the connected components of the forest $T - x$. By Remark 4.55, $\psi|_{T-x}$ permutes these components by isomorphism, so the elements of $\text{Stab}_{\text{Aut}(T)}(x)$ induce a partition of the connected components of $T - x$ into classes of pairwise isomorphic trees, as depicted in Figure 4.10.

Now consider an edge-inversion φ of T , with $\varphi(x) = y$ and $\varphi(y) = x$ for some edge $xy \in E(T)$, and let T_x and T_y denote the two connected components of $T - xy$ that contain x and y , respectively. Applying Lemma 4.49 to $\varphi|_{T-xy}$ shows that T_x and T_y are isomorphic, as illustrated in Figure 4.11.

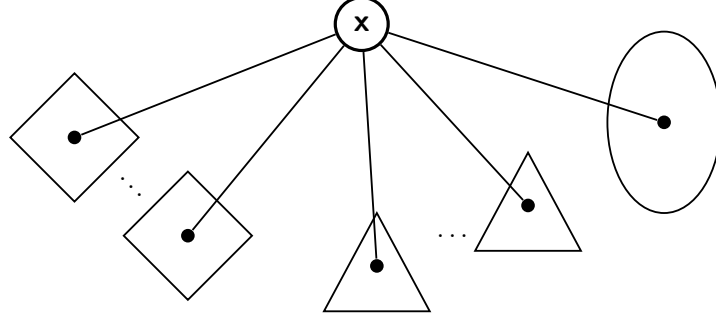


Figure 4.10: Components of $T - x$ partitioned into isomorphic classes by $\text{Stab}_{\text{Aut}(T)}(x)$.

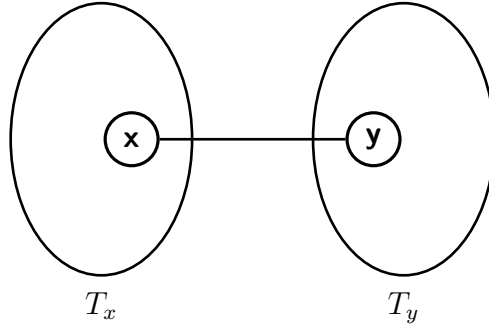


Figure 4.11: Components T_x and T_y of $T - xy$ are swapped by an edge-inversion.

Unicyclic graphs. Automorphisms of unicyclic extensions $T + f$ provide an important family of canonical generators of $\text{Fer}(T)$. We therefore record some structural facts about automorphisms of connected graphs with a unique cycle.

For this paragraph, G is a connected graph with a unique cycle C . For every $u \in V(G)$, denote by $G(u)$ the subgraph induced by the vertices x for which there exists a ux -path that is edge-disjoint from C .

Proposition 4.56. *Let $\varphi \in \text{Aut}(G)$ and $u \in V(C)$.*

- (a) $\text{Aut}(G)$ fixes $V(C)$ set-wise.
- (b) $\text{Aut}_G(C)$ is isomorphic to a normal subgroup of the dihedral group on $|V(C)|$ vertices.
- (c) $\varphi[V(G(u))] = V(G(\varphi(u)))$.
- (d) $\varphi|_{G(u)}: G(u) \rightarrow G(\varphi(u))$ is a graph isomorphism.
- (e) $G(u)$ is a tree on which $\text{Aut}_G(G(u))$ has no edge-inversions.

Proof. Part (a) follows from the fact that φ is an isomorphism, so the image of C must be a cycle of G ; but C is the only cycle in G .

For (b), define D as the image of the monomorphism from Lemma 4.53 applied to $H = C$. The conclusion follows, with normality justified by (a).

For (c), suppose that $x \in V(G(\varphi(u)))$. The automorphism φ^{-1} preserves any $\varphi(u)x$ -path and, by (a), preserves the property of being edge-disjoint from C , so $\varphi^{-1}(x) \in V(G(u))$.

For (d), the restriction of an isomorphism is always an injective graph homomorphism, and by (c), the restriction is surjective. Now suppose that x and y are adjacent in $G - E(C)$. By (c), $\varphi^{-1}(x)$ and $\varphi^{-1}(y)$ both lie in $V(G(u))$, and since φ is an isomorphism, they are adjacent. This proves that the restriction is an isomorphism.

Finally, by the choice of C and the observation that $G(u)$ is a connected induced subgraph of $G - E(C)$, $G(u)$ is a tree. By definition of $\text{Aut}_G(G(u))$, ψ fixes $V(G) \setminus V(G(u))$ pointwise; combined with (a), ψ fixes $V(C)$ set-wise; since ψ fixes $V(G) \setminus \{u\}$ pointwise, ψ fixes u as well. Being an isometry, ψ must preserve the distance from any $x \in V(G(u))$ to u . But an edge-inversion fails to achieve this. $\square$

For any tree T and any $f \in E(T^c)$, the graph $T + f$ is unicyclic, so every element of $A(T + f)$ is subject to Proposition 4.56. In Subsection 4.4.2, we show that these label permutations are canonical generators of $\text{Fer}(T)$.

4.4.2 Labelled copies, automorphisms and feasible edge replacements

Recall from Subsection 4.3.2 that for any finite graph G on a label set X with $|X| = n$, the following subgroup structure holds:

$$1 \leq A(G) (\simeq \text{Aut}(G)) \leq \text{Fer}(G) \leq S_n.$$

A natural question is: for which graphs G is $A(G)$ a normal subgroup of $\text{Fer}(G)$? We divide the analysis into the connected and the disconnected cases. In each case below, examples exist (so the cases are non-vacuous), though constructing them is more delicate than the reader might expect.

Except for trivial cases, $A(T)$ is a normal subgroup of $\text{Fer}(T)$ only when $\text{Aut}(T)$ is trivial.

Proposition 4.57. *Let T be a local amoeba tree with at least three vertices. Then $A(T)$ is a normal subgroup of $\text{Fer}(T)$ if and only if T is asymmetric.⁸*

Proof. Suppose that T is a local amoeba tree on $n \geq 3$ vertices and that $A(T)$ is normal in $\text{Fer}(T)$. Since T is a local amoeba, $\text{Fer}(T) = S_n$. For $n = 3$, the only tree is P_3 , whose automorphism label group has order 2 and is not normal in S_3 , contrary to the hypothesis. For $n = 4$, the only trees are P_4 and $K_{1,3}$, whose automorphism label groups have orders 2 and 6, respectively; neither is normal in S_4 , again contrary to the hypothesis. We may therefore assume that $n \geq 5$. The only normal subgroups of S_n are then the trivial group 1, the alternating group A_n , and S_n itself; Corollary 4.54 rules out the second case, and the third case occurs only for complete and empty graphs. Since T is a tree on at least three vertices, neither possibility applies, and thus $A(T)$ must be the trivial group.

Conversely, if T is asymmetric, then $A(T) \simeq \text{Aut}(T)$ is the trivial subgroup, and thus normality holds immediately, regardless of the local amoeba assumption. $\square$

Proposition 4.58. *If $A(G) = \text{Fer}(G)$, then exactly one of the following holds:*

- G is complete or empty; in either case, G is a local non-global amoeba.
- G has even order and is a perfect matching, that is, $G \cong \frac{n}{2}K_2$; in this case, G is a global non-local amoeba.

⁸A graph is called *asymmetric* if its automorphism group is the trivial group.

We do not know of any local amoebas G for which $A(G)$ is the alternating group of order $|V(G)|$.

Conjecture 4.1. If G is a disconnected local amoeba for which $A(G)$ is normal in $\text{Fer}(G)$, then either G is asymmetric or an empty graph.

For such a G , $\text{Fer}(G) = S_X$ and so has only three normal subgroups. If $A(G)$ is the trivial group, then G is asymmetric; and if $A(G) = \text{Fer}(G)$, then Proposition 4.58 implies that G is empty. The remaining case, $\text{Aut}(G) \simeq A_n$, is open.

Labelled copies obtained from adding or removing a single edge. Recall that $\mathcal{E}_G$ consists of $A(G)$ together with all label permutations that witness a feasible edge replacement $(e \rightarrow f)$ of G , where $e \in E(G)$ and $f \in E(G^c)$. By definition, $\text{Fer}(G) = \langle \mathcal{E}_G \rangle$.

We are interested in three different ways to perform a feasible edge replacement $e \rightarrow f$ on a graph G : applying an automorphism of G itself and then performing $e \rightarrow f$; first removing e , then applying an automorphism of $G - e$ and adding the corresponding f ; and first adding f , then applying an automorphism of $G + f$ before deleting the corresponding e . This motivates the following two sets of label permutations:

$$\begin{aligned} A^+(G) &:= \bigcup_{f \in E(G^c)} A(G + f), \\ A^-(G) &:= \bigcup_{e \in E(G)} A(G - e). \end{aligned}$$

Both sets are subsets of S_X , hence are well-defined as sets of label permutations of $V(G)$, since the graphs G , $G + f$, and $G - e$ share the same vertex set.

The following proposition shows the relationship between these sets and $\mathcal{E}_G$.

Proposition 4.59. For any graph G ,

$$A(G) \cup A^+(G) \cup A^-(G) \subseteq \mathcal{E}_G.$$

Proof. By definition, $A(G) \subseteq \mathcal{E}_G$, so it suffices to handle the other two sets.

Let $\sigma \in A(G + f)$ for some $f \in E(G^c)$, and let $\varphi_\sigma \in \text{Aut}(G + f)$ denote the induced automorphism. Then $e := \varphi_\sigma^{-1}[f]$ is an edge of $G + f$, so either $e \in E(G)$ or $e = f$. In the former case, Definition 4.21 gives

$$E(G_\sigma) = \varphi_\sigma^{-1}[E(G)] = E(G + f) \setminus \{e\} = E(G - e + f).$$

Hence $(e \rightarrow f)$ is a feasible edge replacement of G with $\sigma \in \text{Fer}_G(e \rightarrow f)$, so $\sigma \in \mathcal{E}_G$. If instead $e = f$, then $G_\sigma = G$, so $\sigma \in A(G) \subseteq \mathcal{E}_G$. This proves $A^+(G) \subseteq \mathcal{E}_G$.

Now let $e \in E(G)$ and $\sigma \in A(G - e)$, with induced $\varphi_\sigma \in \text{Aut}(G - e)$, and set $f := \varphi_\sigma^{-1}[e]$. Then f is not an edge of $G - e$: otherwise, applying φ_σ would give $e \in E(G - e)$, a contradiction. If $f = e$, then $G_\sigma = G$, so $\sigma \in A(G) \subseteq \mathcal{E}_G$. Otherwise, $f \notin E(G)$, and

$$E(G_\sigma) = \varphi_\sigma^{-1}[E(G)] = E(G - e) \cup \{f\} = E(G - e + f),$$

so $\sigma \in \text{Fer}_G(e \rightarrow f)$. Hence $\sigma \in \mathcal{E}_G$.

Thus $A^-(G) \subseteq \mathcal{E}_G$, and all three inclusions hold. $\square$

The following example shows that the inclusion in Proposition 4.59 is, in general, strict and that the corresponding generated groups also do not coincide.

Example 4.60. Let G be the graph depicted in Figure 4.12. A direct analysis of the graph shows that $A(G)$ is the trivial group and that $A^+(G)$ and $A^-(G)$ are both equal to

$$\{(0\ 1), (5\ 7), (0\ 6)(1\ 5)(2\ 4)(8\ 10)(9\ 11)\}.$$

Hence $|\langle A(G) \cup A^+(G) \cup A^-(G) \rangle| = 240$. On the other hand, $(v_1v_2 \rightarrow v_7v_8)$ is a feasible edge replacement of G , and the permutation $\sigma := (0\ 10\ 8\ 6\ 4\ 2)(1\ 11\ 9\ 7\ 5\ 3)$ satisfies $\sigma \in \text{Fer}_G(v_1v_2 \rightarrow v_7v_8)$. But $\sigma \notin \langle A(G) \cup A^+(G) \cup A^-(G) \rangle$. Moreover, $\text{Fer}(G) = \langle \mathcal{E}_G \rangle \simeq S_{12}$.

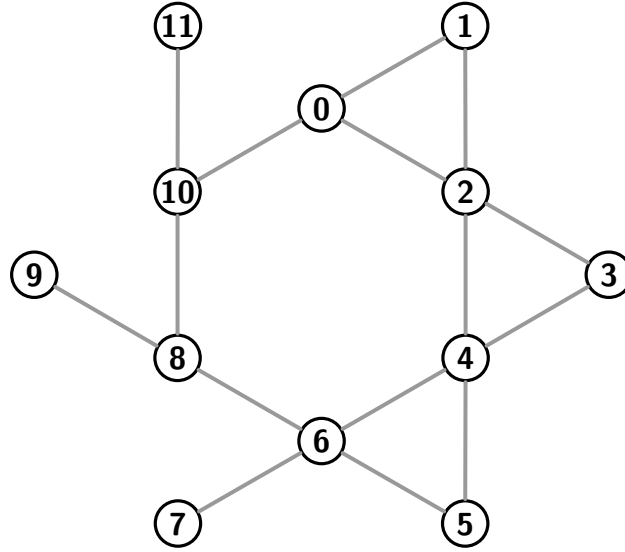


Figure 4.12: A graph G for which $A(G) \cup A^+(G) \cup A^-(G) \subsetneq \mathcal{E}_G$.

The next result shows that, when G is a tree, the considerations simplify considerably: the canonical generators arising from automorphisms of T itself, or from automorphisms of $T - e$ for $e \in E(T)$, are all expressible as products of canonical generators arising from automorphisms of unicyclic extensions $T + f$ for $f \in E(T^c)$.

We first need a lemma that refines the notion of branch permutation.

Lemma 4.61. *Suppose that T is a finite tree, $u \in V(T)$, and $\varphi \in \text{Aut}(T)$ is a branch permutation such that $\varphi(u) \neq u$. Then there is a vertex $c \in V(T)$ such that $\varphi(c) = c$ and $u, \varphi(u)$ lie in different connected components of $T - c$.*

Proof. A classical exercise in graph theory states that for any three vertices in a finite tree, there is a unique vertex m lying on each of the three pairwise paths.

Let c be the vertex fixed by φ closest to u , and apply the preceding fact to the triple $\{c, u, \varphi(u)\}$ to obtain m . Since φ is an isometry on T , the graph distance satisfies

$$d(u, m) = d(\varphi(u), \varphi(m)) \quad \text{and} \quad d(m, c) = d(\varphi(m), c).$$

Also, since T is acyclic,

$$d(\varphi(u), c) = d(\varphi(u), m) + d(m, c)$$

and

$$d(\varphi(u), c) = d(u, c) = d(u, m) + d(m, c),$$

from where

$$d(\varphi(u), m) = d(u, m) = d(\varphi(u), \varphi(m)).$$

But both $\varphi(m)$ and m lie on the path from $\varphi(u)$ to c ; hence $\varphi(m) = m$. By the choice of c , $m = c$, which means that c lies on the path from u to $\varphi(u)$. Removing c therefore disconnects u from $\varphi(u)$, as claimed. $\square$

Proposition 4.62. *If T is a tree with more than one edge, then*

$$\langle A(T) \cup A^-(T) \rangle \subseteq \langle A^+(T) \rangle.$$

Proof. We first show that every involution in $A(T)$ lies in $A^+(T)$; the result for the generated group then follows from the fact that every automorphism of a tree is a product of involutions.

Take a non-identity element $\sigma \in A(T)$ of order two and its induced automorphism $\varphi_\sigma \in \text{Aut}(T)$. By Corollary 4.51, there are precisely two cases.

φ_σ is a branch permutation. Suppose that φ_σ is a branch permutation. Since φ_σ is non-trivial, there exists $u \in V(T)$ with $\varphi_\sigma(u) \neq u$. Let c be the vertex of T as in Lemma 4.61, so that u and $\varphi_\sigma(u)$ lie in different components of $T - c$. The unique path between u and $\varphi_\sigma(u)$ in T passes through c , so $f := u\varphi_\sigma(u) \notin E(T)$. Since φ_σ has order two,

$$\varphi_\sigma[f] = \varphi_\sigma(u)\varphi_\sigma^2(u) = \varphi_\sigma(u)u = f,$$

so φ_σ preserves the edge f . Combined with the fact that φ_σ already preserves all edges of T , this gives $\varphi_\sigma \in \text{Aut}(T + f)$, and hence $\sigma \in A(T + f) \subseteq A^+(T)$.

φ_σ is an edge-inversion. Now assume that φ_σ is an edge-inversion, and let $xy \in E(T)$ be such that $\varphi_\sigma(x) = y$ and $\varphi_\sigma(y) = x$. Let T_x and T_y denote the connected components of $T - xy$ containing x and y , respectively; observe that $\varphi_\sigma|_{T - xy}$ is an automorphism of that exchanges connected components. Since T has at least two edges and $T_x \simeq T_y$, each tree has at least one edge. Hence, we can pick some $u \in V(T_x) \setminus \{x\}$ and set $v := \varphi_\sigma(u)$, so that $v \in V(T_y) \setminus \{y\}$ and, since T is acyclic, $uv \notin E(T)$. Define $f := uv \in E(T^c)$, and notice that $T + f$ is a unicyclic graph. Since φ_σ preserves all adjacencies and non-adjacencies of T , and since $\varphi_\sigma^2 = \text{id}$ implies $\varphi_\sigma(v) = u$, we obtain $\varphi_\sigma[f] = vu = f$. Hence $\varphi_\sigma \in \text{Aut}(T + f)$, so $\sigma \in A(T + f) \subseteq A^+(T)$.

Therefore $\langle A(T) \rangle \subseteq \langle A^+(T) \rangle$.

We next show that every involution in $A^-(T)$ lies in $A^+(T)$. Let $\sigma \in A^-(T)$ be of order two, and let $e = xy \in E(T)$ be such that $\sigma \in A(T - e)$. Set $F := T - e$, a forest with exactly two connected components F_x and F_y containing x and y , respectively. Then $\varphi_\sigma \in \text{Aut}(F)$ either fixes each component set-wise (and then acts as a branch permutation or edge-inversion when restricted to each component) or exchanges them. In either case, $f := \varphi_\sigma(x)\varphi_\sigma(y) \in E(F^c)$.

If $f = e$, then φ_σ preserves e set-wise. Combined with the fact that φ_σ preserves all edges of $F = T - e$, this gives $\varphi_\sigma \in \text{Aut}(T)$, so $\sigma \in A(T)$, and the previous part places $\sigma \in A^+(T)$.

Assume now that $f \neq e$. Since $E(T) = E(F) \cup \{e\}$, it follows that $f \notin E(F)$ and hence $f \notin E(T)$. Thus $T + f$ is a unicyclic graph, and we claim that $\varphi_\sigma \in \text{Aut}(T + f)$. Indeed, if $ab \in E(F)$, then $\varphi_\sigma(a) \varphi_\sigma(b) \in E(F) \subseteq E(T + f)$; and $\varphi_\sigma[e] = \varphi_\sigma(x) \varphi_\sigma(y) = f$. Therefore φ_σ preserves the edge set of $T + f$, and $\sigma \in A(T + f) \subseteq A^+(T)$.

Combining both parts, $\langle A(T) \cup A^-(T) \rangle \subseteq \langle A^+(T) \rangle$, as claimed. $\square$

The containment in Proposition 4.62 can fail if the generated groups are dropped: even when restricting to the two types of tree automorphisms, it is possible that $A(T) \not\subseteq A^+(T)$. Figure 4.13 shows examples of both cases.

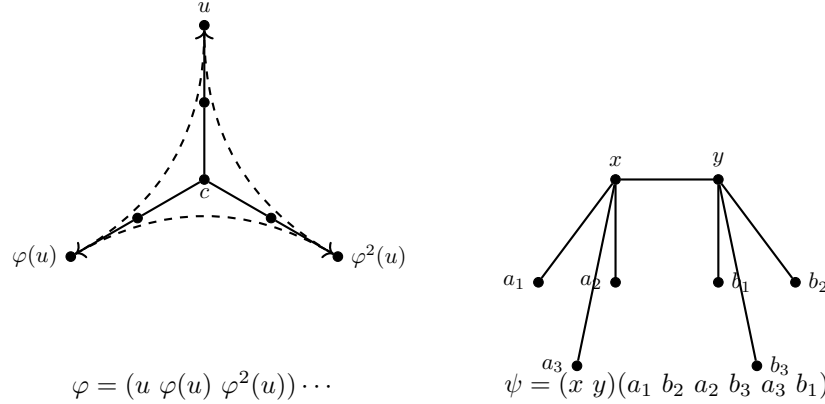


Figure 4.13: Branch permutation (left) and edge-inversion (right) that are **not** elements of $A^+(T)$ for their respective tree T .

All of the above is finite by stipulation. Section 4.5 takes the same combinatorial setup, namely graphs with labelled vertices, the symmetric group on labels, and the **Fer** group as a subgroup of S_X , into the descriptive-set-theoretic context, where X is countably infinite and S_X is a Polish group. The structural questions of this section then translate into questions about Borel and analytic subsets of S_X , and the role of A_n in the proof of Proposition 4.57 is replaced by classification questions in descriptive set theory.

What lies beyond

The forthcoming paper on which this section is based contains additional structural material, devoted to the study of the orbits of $\text{Fer}(T)$ acting on $V(T)$ for a global amoeba tree T , and to the *orbits graph* encoding the interaction between these orbits.

4.5 Infinitary amoebas

There are multiple reasons why studying countable graphs is desirable in this context, but they can all be somewhat summarized by the fact that if X is countable, X^X is a separable, completely metrizable topological space. Some interesting issues arise in the case when X is uncountable, which are explored in Section 4.5.2.

Let G be a graph with an arbitrary label set X , and write $V(G) = \{v_x : x \in X\}$. Definitions 4.21–4.28 extend verbatim by replacing S_n with S_X . In particular, $\text{Fer}(G)$ is generated by the

automorphism label group $A(G)$ and all witness permutations for feasible edge replacements of G . For the first part of this section, assume that X is countably infinite. In this case, X^X is homeomorphic to the Baire space ω^ω , and so we are dealing with separable and completely metrizable spaces (i.e. *Polish spaces*). One can show that S_X , being a G_δ subspace of the Baire space, is also Polish and thus its topology is induced by a complete metric. Indeed, observe that

$$S_X = \bigcap_{\substack{x, y \in X \\ x \neq y}} \{f \in X^X : f(x) \neq f(y)\} \cap \bigcap_{y \in X} \bigcup_{x \in X} \{f \in X^X : f(x) = y\}$$

is a countable intersection of open sets.

Throughout this section, closures of subsets of S_X are taken in the product topology on $S_X \subseteq X^X$.

Definition 4.63 (Local amoeba: infinite case). The graph G is a *local amoeba* if

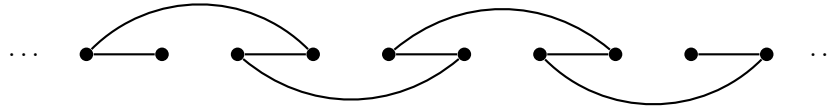
$$\overline{\text{Fer}(G)} = S_X.$$

Equivalently, every permutation of X can be matched on any prescribed finite set of labels by an element of $\text{Fer}(G)$; this equivalence is proved in Proposition 4.65 below.

Many properties of finite amoebas translate to our more general definition, such as the fact that $\text{Fer}(G) = \text{Fer}(G^c)$ for any graph G . Also, any complete graph on any set X is a local amoeba and, by the complementation identity above, so is the empty graph (the graph with no edges). As a first non-trivial example, the bi-infinite path $P_{\mathbb{Z}}$, i.e. the graph defined on $X = \mathbb{Z}$ where the edges are given by consecutive pairs of integers, is interestingly not a local amoeba. In fact in this context, bi-infinite paths play a similar role as finite cycles. Indeed, it is straightforward to verify that $\text{Fer}(P_{\mathbb{Z}}) = A(P_{\mathbb{Z}})$. Notice that the basic open set $[(0 \ 1)(2)]$ is disjoint from $\text{Fer}(P_{\mathbb{Z}})$ since the only automorphism interchanging 0 and 1 is a reflection (which does not fix the label 2), and thus $(0 \ 1) \notin \overline{\text{Fer}(P_{\mathbb{Z}})}$.



Bi-infinite path



Bi-infinite twisted path

Figure 4.14: Two bi-infinite paths that are isomorphic but cannot be transformed into each other using a (possibly infinite) sequence of feasible edge replacements.

Lemma 4.64. Let $\Gamma \leq S_X$ and $\varphi \in S_X$. Then $\varphi \in \overline{\Gamma}$ if and only if for every finite $F \subseteq X$, there is $\sigma \in \Gamma$ such that $\varphi|_F = \sigma|_F$.

Proof. For the forward direction, take φ and F as above. Clearly, $[\varphi|_F]$ is a basic open neighbourhood of φ and hence must intersect Γ . Any σ witnessing this has the sought property.

Conversely, take an arbitrary finite partial function $t: F \rightarrow X$ that is extended by φ . Pick $\sigma \in \Gamma$ with $\varphi|_F = \sigma|_F$. Clearly, $\sigma \in \Gamma \cap [t]$, proving that $\varphi \in \bar{\Gamma}$. $\square$

A group action on X is called n -transitive if $|X| \geq n$ and for any two pairwise distinct n -tuples $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$, there is a group element g such that $gx_i = y_i$ for all $1 \leq i \leq n$. Notice that 1-transitive is just the usual definition of *transitive*, i.e. that there is a single orbit. If for all $n \geq 1$, $\text{Fer}(G)$ acts n -transitively on X , we say $\text{Fer}(G)$ is *highly transitive*.

Proposition 4.65. *The following are equivalent.*

- (1) *The graph G is a local amoeba.*
- (2) *For every finite $F \subseteq X$ and every $\varphi \in S_X$, there is $\sigma \in \text{Fer}(G)$ such that $\varphi|_F = \sigma|_F$.*
- (3) *$\text{Fer}(G)$ is highly transitive.*

Proof. The implication $[1 \implies 2]$ follows from Lemma 4.64. Assume (2) and let $n \geq 1$ and $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ be two pairwise distinct tuples. Let $F := \{x_i : 1 \leq i \leq n\}$, then fix a bijection $X \setminus F \rightarrow X \setminus \{y_i : 1 \leq i \leq n\}$ (which exists since X is countably infinite and these two sets differ from X only in finite subsets) and define φ such that it agrees with said bijection and in addition satisfies $\varphi(x_i) = y_i$. By (2), some $\sigma \in \text{Fer}(G)$ has the desired property.

Finally, assume (3), let φ be an arbitrary permutation on X and $F = \{x_1, \dots, x_n\} \subseteq X$. Apply (3) to the tuples $(x_1, \dots, x_n)$ and $(\varphi(x_1), \dots, \varphi(x_n))$ to get $\sigma \in \text{Fer}(G)$. It follows that σ agrees with φ on F , hence $\varphi \in \overline{\text{Fer}(G)}$ by Lemma 4.64. Since φ was arbitrary, $\overline{\text{Fer}(G)} = S_X$, that is, G is a local amoeba. $\square$

It is worth pointing out that in the example of the bi-infinite path, $\text{Fer}(P_{\mathbb{Z}})$ acts transitively on $P_{\mathbb{Z}}$ but not 2-transitively, as the same aforementioned instance shows: the pairs of labels $(0, 2)$ and $(1, 2)$ witness this failure. In such cases, one could use the largest n for which the group acts n -transitively on $V(G)$ to gauge how close G is to being a local amoeba.

The next result provides an example of a graph where the gap between the automorphism group and the Fer group is as large as possible. Concretely, the countably infinite path on $X = \omega$, denoted P_{ω} where the edges are given by consecutive pairs of naturals, has a trivial automorphism group, but $\text{Fer}(P_{\omega})$ is dense in S_{ω} .

Proposition 4.66. *P_{ω} is a local amoeba.*

Proof. Let $F \subseteq \omega$ be finite and $\varphi \in S_{\omega}$. Define $n := \max(F \cup \varphi[F]) + 1$ and notice that $t := \varphi|_F$ is a bijection between finite subsets of n . In particular, there exists a bijection $s: n \setminus F \rightarrow n \setminus t[F]$. Now define $\sigma: X \rightarrow X$ by

$$\sigma(x) := \begin{cases} s(x), & x \in n \setminus F; \\ t(x), & x \in F; \\ x, & x \geq n. \end{cases}$$

Routine arguments show that $\sigma \in S_X$. Now consider the induced subgraph P_{n+1} and the permutation $\sigma|_{n+1}$. Observe that this permutation fixes the endpoint of the path, n , by definition. Thus by

Lemma 4.38, there is a sequence $f_0, \dots, f_m$ of canonical generators of $\text{Fer}(P_{n+1})$, all of which fix n , such that $\sigma|_{n+1} = f_0 \cdots f_m$.

For each $i \leq m$, we can extend f_i to a permutation $\overline{f_i}$ on X by fixing every label $x > n$. Clearly, each $\overline{f_i}$ is a member of $\text{Fer}(P_\omega)$ and thus $\sigma = \overline{f_0} \cdots \overline{f_m} \in \text{Fer}(P_\omega)$. Finally, the choice of σ makes it evident that $\sigma|_F = t = \varphi|_F$, and thus P_ω is a local amoeba by Proposition 4.65. $\square$

In the rest of this section we restrict to trees, where the topology of the underlying graph offers extra structure (the space of ends) that controls the closed subgroups of $\text{Aut}(T)$ and, in turn, the closure of $\text{Fer}(T)$.

4.5.1 The Freudenthal compactification of a tree and amenability

For the remainder of this section, T denotes a countably infinite, locally finite⁹ tree on a label set X , and $\text{Fer}(T)$ denotes the permutation group of feasible edge replacements as defined in Subsection 4.3.2. The structural theory of finite trees developed in Section 4.4 does not translate verbatim to this setting: Lemma 4.50, in particular, fails for infinite trees, where there is a third type of automorphisms: *translations*, which are automorphisms that fix neither vertex nor edge. The first instance of a classification for automorphisms of locally finite trees that the author is aware of was given by Jacques Tits [Tit70].

The *space of ends* of a locally finite connected graph G is a compactification of G that captures its “directions to infinity.” This notion was introduced by Freudenthal [Fre42] [Fre51]. Halin [Hal73] studied the action of automorphisms on ends, and the topic since became a standard tool in combinatorial and geometric group theory. We follow the presentation of Adams–Lyons [AL91].

Definition 4.67 (Ends). An *end* of a tree T is the equivalence class of an isometry $\pi: P_\omega \rightarrow T$ under the equivalence relation $\pi \sim \pi'$ if there are n, m such that for all $\ell \geq 0$, $\pi(m + \ell) = \pi'(n + \ell)$. The space of ends is called the *boundary* of T and denoted by ∂T .

Equip $\overline{T} := T \cup \partial T$ with the compact topology in which $V(T)$ is discrete and, for $S \subseteq V(T)$, the closure of S in $\overline{T}$ is $\overline{S} := S \cup \{s \in \partial T : \exists \pi \in s \text{ img}(\pi) \subseteq S\}$. Thus a neighbourhood base for $s \in \partial T$ is the collection of all $\overline{S}$ such that there is a finite $K \subseteq T$ with S the component of $T \setminus K$ containing some representative isometry of s and such that there exists $\pi \in s$ with $\text{img}(\pi) \subseteq S$.

The number of ends of a locally finite tree can take any value: a single ray P_ω has one end, the bi-infinite path $P_\mathbb{Z}$ has two ends, a “comb” tree obtained by attaching countably many disjoint rays at each vertex of a fixed ray has $\aleph_0$ ends, and the infinite 3-regular tree has continuum-many ends. Every automorphism of T induces a homeomorphism of ∂T .

We briefly recall the relevant notions of amenability in the context of groups acting on trees; see [Neb88] for a detailed treatment. A locally compact group G is *amenable* if it admits a left-invariant mean on $L^\infty(G)$. Compact groups and abelian groups are amenable; the free group on two generators is not. A key criterion for non-amenability is the existence of a discrete subgroup isomorphic to the free group F_2 .

Informally, this analytic definition is equivalent to several more combinatorial or dynamical characterizations that may be more familiar: G is amenable if and only if it satisfies Følner’s condition (for every compact $K \subseteq G$ and $\varepsilon > 0$ there is a compact $F \subseteq G$ with $|KF \triangle F| < \varepsilon|F|$,

⁹A graph is *locally finite* if every vertex has finitely many incident edges.

i.e. F is almost invariant under translation by K), and if and only if every continuous affine action of G on a compact convex set has a fixed point. When G is discrete, a mean on $L^\infty(G) = \ell^\infty(G)$ is, via indicator functions, the same thing as a finitely additive left-invariant probability measure defined on *all* subsets of G – the formulation most common in the combinatorics literature – so the definition above specializes to that one.

For the automorphism group $\text{Aut}(T)$ of a locally finite tree T , Tits [Tit70] proved that every automorphism φ falls into exactly one of three types:

1. *Rotations*: φ fixes a vertex $v \in V(T)$.
2. *Inversions*: φ swaps the endpoints of an edge $\{x, y\}$.
3. *Translations*: there exists a bi-infinite line $C = \{s_n\}_{n \in \mathbb{Z}}$ and an integer $j \neq 0$ such that $\varphi(s_n) = s_{n+j}$ for all n .

This trichotomy contrasts with the finite case (Lemma 4.50 and Corollary 4.51), where every automorphism of a finite tree is either a branch permutation (a rotation in Tits' terminology) or an edge-inversion.

Nebbia [Neb88] used this classification to characterize amenability for closed subgroups of $\text{Aut}(T)$.

Theorem 4.68 (Nebbia [Neb88, Theorem 1]). *Let T be a locally finite tree and Γ a closed subgroup of $\text{Aut}(T)$. Then Γ is amenable if and only if at least one of the following holds:*

1. Γ fixes a vertex, an edge, or a line (that is, the image of an isometry $P_{\mathbb{Z}} \rightarrow T$) set-wise in T ;
2. Γ fixes an element of cardinality 1 or 2 in ∂T set-wise.

The non-amenable case is witnessed by translations: if Γ contains two translations ϕ and ψ acting on two disjoint bi-infinite lines C_1 and C_2 of T , then $\langle \phi, \psi \rangle$ is a discrete subgroup of $\text{Aut}(T)$ isomorphic to F_2 .

We begin with a fact that motivates the rest of this subsection: if a feasible edge replacement on a finite tree fixes every leaf, then it must be the trivial replacement.

Lemma 4.69. *Let T be a finite tree, and assume that $\sigma \in \text{Fer}_T(e \rightarrow f)$ fixes the set of leaves $L(T)$ pointwise. Then $\sigma = \text{id}_X$ (and thus $e = f$).*

Proof. Let $\varphi_\sigma: T \rightarrow T - e + f$ denote the induced isomorphism. The argument proceeds by induction on $|V(T)|$.

If $|V(T)| \leq 2$, then either T has no leaves (the case $|V(T)| = 1$, where the conclusion is vacuous) or T consists of a single edge whose two endpoints are both leaves and are fixed by hypothesis, so the conclusion is immediate. So suppose that $|V(T)| \geq 3$, and thus T has two non-adjacent leaves; hence, T has a leaf v_ℓ that is not an endpoint of e . Let v_b denote the unique neighbour of v_ℓ in T . Since $\sigma(\ell) = \ell$ and φ_σ preserves degrees, v_ℓ is also a leaf in $T - e + f$, and moreover cannot be an endpoint of f either. Hence the unique neighbour of v_ℓ in $T - e + f$ is the same as in T , namely v_b , and so $\sigma(b) = b$.

Since v_ℓ is not an endpoint of e or f , removing v_ℓ commutes with the edge replacement, and φ_σ restricts to an isomorphism $T - v_\ell \rightarrow (T - v_\ell) - e + f$. Hence the restriction of σ to $X \setminus \{\ell\}$ produces a canonical generator $\sigma' \in \text{Fer}_T(e \rightarrow f)$ of $\text{Fer}(T - v_\ell)$. Every leaf of $T - v_\ell$ is either a leaf of T other than v_ℓ or, if $\deg_T(v_b) = 2$, the vertex v_b ; in both cases, b is fixed by σ . By induction, σ must fix every vertex label of T . $\square$

Unlike the finite case, when T is infinite, fixing the (possibly empty) set of leaves of T does not force triviality of canonical feasible edge replacements. Indeed, the unique infinite tree where every vertex has degree 2 except for a single vertex of degree 3 has feasible edge replacements that are non-trivial (meaning they do move edges).

Recall that a graph is *rigid* if $\overline{\text{Fer}}(G) = A(G)$ (cf. Subsection 4.3.2). For rigid trees, $\overline{\text{Fer}}(T) = \overline{A(T)} = A(T)$ (the last equality holds because $A(T)$ is closed in S_X : each pair $i, j \in X$ contributes the clopen condition that σ either preserves or fails to preserve the adjacency of v_i, v_j , and $A(T)$ is the intersection of these conditions over all pairs), and the amenability of $\overline{\text{Fer}}(T)$ reduces to Theorem 4.68.

Proposition 4.70. *Let T be the (countably infinite) r -regular tree, with $r \geq 2$. Then $\text{Fer}(T) = A(T)$.*

Proof. Let $\sigma \in \text{Fer}_T(e \rightarrow f)$ be a canonical generator with $e = \{u, v\}$ and $f = \{u', v'\}$. Since φ_σ is an isomorphism $T \rightarrow T - e + f$ and T is r -regular, $T - e + f$ is also r -regular. For each $x \in V(T)$,

$$\deg_{T-e+f}(x) = r - 1_{\{x \in \{u, v\}\}} + 1_{\{x \in \{u', v'\}\}}.$$

Equating this to r for every x forces $1_{\{x \in \{u, v\}\}} = 1_{\{x \in \{u', v'\}\}}$, that is, $\{u, v\} = \{u', v'\}$, i.e., $e = f$. Hence the only canonical generators of $\text{Fer}(T)$ are those induced by automorphisms. $\square$

In particular, the infinite r -regular tree is rigid for every $r \geq 2$. For $r \geq 3$, a standard fact in tree amenability is that $\text{Aut}(T)$ contains translations on disjoint lines, whence by the discussion following Theorem 4.68, $A(T)$ contains a copy of F_2 and is non-amenable. Thus $\overline{\text{Fer}}(T) = A(T)$ is non-amenable for these trees.

In the finite theory, a key structural result is that the canonical generators of $\text{Fer}(T)$ arising from automorphisms of T or of forests $T - e$ are accounted for by automorphisms of unicyclic graphs $T + f$. At the group level, $\langle A(T) \cup A^-(T) \rangle \subseteq \langle A^+(T) \rangle$ (Proposition 4.62). The element-level inclusion $A(T) \cup A^-(T) \subseteq A^+(T)$ fails when $\text{Aut}(T)$ contains elements of order > 2 (see Figure 4.13), but the group-level containment still holds because every tree automorphism decomposes as a product of involutions, each of which preserves a non-edge.

For infinite trees, translations introduce a genuinely new phenomenon. A translation ϕ_σ on a bi-infinite line $C \subseteq T$ has infinite order and moves every vertex of C , so no non-edge f is preserved by ϕ_σ . Thus $\sigma \notin A(T + f)$ for any $f \in E(T^c)$.

That said, an infinite unicyclic graph $U = T + f$ (where f is a non-edge of T) has a finite cycle C with finitely many infinite trees hanging off the vertices of C . Every automorphism of U either preserves or reverses C through a dihedral action, and the remaining structure is essentially a forest automorphism. In the closure $\overline{A(U)}$, the cycle contributes at most a finite group factor: given that C is finite, the restriction $\sigma|_{V(C)}$ is determined by finitely many values, and the “infinite content” of any element of $\overline{A(U)}$ lies in its action on the branches, which are trees. In this sense, infinite unicyclic graphs are indistinguishable from trees at infinity, and any asymptotic property of $\overline{\text{Fer}}(T)$ that depends only on the behaviour of permutations far from a fixed finite set is unaffected by adding a single edge.

Combining the preceding observations yields a partial picture of the amenability of $\overline{\text{Fer}}(T)$ for a countably infinite, locally finite tree T .

1. **Local amoeba trees** (e.g., P_ω): $\overline{\text{Fer}(T)} = S_X$, which, by a standard fact, is amenable as a Polish group.
2. **Rigid trees**: $\overline{\text{Fer}(T)} = A(T)$, and amenability is characterized by Nebbia's Theorem 4.68. The infinite r -regular tree, with $r \geq 3$, falls into this category by Proposition 4.70, and its $\overline{\text{Fer}(T)}$ is non-amenable.
3. **Intermediate trees**: non-regular trees that are not local amoebas, such as the infinite tree with a unique vertex v_0 of degree 3 and all other vertices of degree 2. Here $A(T) \subsetneq \overline{\text{Fer}(T)} \subsetneq S_X$ (to see the latter, consider σ fixing 0 and realizing $a \mapsto a_1$ where a and a_1 are consecutive on some ray; the basic open set $[\sigma|_{\{v_0, a\}}]$ is disjoint from $\text{Fer}(T)$), and the amenability of $\overline{\text{Fer}(T)}$ is not determined by either extreme case.

4.5.2 Baire-measurable colourings

For uncountable X , the topology on S_X is no longer second-countable, and in particular not sequential: a permutation $\varphi \in S_X$ may lie in the closure of a set $\Gamma \subseteq S_X$ without being a limit of any sequence in Γ . Example 4.71 below exhibits this phenomenon for the automorphism group of a natural graph, foreshadowing the kind of complications one must accommodate when adapting finite combinatorial arguments to the uncountable Baire-category setting.

Example 4.71. Let G be the graph on the vertex set 2^ω with edges $\{x, y\}$ where $x|_{\omega \setminus 1} \neq y|_{\omega \setminus 1}$.

Consider the map $s: 2^\omega \rightarrow 2^\omega$ that flips the first bit of an infinite binary sequence, that is, for every $x \in 2^\omega$, $s(i \frown x) := (1 - i) \frown x$. For each finite s -invariant set $F \subseteq 2^\omega$, define

$$\sigma_F(x) := \begin{cases} s(x), & x \in F; \\ x, & x \notin F. \end{cases}$$

It is straightforward to show that $\{\sigma_F : F \in [2^\omega]^{<\omega}\} \cup \{s\} \subseteq A(G) = \text{Fer}(G)$. We claim that $s \in \overline{\{\sigma_F : F \in [2^\omega]^{<\omega}, F = s[F]\}}$ but that no sequence of σ_F converges to s . In other words, s belongs to $\text{Fer}(G)$ but is not a finite product of permutations of the form σ_F , despite being topologically close to them.

To prove membership in the closure, let $F \subseteq 2^\omega$ be finite and consider the basic open neighbourhood $\{\varphi \in S_{2^\omega} : \varphi|_F = s|_F\}$ of s . The set $F' := F \cup s[F]$ is finite and s -invariant, and $\sigma_{F'}$ agrees with s on every element of F (since $F \subseteq F'$). Hence $\sigma_{F'}$ belongs to this neighbourhood, proving that $s \in \overline{\{\sigma_F : F \in [2^\omega]^{<\omega}, F = s[F]\}}$.

Now, consider any sequence $\{F_n : n < \omega\}$ of finite subsets of the Cantor space. Since each F_n is finite, the union $\bigcup_n F_n$ is countable; as 2^ω is uncountable, there exists a binary sequence x that is not in any F_n . Therefore, the pointwise limit $\lim_{n \rightarrow \infty} \sigma_{F_n}(x) = x$ differs from $s(x)$ at the first bit and thus cannot be equal.

In the finite setting, a 2-edge-colouring of K_n contains a balanced copy of a balanceable graph G provided each colour class has more than $\text{bal}(n, G)$ edges. The natural Baire-category transposition of “many edges of each colour” is “non-meagre colour classes,” and the goal of this subsection is to show that two of the central finite results of the previous sections, namely Theorem 4.19 and Theorem 4.41, have infinitary analogues.

The proofs share the following localization lemma. A subtlety, absent in the finite theory, is that the localization is necessarily *off-diagonal*: it produces a pair of disjoint open sets whose cross-pair set carries a comeagre colour class, rather than a single open set on whose full pair-set the colour is comeagre. This phenomenon is unavoidable. For example, let $X = (0, 1)$ and define a colouring

$$c: [X]^2 \rightarrow 2 \quad \text{by} \quad c(x, y) = 0 \iff |\{x, y\} \cap (0, 1/2)| = 1.$$

Then both colour classes are open and non-meagre. However, for every non-empty open set $U \subseteq X$, either:

- $U \subseteq (0, 1/2)$ or $U \subseteq (1/2, 1)$, in which case $[U]^2$ is monochromatic; or
- U meets both halves of $(0, 1)$, in which case both colours occur on non-meagre open subsets of $[U]^2$.

Thus no colour class is comeagre on every $[U]^2$.

Lemma 4.72. *Let X be a perfect Polish space, and let $\pi: X^2 \setminus \Delta \rightarrow [X]^2$ denote the quotient map. Given disjoint non-meagre sets $A, B \subseteq [X]^2$ with the Baire property, there exist pairwise disjoint non-empty open sets $U_A^1, U_A^2, U_B^1, U_B^2 \subseteq X$ such that A is comeagre in $\pi[U_A^1 \times U_A^2]$ and B is comeagre in $\pi[U_B^1 \times U_B^2]$.*

Proof. Since A and B have the Baire property, there exist open sets $O_A, O_B \subseteq [X]^2$ such that $A \Delta O_A$ and $B \Delta O_B$ are meagre. Because A and B are non-meagre, the sets O_A and O_B are non-empty. Moreover, $O_A \cap O_B \subseteq (A \Delta O_A) \cup (B \Delta O_B)$, since $A \cap B = \emptyset$. Hence $O_A \cap O_B$ is meagre as well. But $O_A \cap O_B$ is open in the Polish space $[X]^2$, so by the Baire category theorem, $O_A \cap O_B = \emptyset$.

Let

$$\mathcal{O}_A := \pi^{-1}[O_A], \quad \mathcal{O}_B := \pi^{-1}[O_B].$$

These are disjoint non-empty open subsets of $X^2 \setminus \Delta$. Choose $(\alpha_1, \alpha_2) \in \mathcal{O}_A$. Since $\alpha_1 \neq \alpha_2$ and X is metrizable, there exist disjoint open neighbourhoods

$$V_A^1 \ni \alpha_1, \quad V_A^2 \ni \alpha_2$$

such that $V_A^1 \times V_A^2 \subseteq \mathcal{O}_A$. Similarly, choose disjoint non-empty open sets $V_B^1, V_B^2 \subseteq X$ with $V_B^1 \times V_B^2 \subseteq \mathcal{O}_B$. Since X is perfect, every non-empty open set is infinite. Pick points $a_1 \in V_A^1$ and $a_2 \in V_A^2$. Since $V_A^1 \cap V_A^2 = \emptyset$, we have $a_1 \neq a_2$; then $b_1 \in V_B^1 \setminus \{a_1, a_2\}$, possible by perfectness; finally $b_2 \in V_B^2 \setminus \{a_1, a_2, b_1\}$, again by perfectness. The four points $a_1, a_2, b_1, b_2 \in X$ are pairwise distinct by construction. Using metrizability again, choose pairwise disjoint open neighbourhoods

$$U_A^1 \ni a_1, \quad U_A^2 \ni a_2, \quad U_B^1 \ni b_1, \quad U_B^2 \ni b_2$$

such that

$$U_A^1 \subseteq V_A^1, \quad U_A^2 \subseteq V_A^2, \quad U_B^1 \subseteq V_B^1, \quad U_B^2 \subseteq V_B^2.$$

Then

$$\pi[U_A^1 \times U_A^2] \subseteq O_A, \quad \pi[U_B^1 \times U_B^2] \subseteq O_B.$$

Now

$$\pi[U_A^1 \times U_A^2] \setminus A \subseteq O_A \setminus A \subseteq A\Delta O_A,$$

and $A\Delta O_A$ is meagre in $[X]^2$. Since U_A^1 and U_A^2 are disjoint, $U_A^1 \times U_A^2 \subseteq X^2 \setminus \Delta$, and therefore $\pi[U_A^1 \times U_A^2]$ is open in $[X]^2$. Since $\pi[U_A^1 \times U_A^2]$ is open in $[X]^2$, the subset $\pi[U_A^1 \times U_A^2] \setminus A$ is meagre in $\pi[U_A^1 \times U_A^2]$ as well. Hence A is comeagre in $\pi[U_A^1 \times U_A^2]$. The same argument applies to B . $\square$

We recall the following classical theorem of Galvin.

Fact 4.73 (Galvin). *Let X be a perfect Polish space and let $c: [X]^2 \rightarrow k$ be a Baire-measurable colouring with finitely many colours. Then there exists a Cantor set $C \subseteq X$ such that $[C]^2$ is monochromatic.*

A proof may be found in [Kec95, Theorem 19.6]. We shall also use the following product form of Mycielski's theorem.

Fact 4.74 (Product Mycielski). *Let X, Y be perfect Polish spaces, let $U \subseteq X$ and $V \subseteq Y$ be non-empty open sets, and let $R \subseteq U \times V$ have the Baire property and be comeagre in $U \times V$. Then there exist Cantor sets $C \subseteq U$, $D \subseteq V$ such that $C \times D \subseteq R$.*

Proof sketch. Write $(U \times V) \setminus R = \bigcup_n M_n$ with each M_n nowhere dense. A standard fusion construction simultaneously builds binary trees of open sets

$$\{U_s : s \in 2^{<\omega}\} \subseteq U, \quad \{V_s : s \in 2^{<\omega}\} \subseteq V,$$

with shrinking diameters and pairwise disjoint successors, while ensuring that for every k and every $s, t \in 2^k$, $\overline{U_s} \times \overline{V_t} \cap M_k = \emptyset$. The resulting limit sets C, D are Cantor sets satisfying $C \times D \subseteq R$. $\square$

The next lemma is the key extraction step.

Lemma 4.75 (Joint Cantor extraction). *Let X be a perfect Polish space, let $U^1, U^2 \subseteq X$ be disjoint non-empty open sets, and let $c: [X]^2 \rightarrow 2$ be a Baire-measurable colouring such that $c^{-1}\{\text{red}\}$ is comeagre in $\pi[U^1 \times U^2]$. Then there exist Cantor sets*

$$D^1 \subseteq U^1, \quad D^2 \subseteq U^2$$

such that:

- (i) $[D^1]^2$ is monochromatic;
- (ii) $[D^2]^2$ is monochromatic;
- (iii) $c(x, y) = \text{red}$ for all $x \in D^1$ and $y \in D^2$.

Proof. Define $R := \{(x, y) \in U^1 \times U^2 : c(x, y) = \text{red}\}$. Since $c^{-1}\{\text{red}\}$ is comeagre in $\pi[U^1 \times U^2]$, and $\pi: U^1 \times U^2 \rightarrow \pi[U^1 \times U^2]$ is continuous and open, the set R is comeagre in $U^1 \times U^2$. Moreover, R has the Baire property because c is Baire measurable.

Applying Fact 4.74, there exist Cantor sets

$$P^1 \subseteq U^1, \quad P^2 \subseteq U^2$$

such that $P^1 \times P^2 \subseteq R$. Equivalently, $c(x, y) = \text{red}$ for all $x \in P^1$, $y \in P^2$.

Now apply Fact 4.73 to the restriction $c|_{[P^1]^2}$. There exists a Cantor set $D^1 \subseteq P^1$ such that $[D^1]^2$ is monochromatic.

Similarly, applying Fact 4.73 to $c|_{[P^2]^2}$, there exists a Cantor set

$$D^2 \subseteq P^2$$

such that $[D^2]^2$ is monochromatic. Since $D^1 \times D^2 \subseteq P^1 \times P^2 \subseteq R$, every cross-pair between D^1 and D^2 is red. $\square$

Remark 4.76. The statement of Lemma 4.75 is symmetric in the two colours: its hypothesis and conclusion exchange under the swap $\text{red} \leftrightarrow \text{blue}$. Hence the lemma may equally be applied with blue in the role of the comeagre colour, the conclusion then being that every cross-pair is blue.

We will also use a trivial discrete intermediate value theorem.

Lemma 4.77 (Discrete IVT). *Let $(\ell_i)_{i=0}^t$ be a sequence of integers with $\ell_0 = a$, $\ell_t = b$, and $|\ell_{i+1} - \ell_i| \leq 1$ for all i . Then every integer between a and b inclusive is equal to some ℓ_i .*

An infinitary analogue of constant balancing. Theorem 4.19 characterizes graphs of constant balancing number as those admitting both a matching $\frac{k}{2}K_2$ and a star $K_{1,k/2}$. The result below is the natural Baire-category analogue: every non-meagre Baire-measurable 2-colouring of $[X]^2$ on a perfect Polish space contains both a monochromatic Cantor clique in one colour and a Cantor-indexed star in the other colour.

Theorem 4.78. *Let X be a perfect Polish space and suppose that $c: [X]^2 \rightarrow 2$ is a Baire-measurable colouring whose colour classes are non-meagre. Then there exist a colour $i < 2$, a vertex $v_0 \in X$, and a Cantor set $C \subseteq X$ such that:*

- (i) $c(v_0, x) = i$ for every $x \in C$;
- (ii) $[C]^2$ is monochromatic in colour $1 - i$.

In particular, $[C]^2$ contains a matching of cardinality continuum in colour $1 - i$.

Proof. Apply Lemma 4.72 with

$$A := c^{-1}\{\text{red}\}, \quad B := c^{-1}\{\text{blue}\},$$

to obtain pairwise disjoint non-empty open sets $U_A^1, U_A^2, U_B^1, U_B^2 \subseteq X$ such that red is comeagre in $\pi[U_A^1 \times U_A^2]$ and blue is comeagre in $\pi[U_B^1 \times U_B^2]$. Apply Lemma 4.75 on the A -side. We obtain Cantor sets

$$D_A^1 \subseteq U_A^1, \quad D_A^2 \subseteq U_A^2$$

such that:

- every cross-pair between D_A^1 and D_A^2 is red;
- each of $[D_A^1]^2$ and $[D_A^2]^2$ is monochromatic.

Let $j_A^1, j_A^2 \in \{\text{red}, \text{blue}\}$ denote the colours of $[D_A^1]^2$ and $[D_A^2]^2$ respectively.

Case 1. At least one of j_A^1, j_A^2 equals blue.

Concretely:

- if $j_A^2 = \text{blue}$, fix $v_0 \in D_A^1$ and let $C := D_A^2$;
- if $j_A^1 = \text{blue}$ and $j_A^2 = \text{red}$, fix $v_0 \in D_A^2$ and let $C := D_A^1$;
- if $j_A^1 = j_A^2 = \text{blue}$, either choice works; pick $v_0 \in D_A^1$ and $C := D_A^2$.

In each configuration, every edge between v_0 and a point of C is a cross-pair between D_A^1 and D_A^2 , hence red, while $[C]^2$ is monochromatic blue. Set $i := \text{red}$.

Case 2. $j_A^1 = j_A^2 = \text{red}$.

By Remark 4.76, we may apply Lemma 4.75 on the B -side with blue in the role of the comeagre colour. We obtain Cantor sets

$$D_B^1 \subseteq U_B^1, \quad D_B^2 \subseteq U_B^2$$

such that:

- every cross-pair between D_B^1 and D_B^2 is blue;
- each of $[D_B^1]^2$ and $[D_B^2]^2$ is monochromatic.

Let j_B^1, j_B^2 denote the corresponding clique colours.

Subcase 2.1. At least one of j_B^1, j_B^2 equals red.

By the analogue of Case 1 with the roles of red and blue swapped:

- if $j_B^2 = \text{red}$, fix $v_0 \in D_B^1$ and let $C := D_B^2$;
- if $j_B^1 = \text{red}$ and $j_B^2 = \text{blue}$, fix $v_0 \in D_B^2$ and let $C := D_B^1$;
- if $j_B^1 = j_B^2 = \text{red}$, pick $v_0 \in D_B^1$ and $C := D_B^2$.

In each configuration, every edge between v_0 and a point of C is a cross-pair between D_B^1 and D_B^2 , hence blue, while $[C]^2$ is monochromatic red. Set $i := \text{blue}$.

Subcase 2.2. $j_B^1 = j_B^2 = \text{blue}$.

In this configuration $D_A^1 \cup D_A^2$ is an all-red Cantor clique and $D_B^1 \cup D_B^2$ is an all-blue Cantor clique, but the cross-colour between the two sides has not yet been controlled. We extract a controlled cross between D_A^1 and D_B^1 via the Baire property and Fact 4.74.

Since c is Baire-measurable, the two colour classes partition $\pi[D_A^1 \times D_B^1]$ into sets with the Baire property, at least one of which is non-meagre in $D_A^1 \times D_B^1$. Let $k \in \{\text{red}, \text{blue}\}$ denote the non-meagre colour. Since D_A^1 and D_B^1 are Cantor sets, they are themselves perfect Polish spaces in the subspace topology. Thus there exist relatively open non-empty sets $W_A \subseteq D_A^1$ and $W_B \subseteq D_B^1$ such that $c^{-1}\{k\}$ is comeagre in $W_A \times W_B$. Applying Fact 4.74 inside the perfect Polish spaces D_A^1 and D_B^1 , we obtain Cantor sets

$$\tilde{D}_A \subseteq W_A \subseteq D_A^1, \quad \tilde{D}_B \subseteq W_B \subseteq D_B^1$$

such that $\tilde{D}_A \times \tilde{D}_B \subseteq c^{-1}\{k\}$. Hence every cross-pair between $\tilde{D}_A$ and $\tilde{D}_B$ has colour k .

If $k = \text{red}$, fix any $v_0 \in \tilde{D}_A$ and let $C := \tilde{D}_B$. Then every edge between v_0 and C is red, while $[C]^2 \subseteq [D_B^1]^2$ is monochromatic blue. Set $i := \text{red}$.

If $k = \text{blue}$, fix any $v_0 \in \tilde{D}_B$ and let $C := \tilde{D}_A$. Then every edge between v_0 and C is blue, while $[C]^2 \subseteq [D_A^1]^2$ is monochromatic red. Set $i := \text{blue}$.

The cases above exhaust all possibilities for (j_A^1, j_A^2) and (j_B^1, j_B^2) , and in every case the resulting v_0, C, i satisfy conditions (i) and (ii). Finally, if C is any Cantor set such that $[C]^2$ is monochromatic in colour $1 - i$, then $[C]^2$ contains a matching of cardinality continuum in colour $1 - i$: fixing a homeomorphism $\varphi: 2^\omega \rightarrow C$, the family

$$\{\{\varphi(0 \smallfrown x), \varphi(1 \smallfrown x)\} : x \in 2^\omega\}$$

is a pairwise vertex-disjoint family of monochromatic edges of cardinality $2^{\aleph_0}$. $\square$

An infinitary analogue of $\text{bal}(n, G) = \text{ex}(n, \mathcal{H}_G)$. Theorem 4.41 expresses the balancing number of a global amoeba as a Turán-type threshold. Its upper-bound direction asserts that sufficiently dense red and blue edge sets force a balanced copy of the target graph. The theorem below is the corresponding Baire-category statement: replacing positive density by non-meagreness of the colour classes forces a balanced copy of every finite global amoeba.

Theorem 4.79. *Let X be a perfect Polish space, let G be a finite global amoeba on $n \geq 1$ vertices and m edges, and let $c: [X]^2 \rightarrow 2$ be a Baire-measurable colouring with non-meagre colour classes. Then K_X , with the colouring c , contains a balanced copy of G .*

Proof. The result is trivial when $m \leq 1$, so assume that $m \geq 2$.

By the global amoeba property of G , there exists an integer $N \geq n$ such that any two copies of G in K_{2N} are connected by a chain of feasible edge replacements.

Apply Lemma 4.72 to the colour classes to obtain pairwise disjoint non-empty open sets

$$U_A^1, U_A^2, U_B^1, U_B^2$$

such that red is comeagre in $\pi[U_A^1 \times U_A^2]$ and blue is comeagre in $\pi[U_B^1 \times U_B^2]$.

Apply Lemma 4.75 on the A -side. We obtain Cantor sets

$$D_A^1 \subseteq U_A^1, \quad D_A^2 \subseteq U_A^2$$

such that:

- every cross-pair between D_A^1 and D_A^2 is red;
- each of $[D_A^1]^2$ and $[D_A^2]^2$ is monochromatic.

Let $j_A^1, j_A^2 \in \{\text{red}, \text{blue}\}$ denote the colours of $[D_A^1]^2$ and $[D_A^2]^2$ respectively.

Case 1. $\{j_A^1, j_A^2\} = \{\text{red}, \text{blue}\}$. Without loss of generality,

$$j_A^1 = \text{red}, \quad j_A^2 = \text{blue}.$$

Choose sets $V_R \subseteq D_A^1$ and $V_B \subseteq D_A^2$ each of cardinality N , and set

$$V := V_R \cup V_B.$$

Then:

- every edge inside V_R is red;
- every edge inside V_B is blue;
- every cross-edge between V_R and V_B is red.

Choose copies $G_R, G_B \subseteq K_V$ of G with

$$V(G_R) \subseteq V_R, \quad V(G_B) \subseteq V_B.$$

Then all edges of G_R are red and all edges of G_B are blue. By the choice of N , there exists a chain $G_R = H_0, H_1, \dots, H_t = G_B$ of copies of G in K_V such that each H_{i+1} is obtained from H_i by a feasible edge replacement.

Let ℓ_i denote the number of red edges of H_i . Then $\ell_0 = m$, $\ell_t = 0$, and $|\ell_{i+1} - \ell_i| \leq 1$ for every i . By Lemma 4.77, some H_i satisfies $\ell_i \in \{\lfloor m/2 \rfloor, \lceil m/2 \rceil\}$, hence is balanced.

Case 2. $j_A^1 = j_A^2 = \text{blue}$.

Choose sets $V_1 \subseteq D_A^1$ and $V_2 \subseteq D_A^2$ each of cardinality N , and set $V := V_1 \cup V_2$. Then:

- every edge inside V_1 or V_2 is blue;
- every cross-edge between V_1 and V_2 is red.

Let $G^{(1)}$ be a copy of G inside V_1 . Then every edge of $G^{(1)}$ is blue, so the number of red edges equals 0. Next choose a partition $V(G) = L \sqcup R$ realizing $\text{MaxCut}(G)$. By the classical probabilistic argument of Erdős, $\text{MaxCut}(G) \geq \frac{m}{2}$, hence $\text{MaxCut}(G) \geq \lceil m/2 \rceil$. Choose an injection $\iota: V(G) \rightarrow V$ with $\iota[L] \subseteq V_1$ and $\iota[R] \subseteq V_2$. Let $G^{(2)}$ denote the image copy of G under ι . Then, $G^{(2)}$ has exactly $\text{MaxCut}(G)$ red edges.

Now connect $G^{(1)}$ and $G^{(2)}$ by a sequence of feasible edge replacements inside K_V . The number of red edges begins at 0 and ends at at least $\lceil m/2 \rceil$, changing by at most 1 at each step. Lemma 4.77 therefore yields a balanced copy of G .

Case 3. $j_A^1 = j_A^2 = \text{red}$. Applying Lemma 4.75 on the B -side (with blue in the role of the comeagre colour, by Remark 4.76), we obtain Cantor sets $D_B^1 \subseteq U_B^1$ and $D_B^2 \subseteq U_B^2$ such that:

- every cross-pair between D_B^1 and D_B^2 is blue;
- each of $[D_B^1]^2$ and $[D_B^2]^2$ is monochromatic.

Let j_B^1, j_B^2 denote the corresponding clique colours.

Subcase 3.1. $\{j_B^1, j_B^2\} = \{\text{red}, \text{blue}\}$. This is symmetric to Case 1 with the roles of red and blue reversed.

Subcase 3.2. $j_B^1 = j_B^2 = \text{red}$. This is symmetric to Case 2 with the roles of red and blue reversed.

Subcase 3.3. $j_B^1 = j_B^2 = \text{blue}$. Choose sets $V_R \subseteq D_A^1$ and $V_B \subseteq D_B^1$ each of cardinality N . Then every edge inside V_R is red and every edge inside V_B is blue. Choose copies $G_R \subseteq K_{V_R}$ and $G_B \subseteq K_{V_B}$ of G . Then all edges of G_R are red and all edges of G_B are blue. By the choice of N , there exists a sequence of feasible edge replacements $G_R = H_0, \dots, H_t = G_B$ inside $K_{V_R \cup V_B}$. The number of red edges decreases from m to 0 in steps of size at most 1, so Lemma 4.77 yields a balanced copy. The cases above exhaust all possibilities, completing the proof. $\square$

Remark 4.80. Theorem 4.78 and Theorem 4.79 are different in character. The former produces explicit infinitary structure: a Cantor-indexed monochromatic star together with a monochromatic Cantor clique in the opposite colour in every non-meagre Baire-measurable colouring. The latter instead uses the finite global amoeba structure of the target graph G to interpolate between monochromatic copies via feasible edge replacements.

Both arguments rely on the localization lemma together with Galvin's theorem and product forms of Mycielski's theorem.

Chapter 5

Closing Remarks and Open Problems

5.1 Further questions on Borel dichromatic number

This section collects several natural questions suggested by the results presented in Section 2.3, the work of Raghavan and Xiao [RX24], and suggestions of Thornton.

Question 5.1 (Minimality of the Raghavan–Xiao basis.). Raghavan and Xiao [RX24] produce a continuum-size family of directed graphs that serves as a basis for Borel directed graphs with uncountable Borel dichromatic number. This family consists of pairwise Borel-incomparable directed graphs. *Does every basis for this class have size $\mathfrak{c}$?* Raghavan has noted (personal communication) that the directed graphs in his family are pairwise non-homomorphic, but it is unknown whether a smaller basis exists.

Question 5.2 (Structural restrictions.). The result presented here applies to the full class of Borel directed graphs. *Does a countable basis exist when one restricts to structurally simpler classes, say locally finite acyclic Borel directed graphs, or directed graphs generated by a single Borel function?* In the undirected case, Miller [Mil08] proved that there is no countable $\leq_B$ -basis for graphs of the form G_f (where f is a Borel function) with $\chi_B(G_f) \geq 3$. The analogous question for directed graphs generated by a single Borel function and dichromatic number remains open.

Question 5.3 (Borel dichromatic number and Borel order dimension.). Raghavan and Xiao [RX24] established a close connection between Borel dichromatic number and a notion of Borel order dimension for Borel quasi-orders. *Does the complexity result of Proposition 2.14 yield new complexity results for Borel order dimension?*

Question 5.4 (Dichotomies for complete multipartite graphs.). It is known that checking whether a Borel graph in which every connected component is complete k -partite admits a Borel k -colouring is Π_1^1 , but no satisfying dichotomy theorem is known for this problem. Thornton has suggested (personal communication) that a finite basis may exist for each k , though its size appears to grow faster than $n!$. *Is there an explicit finite basis for each k ? What is its growth rate?*

Question 5.5 (Bounded width CSPs.). Among the constraint satisfaction problems studied in the Borel setting, the *bounded width* problems occupy a distinguished position. Thornton [Tho22a] showed that the structures whose every solvable Borel instance has a Borel solution are exactly the width 1 structures, and proved partial complexity results for certain bounded width structures. Recent work of Grebík and Vidnyánszky [GV25] shows that the split between easy and hard problems lies at a different place in the Borel setting than in the classical CSP dichotomy: in particular, there is no dichotomy for any Borel CSP that is not of bounded width. However, sharp complexity bounds within bounded width remain known only in very special cases. *Can the exact complexity dividing line within bounded width CSPs be determined?* This problem appears to be very difficult; even identifying good illustrative examples of bounded width CSPs beyond the well-studied special cases remains open.

Question 5.6 (Measurable arboricity.). In the undirected setting, the *vertex arboricity*, the natural undirected analogue of dichromatic number, of a graph (the minimum number of acyclic sets needed to cover the vertex set) is the undirected analogue of the dichromatic number. Several basic questions about arboricity in measurable combinatorics remain open. For instance: *Can the μ -measurable arboricity of a probability-measure-preserving graph differ from its (classical) arboricity by more than 1?* And: *What is the μ -measurable arboricity of the k -th power of the Bernoulli shift graph of F_2 ?*

5.2 Uncountable sets and an infinite linear order game

When does the converse of Proposition 2.31 hold? By the results of Section 2.4 such orderings must be ω_1 -free and ω_1^* -free. Separable and even ccc linear orderings have this property. The separable case (the reals) was established in [BC23]. Then, the next step is to study dense linear orders that are ω_1 -free, ω_1^* -free but not separable. A classical example of such ordering is the Aronszajn ordering.

Definition 5.1. An *Aronszajn line* (*A-line*) is an uncountable linearly ordered set that is ω_1 -free, ω_1^* -free and contains no uncountable subset of the reals.

Following [Tod95], define a proper decomposition for A-lines. Let A be a dense Aronszajn line. Start with $D_0 := \emptyset$. Given a countable $D_\alpha \subseteq A$, consider the equivalence relation on $A \setminus D_\alpha$: $x \sim_\alpha y$ if and only if there is no $z \in D_\alpha$ such that $x < z < y$. Each $\sim_\alpha$ class is convex. Let T_α be the set of all $\sim_\alpha$ classes and let D'_α be a selector of the $\sim_\alpha$ classes. Since D_α is countable and dense in $D_{\alpha+1} := D_\alpha \cup D'_\alpha$, then $D_{\alpha+1}$ must be countable because A does not contain uncountable separable subsets. Given D_α for $\alpha < \gamma$ and γ limit, let $D_\gamma = \bigcup_{\alpha < \gamma} D_\alpha$. Since A does not contain copies of ω_1 and ω_1^* , $A = \bigcup_{\alpha < \omega_1} D_\alpha$. Also, $(T, \supseteq)$ is an Aronszajn tree where $T = \bigcup_{\alpha < \omega_1} T_\alpha$.

Question 5.7. Is there a dense Aronszajn line A where Player II has a winning strategy in $BG(A, S)$ for all $S \subseteq A$?

Another class of lines where a similar decomposition can be performed is the class of nowhere separable Souslin continua.

Definition 5.2. A *Souslin continuum* is a non-separable, dense, complete linearly ordered set that has the countable chain condition (ccc), i.e., every family of pairwise disjoint open intervals is countable. A Souslin continuum is *nowhere separable* if no non-empty open interval is separable.

Define a *proper decomposition* for the nowhere separable Souslin continuum L as follows. Let $D_0 := D \neq \emptyset$ be any countable set. Given a countable $D_\alpha \subseteq L$ consider the equivalence relation on $L \setminus \overline{D_\alpha}$: $x \sim_\alpha y$ if and only if there is no $z \in D_\alpha$ such that $x < z < y$. Again, each $\sim_\alpha$ class is convex. Let T_α be the set of all $\sim_\alpha$ classes and let D'_α be a selector of the $\sim_\alpha$ classes. Since L has the ccc, then T_α is countable. Let $D_{\alpha+1} = D_\alpha \cup D'_\alpha$. For $\gamma < \omega_1$ limit let $D_\gamma = \bigcup_{\alpha < \gamma} D_\alpha$.

Lemma 5.3. *If $\{D_\alpha : \alpha < \omega_1\}$ is a proper decomposition of a nowhere separable Souslin continuum L , then*

$$L = \bigcup_{\alpha < \omega_1} \overline{D_\alpha}.$$

Proof. Suppose that $x \in L$ and $x \notin \bigcup_{\alpha < \omega_1} \overline{D_\alpha}$. For each $\xi < \omega_1$ let $I_\xi^0 := [x]_\xi$ (the $\sim_\xi$ class of x). Note that $I_\eta^0 \subseteq I_\xi^0$ for all $\xi < \eta$.

Claim 5.3.1. *For all $\alpha < \omega_1$, $|T_\alpha| > 1$.*

Proof of Claim. Let $x_0 \in D_\alpha$. Since L is nowhere separable, $\overline{D_\alpha}$ is nowhere-dense so pick $y_1 \in (L \setminus \overline{D_\alpha}) \cap (-\infty, x_0)$ and $y_2 \in (L \setminus \overline{D_\alpha}) \cap (x_0, \infty)$. Then, $[y_1]_\alpha, [y_2]_\alpha \in T_\alpha$. $\dashv$

For each $\xi < \omega_1$ let $I_{\xi+1}^1 \in T_{\xi+1}$ such that $I_{\xi+1}^1 \cap I_{\xi+1}^0 = \emptyset$. Then, $\{I_{\xi+1}^1 : \xi < \omega_1\}$ is an uncountable pairwise disjoint family of open sets, contradicting ccc. $\square$

Settling the following conjecture, presumably by means of a proper decomposition argument, remains a question of great interest.

Conjecture 5.1. Let L be a nowhere separable Souslin continuum. Then Player II has a winning strategy for $BG(L, S)$ if and only if S is countable.

5.3 Minimal rigid graphs

A conjecture of Nešetřil, on which both Nešetřil and Sabidussi made substantial progress, and finally recently resolved by Schweitzer and Schweitzer [SS17], asserts that there are exactly 18 *minimal asymmetric graphs*: graphs G satisfying $\text{Aut}(G) = 1$ such that every proper induced subgraph of G on at least two vertices has a nontrivial automorphism. Most graphs are asymmetric¹, and every graph is an induced subgraph of some asymmetric graph, so one might expect the class of minimal asymmetric graphs to be very rich. That only finitely many exist is quite surprising.

Theorem 1.8 (Schweitzer–Schweitzer [SS17]). *There exists a set $\mathcal{B}$ of exactly 18 finite graphs, the minimal asymmetric graphs, such that every finite asymmetric graph G contains some $B \in \mathcal{B}$ as an induced subgraph; that is, there is an injective homomorphism $\varphi: B \hookrightarrow G$ that preserves non-adjacency. Moreover, the elements of $\mathcal{B}$ are pairwise incomparable under the induced subgraph ordering.*

¹Precisely, Erdős and Rényi [ER63] showed that the proportion of labelled graphs on n vertices with trivial automorphism group tends to 1 as $n \rightarrow \infty$.

It is natural to ask whether an analogous result holds for finite rigid² graphs, that is, graphs satisfying $\text{Fer}(G) = A(G)$. Since $A(G) \leq \text{Fer}(G)$ always holds, rigidity is the condition that Fer is as small as possible. Say that a graph G is *minimal rigid* if $\text{Fer}(G) = A(G)$ and every proper induced subgraph H of G on at least two vertices satisfies $\text{Fer}(H) \supsetneq A(H)$.

Unfortunately, the naïve translation of the Schweitzer–Schweitzer theorem to this setting fails. Every odd cycle C_{2n+1} is rigid: since C_{2n+1} is regular, no degree-preserving edge replacement exists, so $\text{Fer}(C_{2n+1}) = A(C_{2n+1})$ vacuously. On the other hand, every proper induced subgraph of C_{2n+1} on at least two vertices is a disjoint union of paths, and paths are non-rigid (they are, in fact, local amoebas). Thus each C_{2n+1} is minimal rigid. Moreover, distinct odd cycles are pairwise incomparable under the induced subgraph ordering, so the family $\{C_{2n+1} : n \geq 1\}$ is an infinite antichain of minimal rigid graphs.

The issue is that the rigidity of odd cycles is, in a sense, uninteresting: it is forced entirely by the degree sequence. A regular graph admits no degree-preserving edge replacement at all, so the equation $\text{Fer}(G) = A(G)$ holds for trivial reasons. This motivates the following refinement. A graph G is *genuinely rigid* if $\text{Fer}(G) = A(G)$ and there exists at least one degree-preserving edge replacement (e, f) for G , that is, a pair consisting of an edge e and a non-edge f such that $G - e + f$ has the same degree sequence as G , in particular $G - e + f \not\cong G$ for every such pair. In other words, the degree sequence permits a nontrivial feasible edge replacement to exist, but the deeper structure of G prevents it.

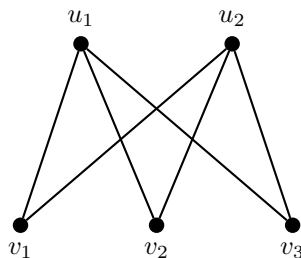


Figure 5.1: The complete bipartite graph $K_{2,3}$, the smallest genuinely rigid graph.

The smallest genuinely rigid graph is $K_{2,3}$ (see Figure 5.1). The degree sequence of $K_{2,3}$ is $(3, 3, 2, 2, 2)$, and there exist pairs (e, f) preserving this sequence; however, every such replacement $K_{2,3} - e + f$ contains an odd cycle and is therefore not bipartite, hence not isomorphic to $K_{2,3}$.

Question 5.8. Are there finitely many minimal genuinely rigid graphs? That is, are there finitely many genuinely rigid graphs G such that no proper induced subgraph of G on at least two vertices is genuinely rigid?

A computational search reveals that all 37 genuinely rigid graphs on at most 7 vertices have nontrivial automorphism groups, and none of them are vertex-transitive. We also verified that all 18 minimal asymmetric graphs from [SS17] satisfy $\text{Fer}(G) \supsetneq A(G)$, so the two classes of minimal graphs are disjoint. A positive answer to Question 5.8 would further connect the Fer group to classical structural graph theory, and a classification of the minimal genuinely rigid graphs would be a natural part of the basis framework developed in this thesis.

²See Definition 4.30.

5.4 Infinitary amoebas

I conclude with several questions motivated by the preceding discussion.

Question 5.9. Characterize the countably infinite, locally finite trees that are local amoebas. Is having a vertex of degree 1 both necessary and sufficient? Or is an additional condition on the growth rate or branching structure required?

Question 5.10. Does the finite characterization of global amoebas (Theorem 4.34) extend to infinite trees? Specifically, if T is an infinite local amoeba tree, is $T \sqcup K_1$ necessarily also a local amoeba?

Question 5.11. Is $\overline{\text{Fer}(T)}$ always amenable for countably infinite, locally finite non-rigid trees? Feasible edge replacements are inherently local (each canonical generator changes a single edge), so the closure may be unable to produce the “independent translations on disjoint lines” that generate free subgroups in $\text{Aut}(T)$. A positive answer would show that the Fer group, despite potentially being much larger than $\text{Aut}(T)$, is always tamer in terms of amenability.

Question 5.12. For trees T in the intermediate case ($A(T) \subsetneq \overline{\text{Fer}(T)} \subsetneq S_X$), what is the structure of $\overline{\text{Fer}(T)}$ as a closed subgroup of S_X ? Is it determined by the degree profile and branching structure of T ? Are there combinatorial invariants of T (number of ends, growth rate, degree sequence) that classify $\overline{\text{Fer}(T)}$ up to isomorphism?

Question 5.13. What is the Borel complexity of the local amoeba property among countable trees? The set of canonical generators of $\text{Fer}(T)$ is definable from T by a local condition (a single edge change preserving isomorphism type), so $\text{Fer}(T)$ is an analytic, possibly Borel, subgroup of S_ω . Is it always Borel? Is the property “ T is a local amoeba” (equivalently, $\overline{\text{Fer}(T)} = S_\omega$) itself Borel, or does it require a Σ_1^1 or Π_1^1 quantifier?

Question 5.14. Does the group-level containment $\langle A(T) \cup A^-(T) \rangle \subseteq \langle A^+(T) \rangle$ from Proposition 4.62 hold for infinite trees? The main obstacle is translations: a translation on a bi-infinite line preserves no non-edge and has infinite order, so it cannot belong to $A(T + f)$ for any f . Can it nevertheless be expressed as a product of elements that do?

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